

Well Completion Overview

Jakarta, 25 Sept 2026

Mohamad Ressa Jodhitya

**Sr. Completion & Well Test Engineer (Completions Specialist) – Pertamina Hulu
Energi Offshore South East Sumatera (PHE OSES)**

By:



Resume



20+ years experience in Drilling, Completion and Well Intervention engineering and supervisory (operations) with extensive knowledge in completions including intelligent completions system (IWS), re-completions, well abandonment, well integrity and well services/well intervention engineering – both onshore and offshore

EDUCATIONAL BACKGROUND

- Magister Teknik Perminyakan, Institut Teknologi Bandung, August 2010 – March 9th, 2012
- Sarjana Teknik Perminyakan, Universitas Pembangunan Nasional “Veteran”, Yogyakarta, August 2000 – April 1st, 2005

WORK EXPERIENCES

- Aug 2023 – Now: Sr. Completions & Well Testing Engineer (PHE OSES)
- Mar – July 2024: Completions Engineering Specialist (PEP-C)
- Apr – July 2023: Completions Engineering Specialist (PHE TEJ)
- Nov 2022 – Mar 2023: Sr. Completions & Well Testing Engineer (PHE OSES)
- Aug 2021 – Sept 2022: Completions Engineering Specialist (PHE WMO)
- Mar 2020 – July 2021: Sr. Completions & Well Testing Engineer (PHE OSES)
- Jan 2014 – Jan 2020: Completion Engineer (BP Berau Ltd.)
- Sept 2012 – Jan 2014: Completion Engineer (Mobil Cepu Ltd.)
- Dec 2009 – Sept 2012: Sr. Well Services & Completion Engineer (ConocoPhillips Indonesia Inc. Ltd.)
- June 2008 – Dec 2009: Drilling Supervisor (Chevron Pacific Indonesia)
- Sept 2006 – May 2008: Well Operations Engineer (ConocoPhillips Indonesia Inc. Ltd.)
- Feb – Aug 2006: Sales Engineer (PT Catur Khita Persada)
- Oct – Dec 2005: Underbalance Drilling Engineer (Weatherford Indonesia)

Agenda

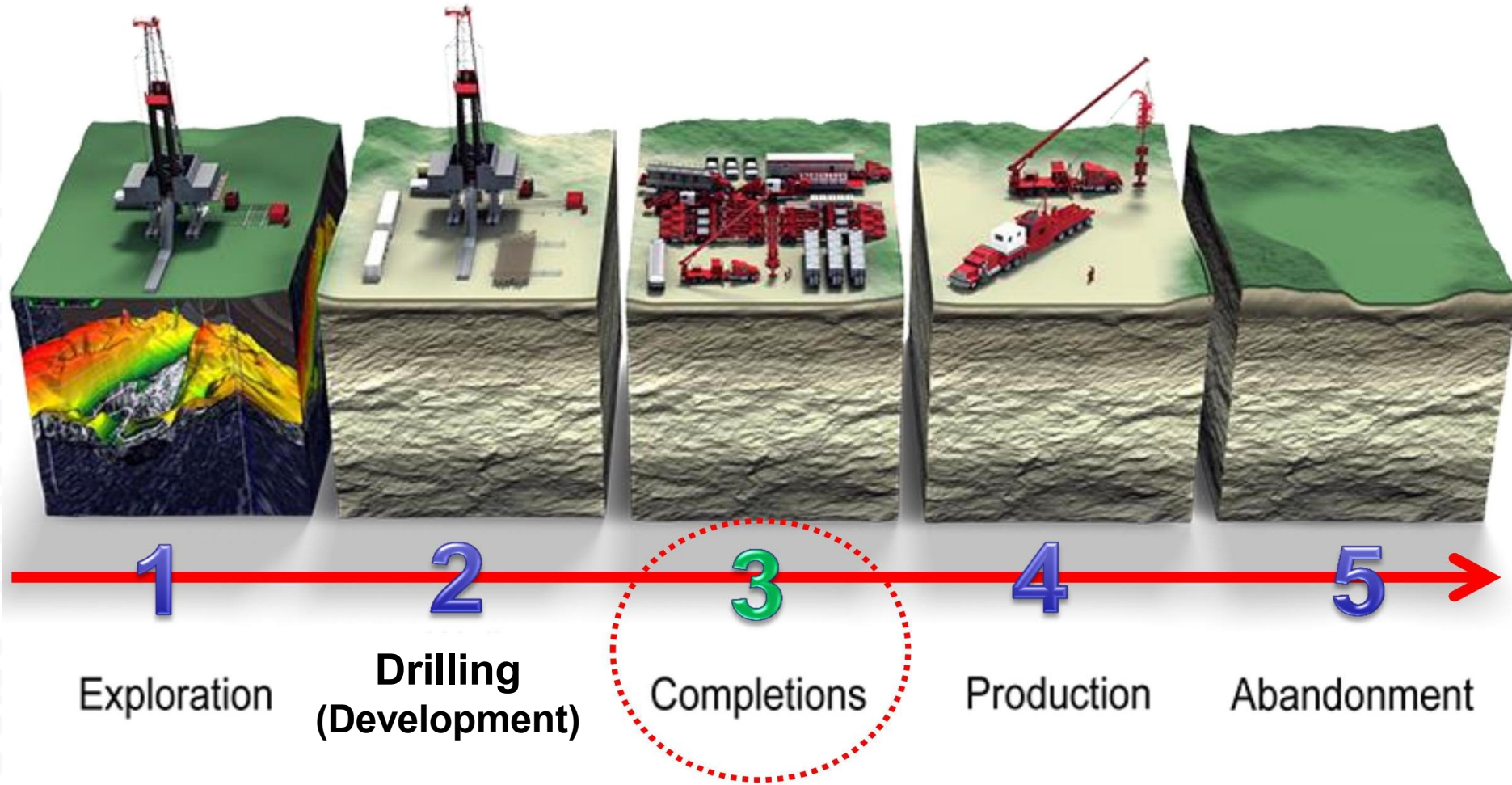
1. Introduction to Well Completions (What is Well Completion?)
2. Production Well System
3. Summary Of Completion Process
4. Completion Design
5. Types of Well Completion
6. Well Completion Equipment and Accessories
7. Completions Operation Sequences
8. Material Selection for Tubular
9. Well Head & Tree Design Criteria (API 6A 21st Ed.)
10. Quiz Time!!!



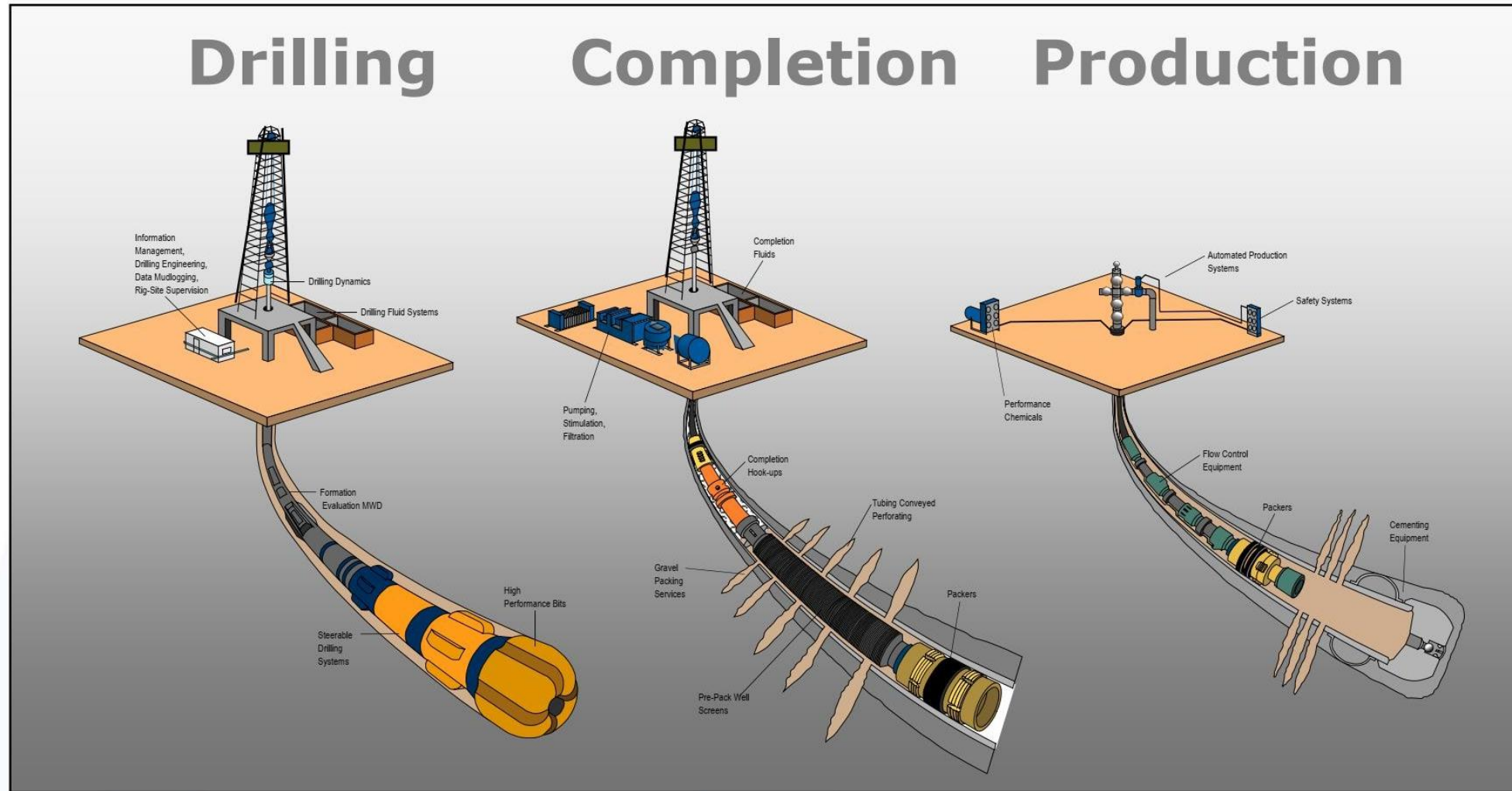
1. Introduction to Well Completions (What is Well Completion?)

- After careful interpretation and consideration on well test data (coring, logging, etc), a decision is made whether to set production casing and complete the well or to plug and abandon it.
- Decision to abandon is made when the well is not capable to produce oil or gas in commercial quantities.
- However, sometimes wells that were plugged and abandoned at one time in the past may be reopened and produced if the price of oil or gas has become more favorable.
- “Completing a well” means installing equipment in the well to allow a safe and controlled flow of petroleum (HC) from the well.
- A series of activities to prepare an oil well or a gas well, so that the well can be flowed in a controlled manner. All wells have to be completed.
- In addition to the casing that lines the wellbore, *tubing* and a system of flow valves must be installed.
- Cannot operate alone – must joint effort with other sub-disciplines such as production engineering and reservoir engineering.

Well Life Cycle



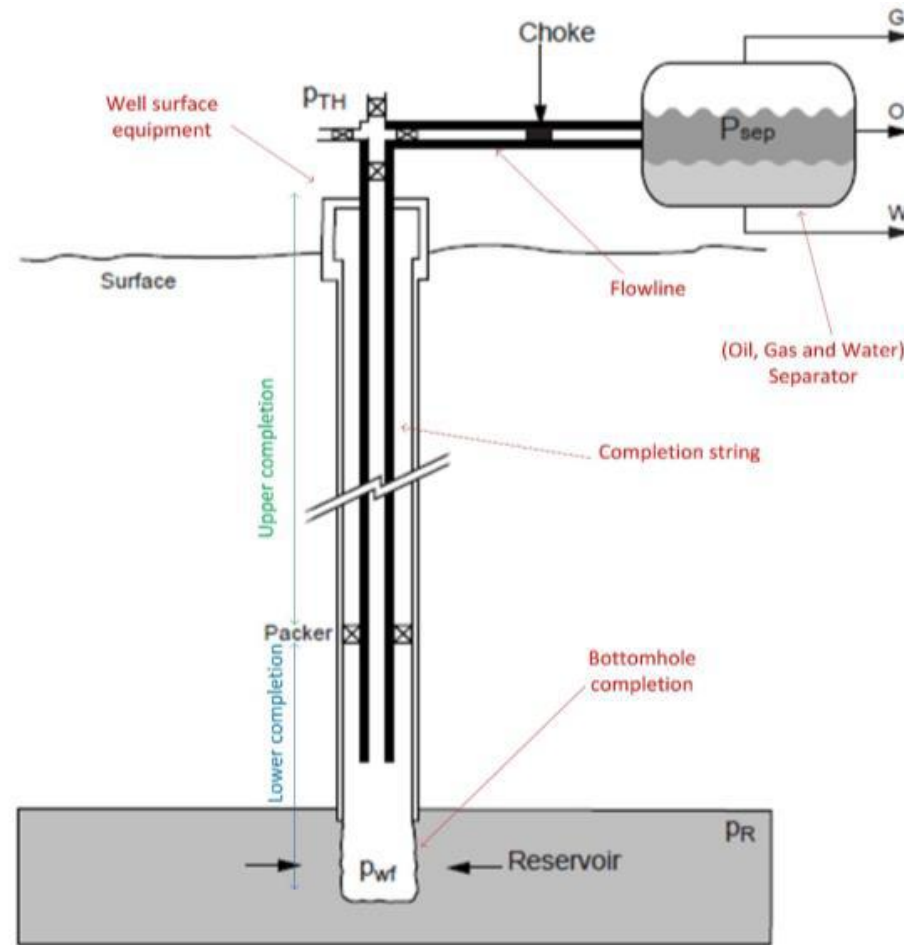
Well Life Cycle



2. Production Well System

When Drilling:

$$P_{\text{Wellbore}} > P_R$$



$$P_R > P_{wf}$$



3. Summary of Completion Process

WHY COMPLETION ENGINEERING MATTERS?



KEY QUALIFICATIONS / EXPERTISE

- Lead and support Gravel Pack design and execution for optimal well performance.
- Design tubing strings to meet well performance requirements.
- Support well delivery through effective completion and material management.
- Support Preparing Completion Design and Program
- Support pre-job analysis for intervention Design & Program

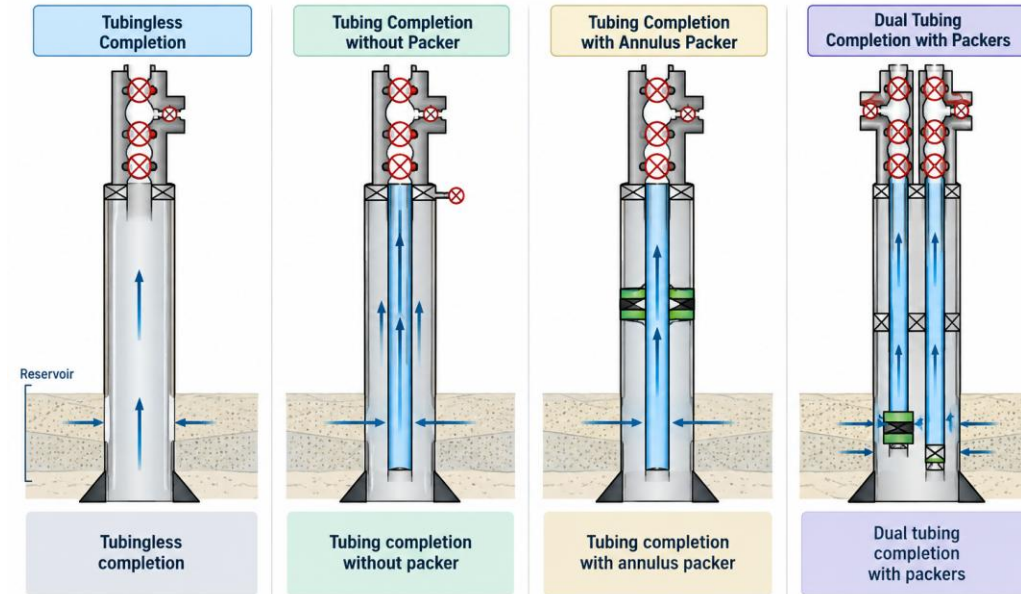
THE IMPACT

- SAFER OPERATIONS**
Reduce risk, protect people and assets
- RELIABLE PRODUCTION**
Unlock reservoir potential
- COST EFFICIENCY**
Lower well cost and optimize capital spending
- LONG TERM VALUE**
Sustainable and profitable assets

**RIGHT COMPLETION.
BRIGHTER TOMORROW.**

UPPER COMPLETION METHODS

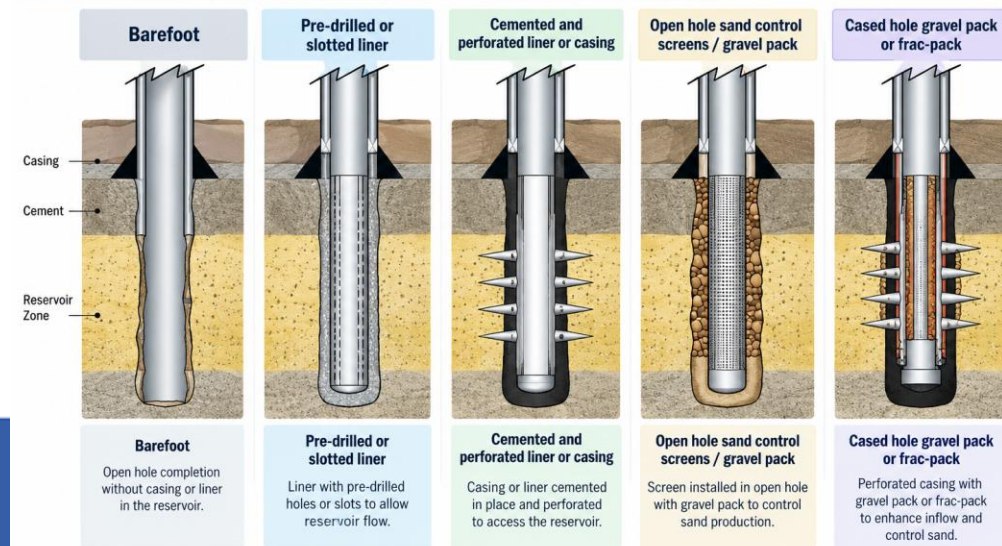
COMMON COMPLETION CONFIGURATIONS



RESERVOIR COMPLETION METHODS

COMMON RESERVOIR COMPLETION CONFIGURATIONS

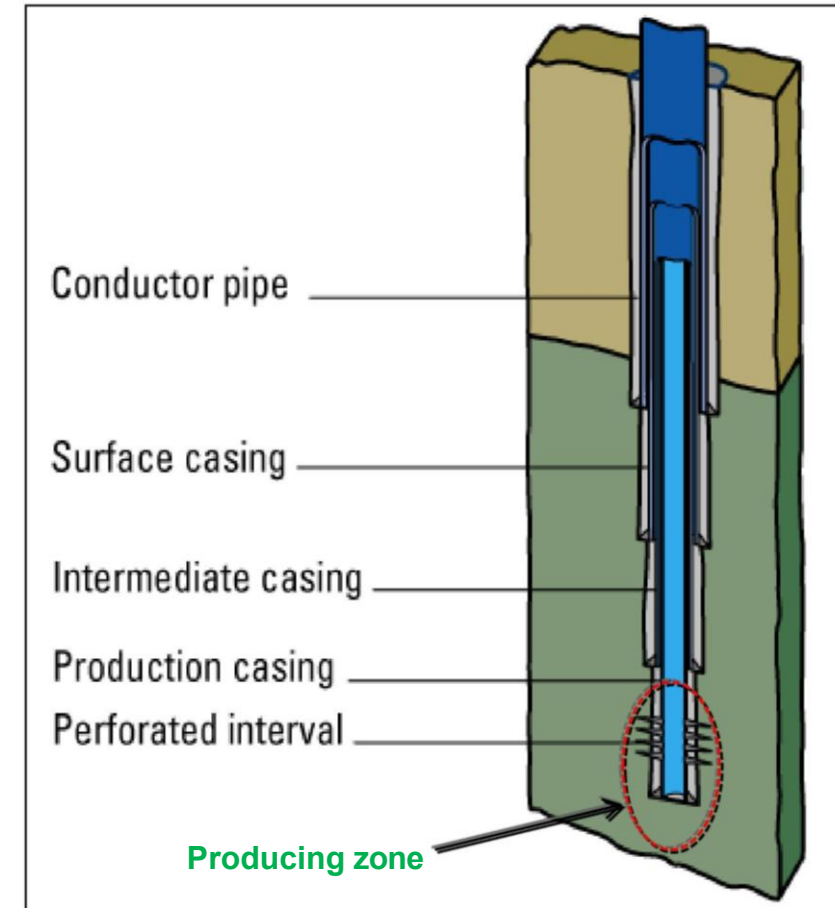
MATCHING RESERVOIR CONDITIONS
WITH THE RIGHT COMPLETION SOLUTION
FOR SUSTAINABLE PRODUCTION



Summary of Completion Process

Setting production casing

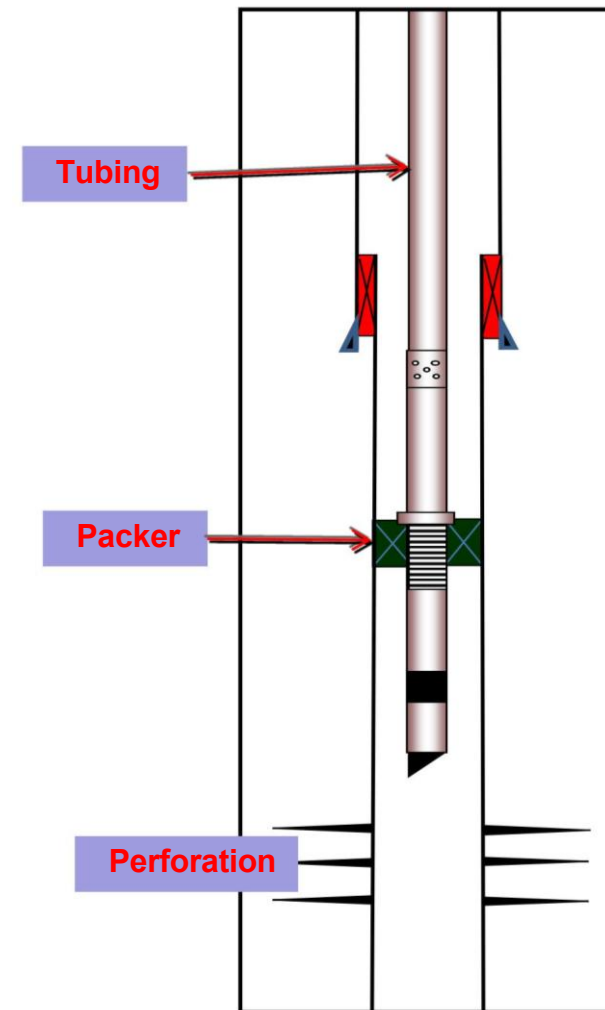
- Production casing is the final casing in a well.
- The hole is drilled beyond the producing interval.
- Production casing is set and cemented through the pay zone.
- The casing and cement actually seal off the producing zone



Summary of Completion Process

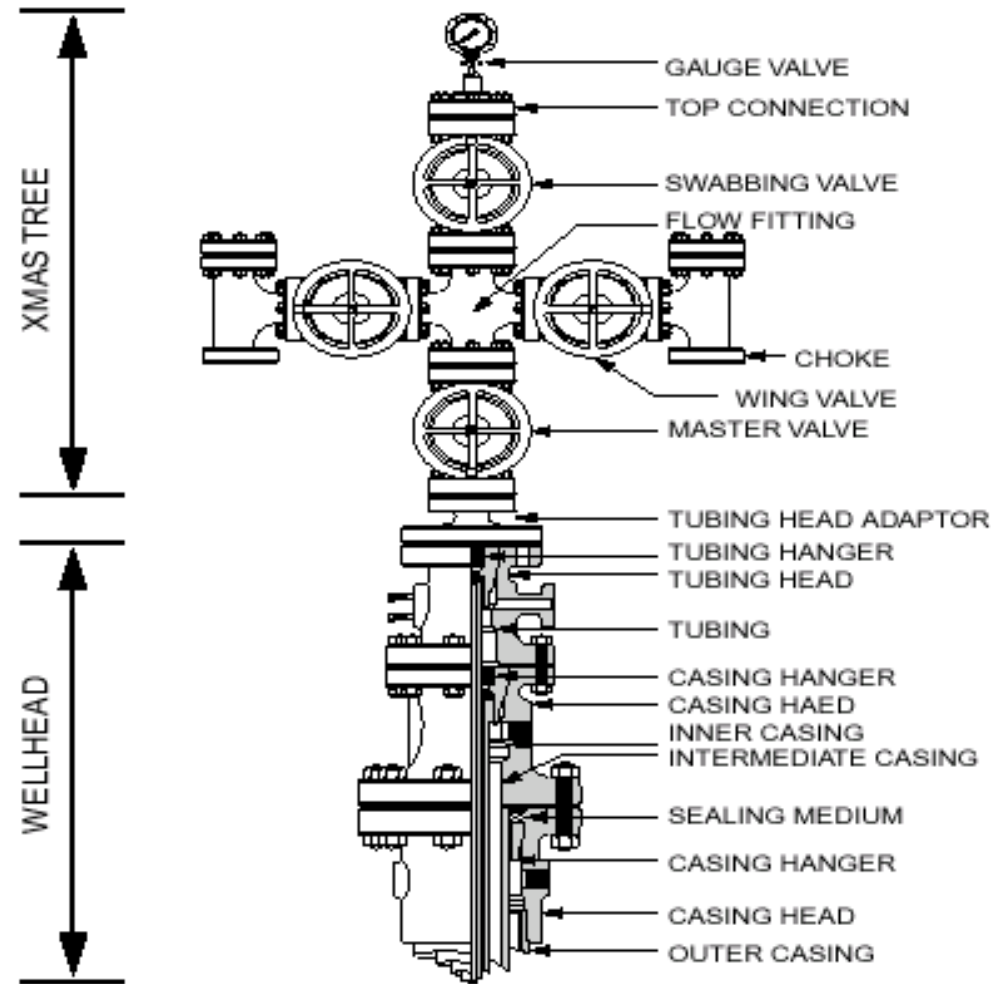
Installing the Tubing and packer

- Tubing is run into the well inside production casing to help oil or gas or condensate to flow to the surface.
- Packer is placed at a depth just above the producing interval (perforation)
- packer seal between outside of tubing and inside of casing to protect the casing from damage during the production (corrosion, H_2S , CO_2 ...etc)

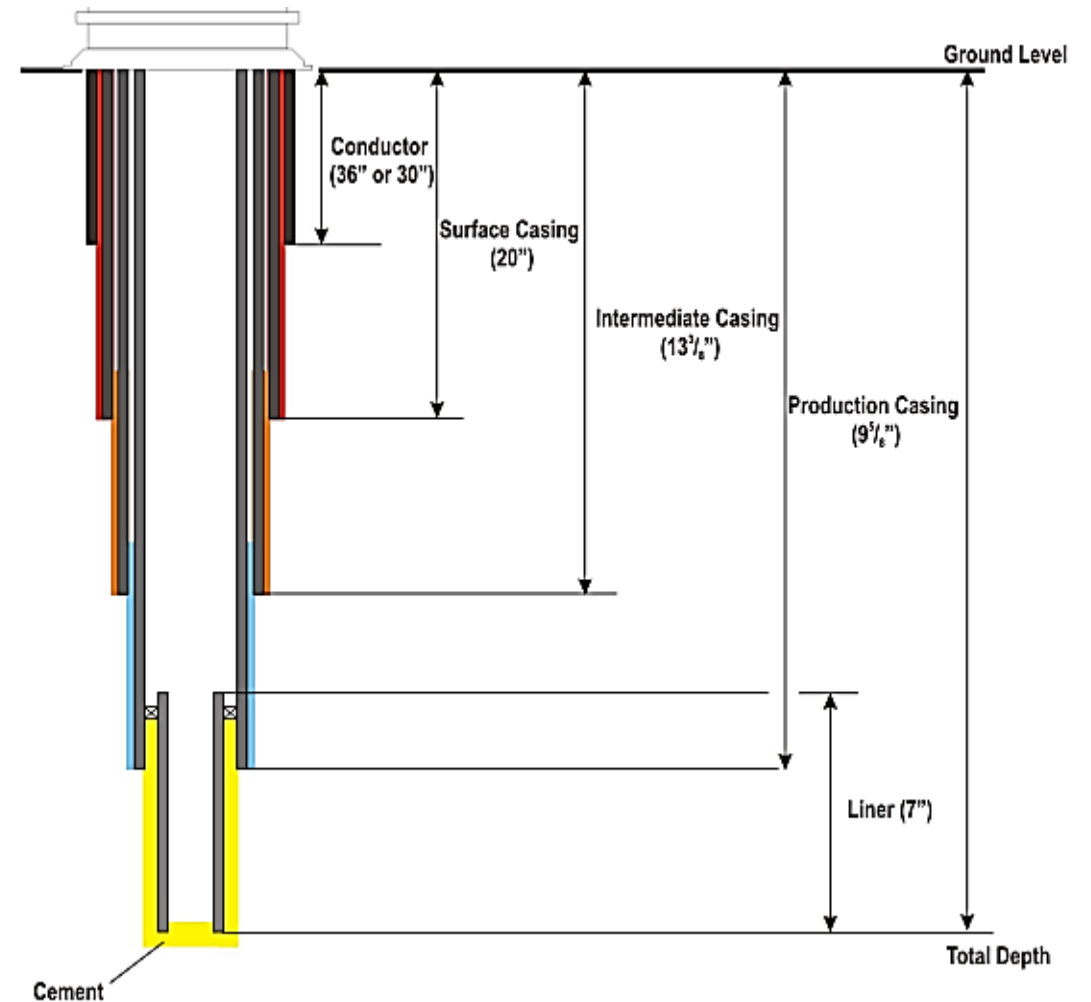
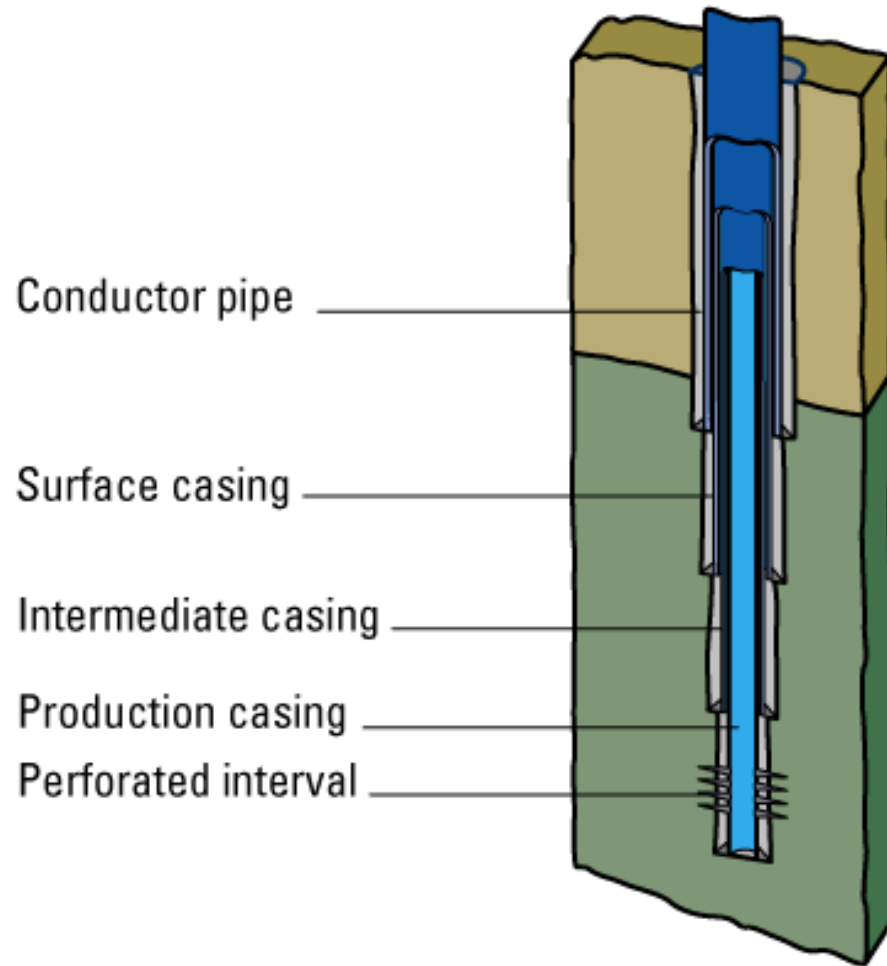


Installing the Christmas Tree

- A collection of valves called a Christmas tree is installed on the surface at the top of the casing hanger.
- As the well's production flows up the tubing, it enters the christmas tree.
- So, the production can be controlled by opening or closing valves on the christmas tree.



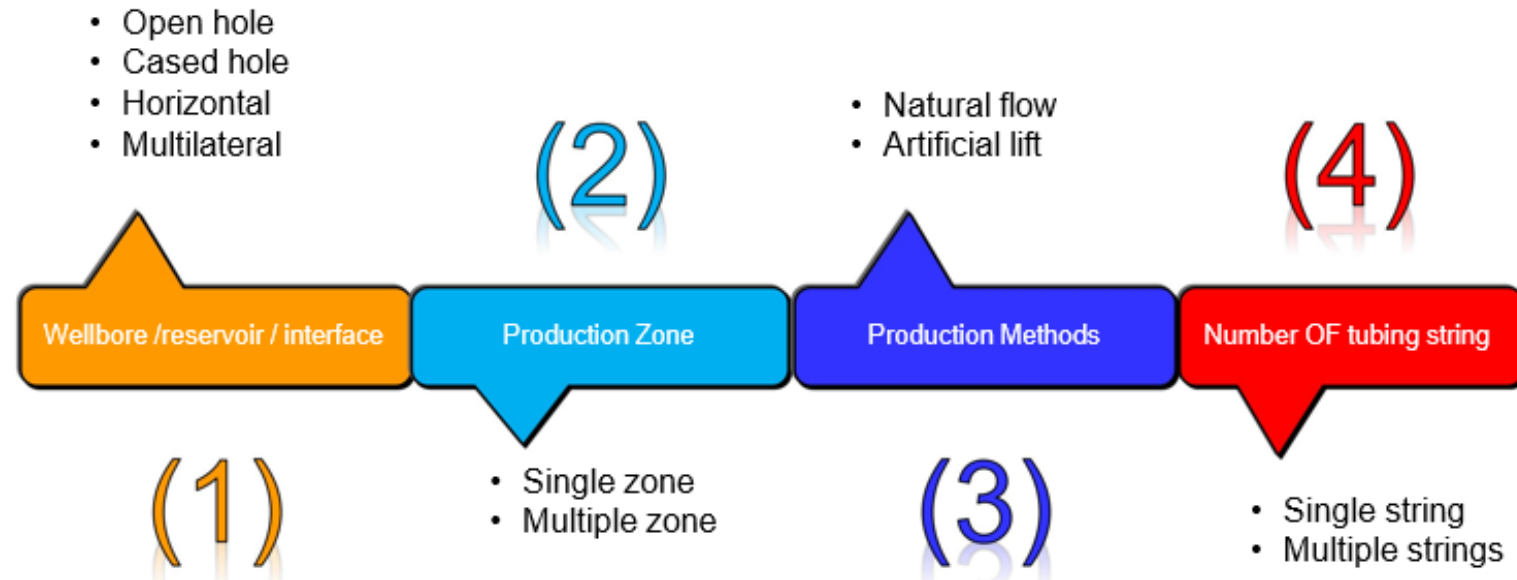
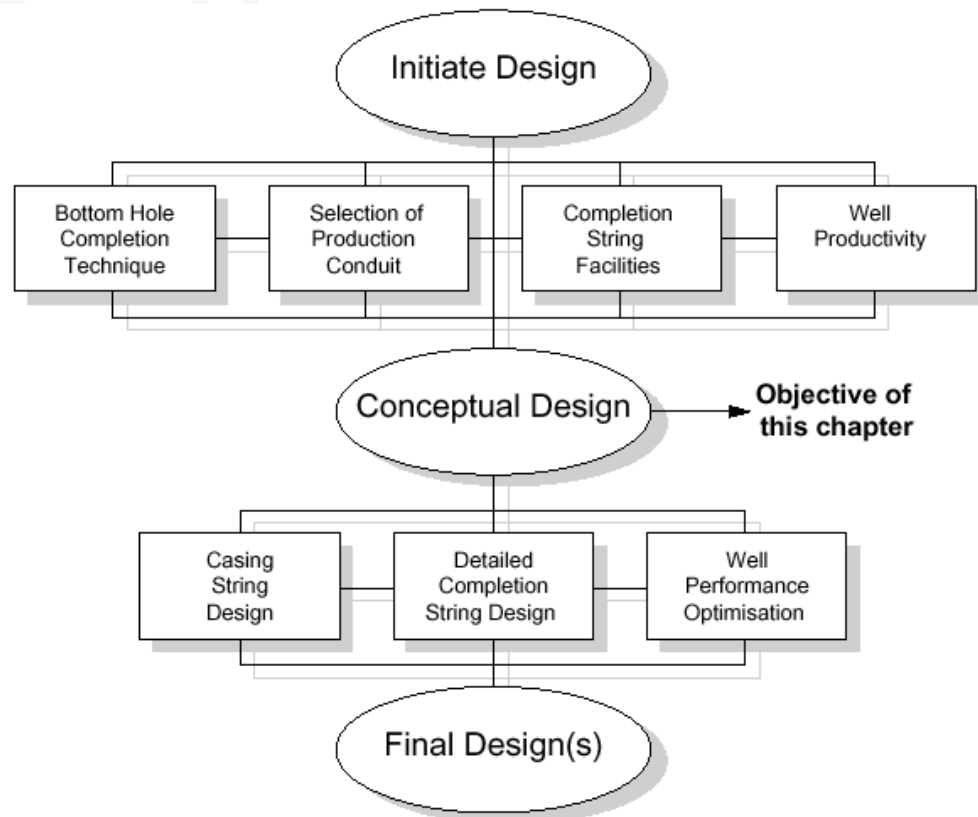
Typical Casing Program





4. Completion Design

Completion Design



The choice of completion concept is dictated by:

The reservoir model and operator's drainage strategy.

- Horizontal wells
- Single zone/multiple zone
- Co-mingled production/flow control
- Pressure maintenance

Reservoir properties

- Open hole/Cased hole completion
- Sand control measures
- Stimulation
- Artificial lift

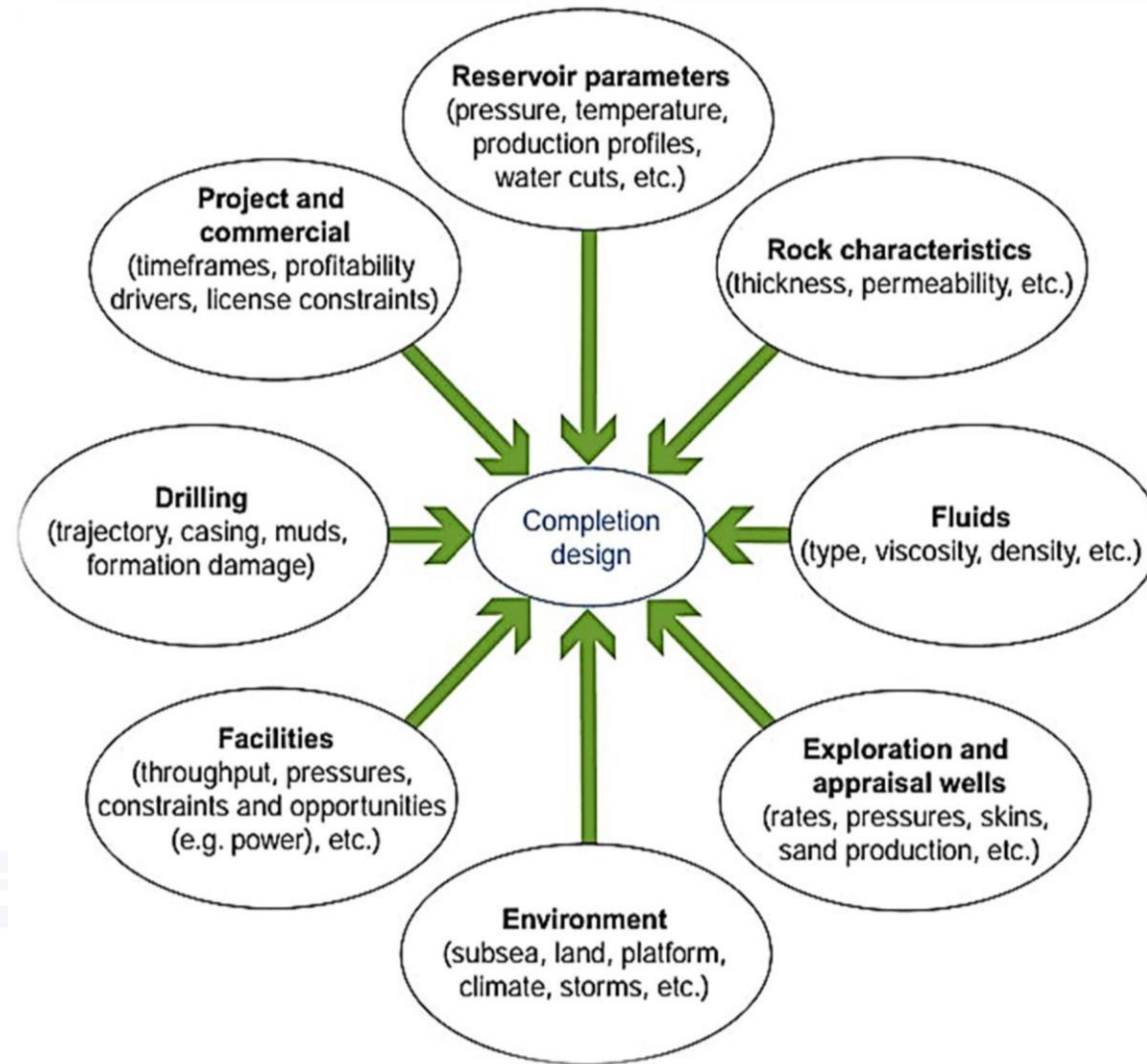
The physical and chemical qualities of the fluids.

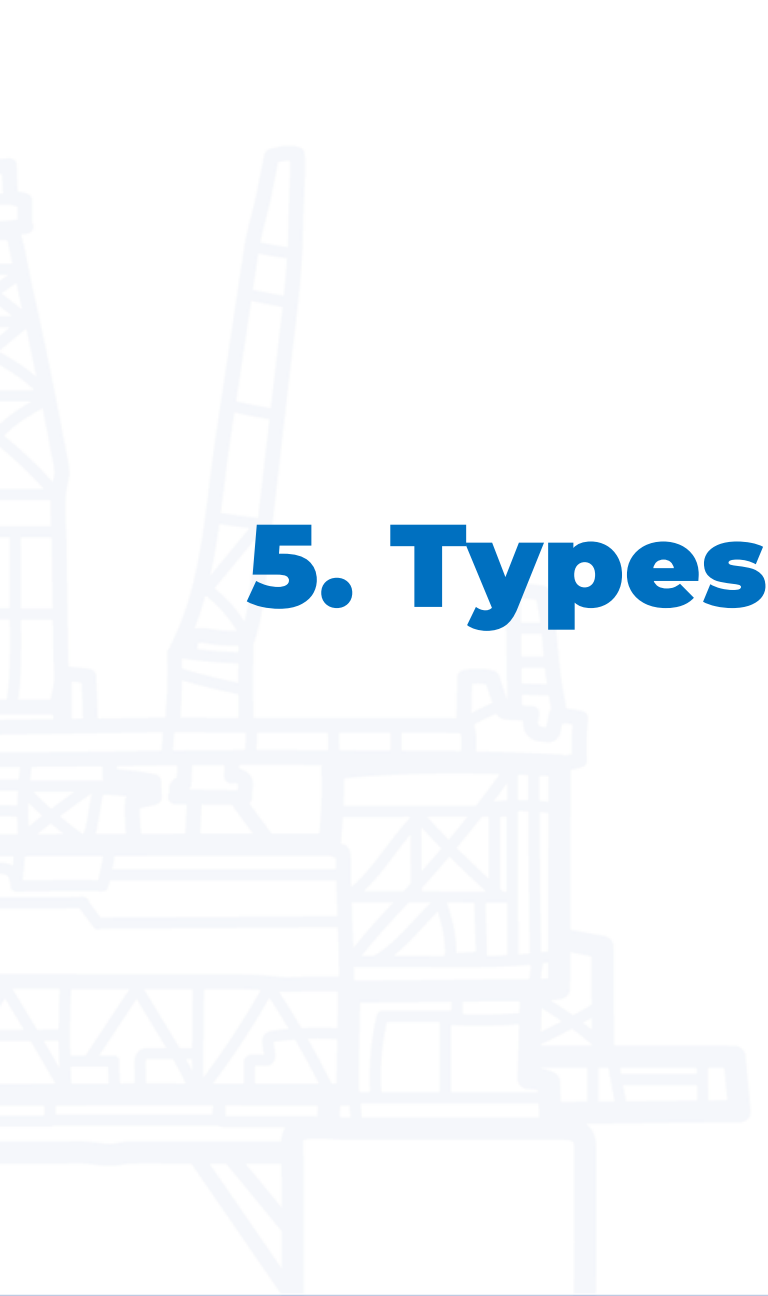
- Completion metallurgy
- Elastomer qualities
- Deposits/Scale
- Chemical injection

Other development considerations

- Expandable tubulars
- Future well interventions
- Well trajectory
- Intelligent Wells
- Multilateral

Data sources for completion design





5. Types of Well Completion

Open Hole Completion

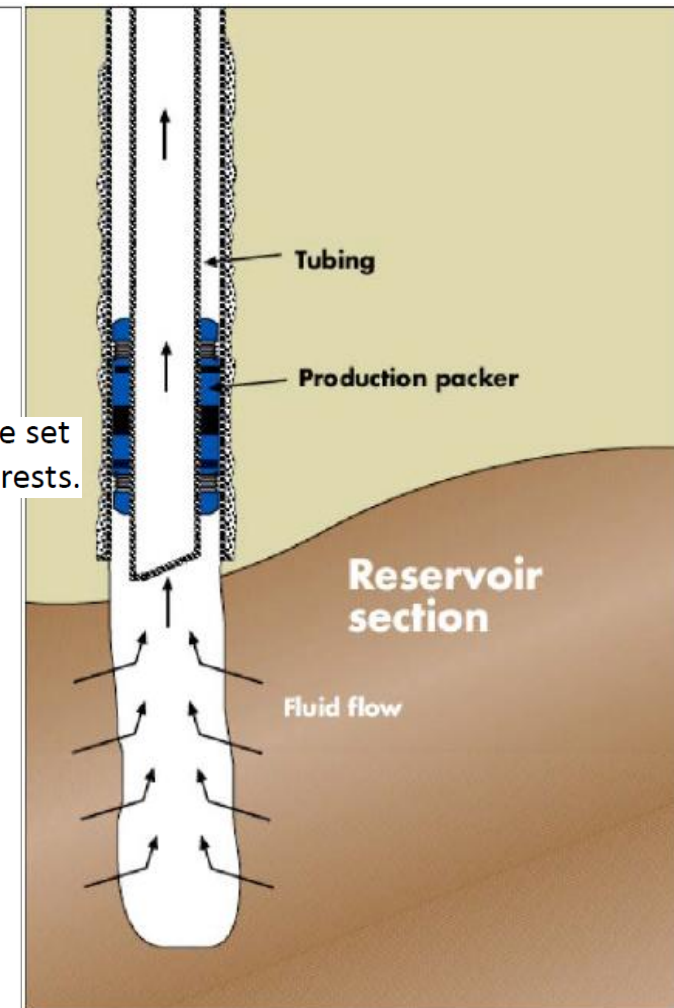
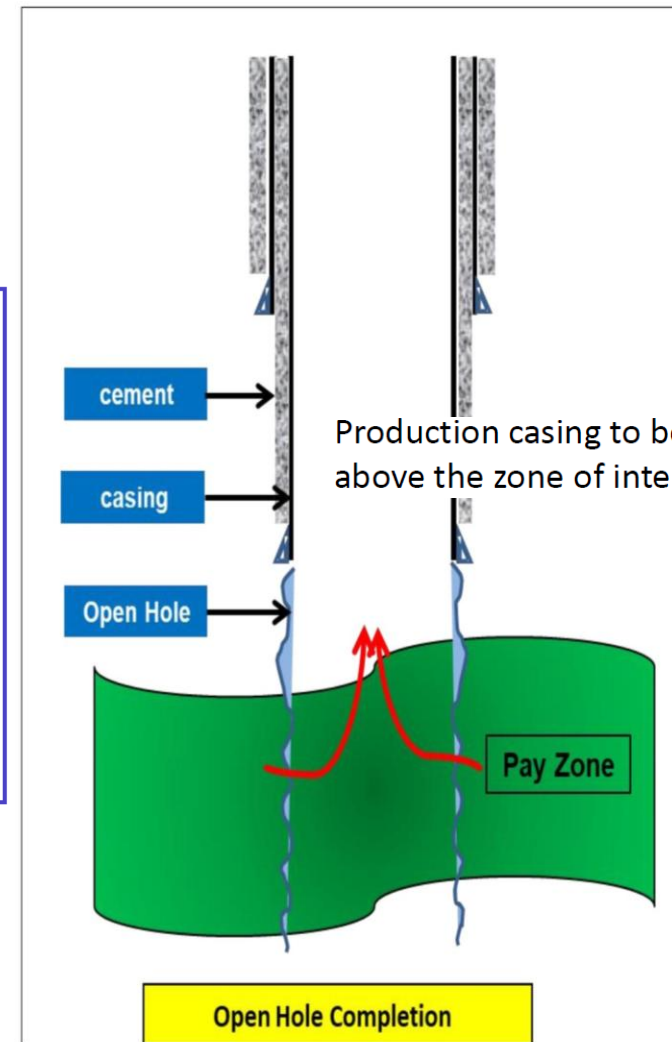
- In this style of completion, the production casing shoe would be set just above the reservoir section and cemented in place, then the reservoir section is drilled with non-damaging drilling fluids.

Advantages

- Low cost and simple (especially for long intervals)
- Save perforation cost
- Maximum diameter across reservoir section
- Easy to deepen well
- Prevent formation damage by cement

Disadvantages

- Mud filter cake will reduce productivity unless it was cleaned up
- No protection against wellbore collapse of the producing interval
- Limited zonal isolation
- Separate zone within the section are difficult to be stimulated selectively



Open Hole Completion

Cased Hole & Perforated Completion

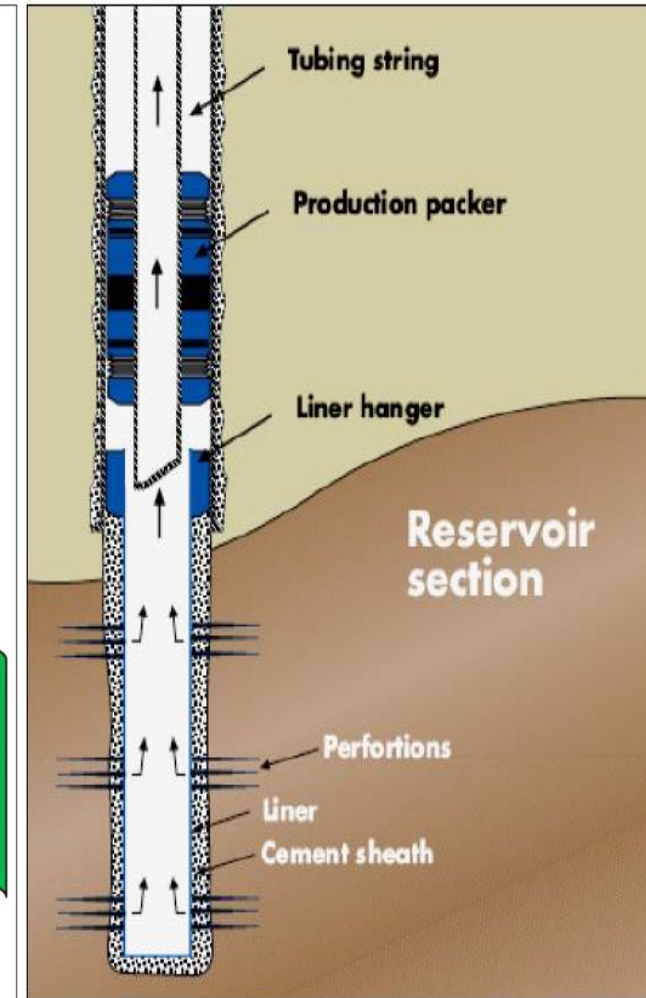
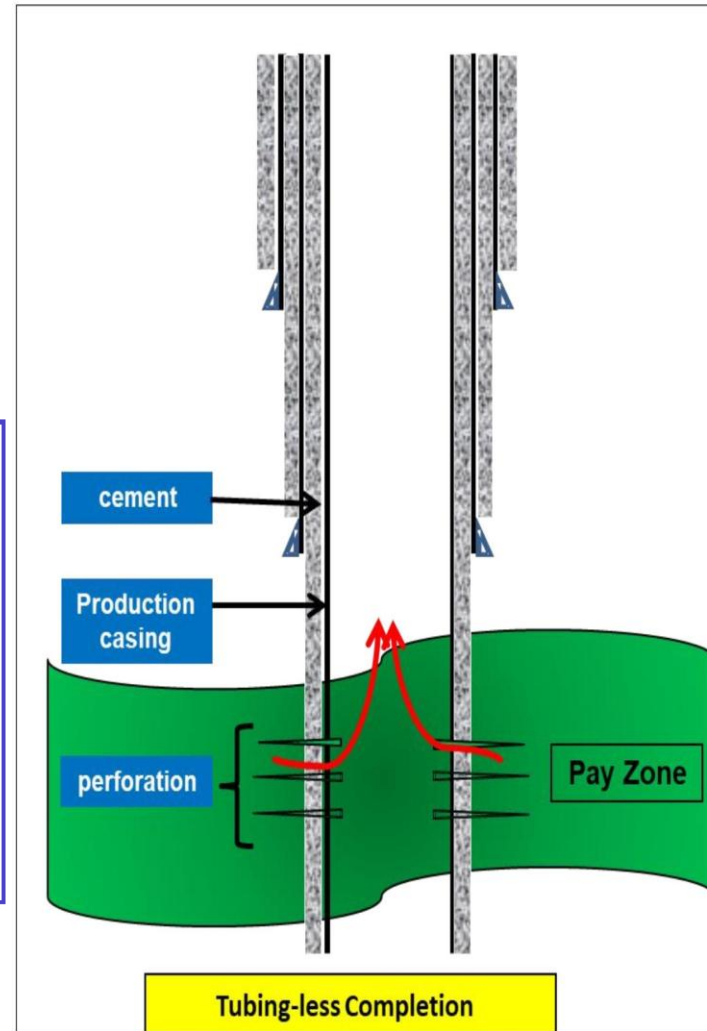
- The majority of well completions are designed with cemented production casing string which is then perforated to establish communication with the formation
- A small diameter liner may be used across the producing intervals "hang off" on the inside of the bottom section of the main casing string by using a Liner Hanger System
- The perforations penetrate the cement and damaged regions around the wellbore area.

Advantages

- No need to clean up filter cake
- Perforations bypass the damaged zone
- Good zonal isolation
- Multiple / Selective completion
- Good well integrity if properly cemented
- Protection against hole collapse

Disadvantages

- Possible in-flow restriction due to lack of 360° perforation coverage
- Permeability impairment due to crushed zone & perforation debris (can be minimized by perforating underbalance)
- Liner cementing can be difficult at higher angles
- Cementing, Perforating and additional rig time can be costly for offshore operation

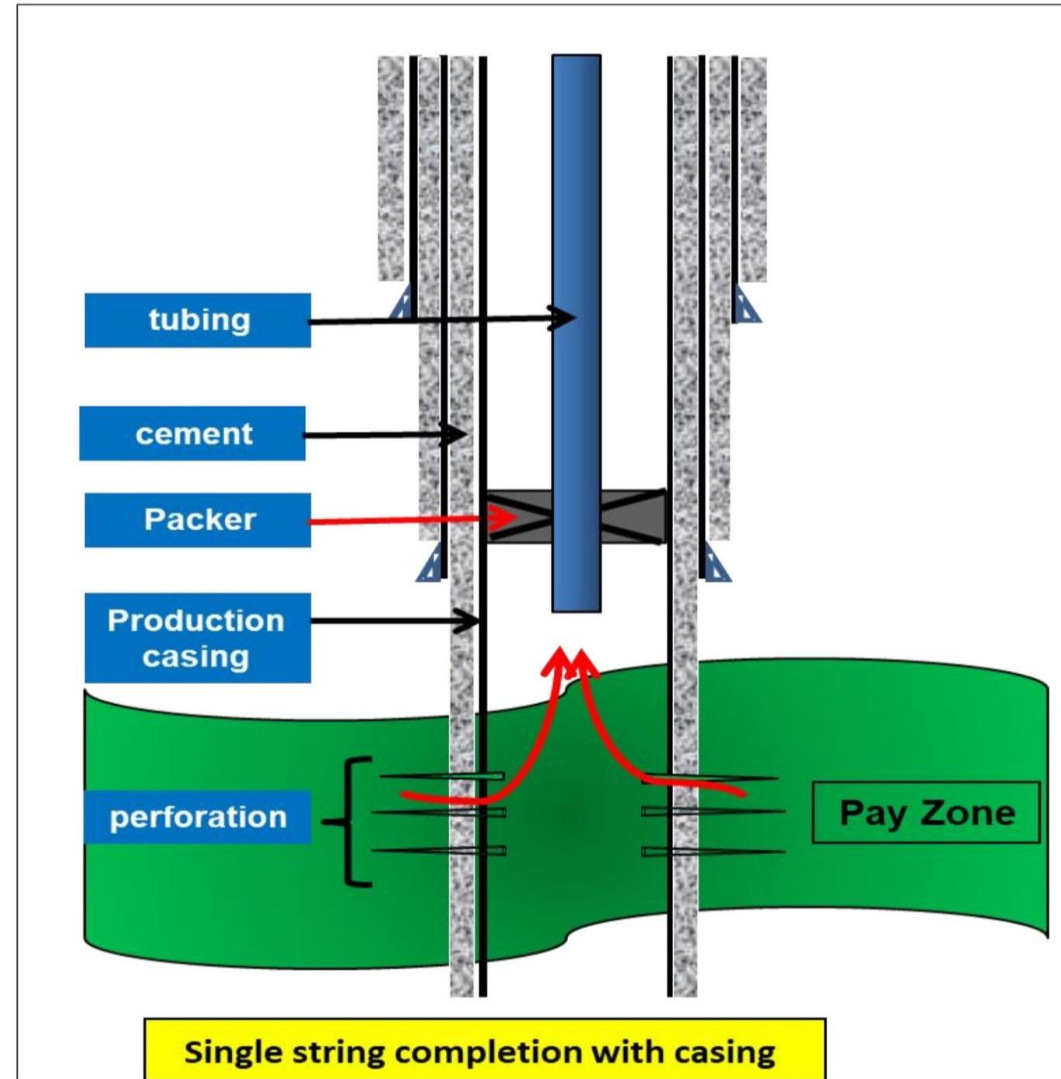


Cased and Perforated Completion

Single string completion – with casing

Single string completion with casing

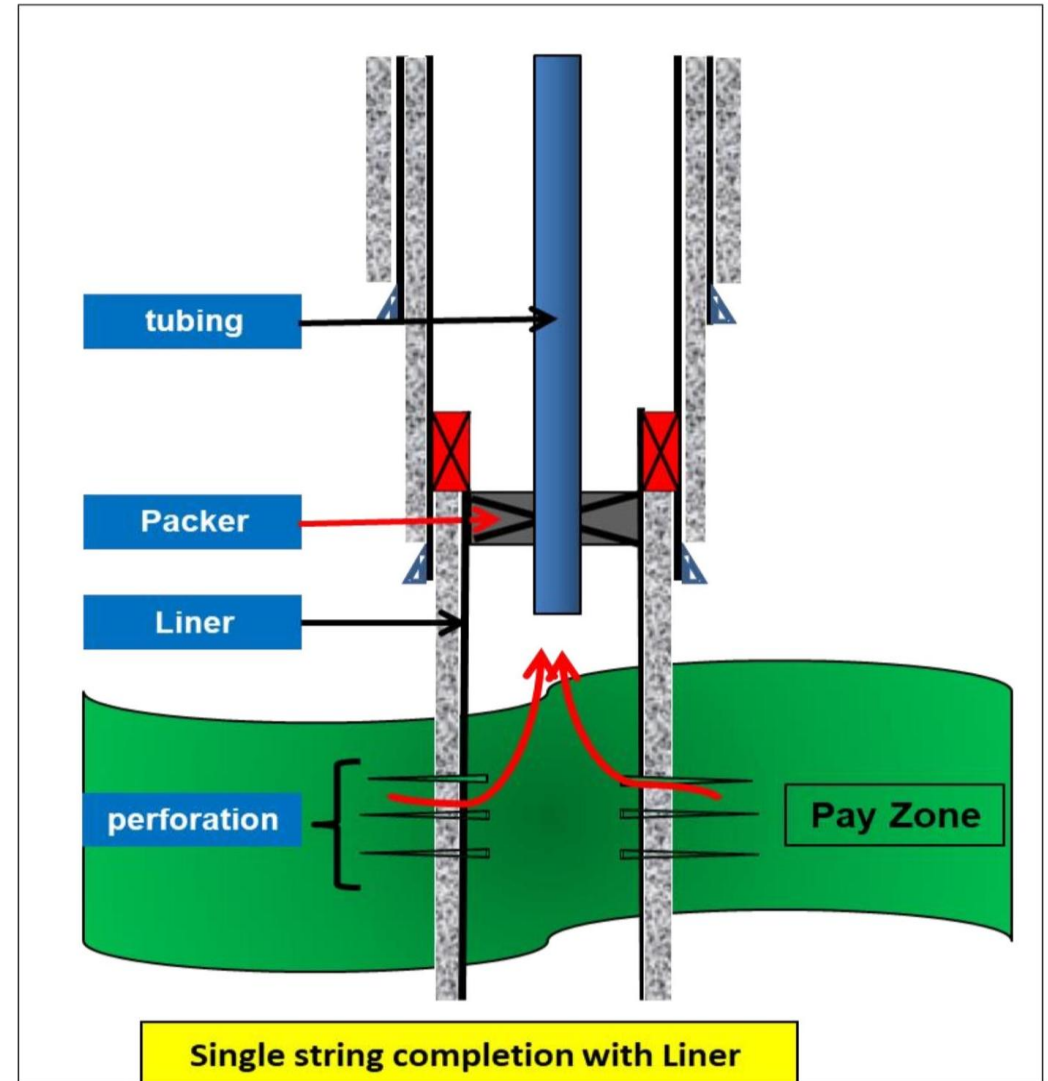
- It is the simplest way of completing the well.
- In this method well is completed for single zone with single tubing



Single string completion – with liner

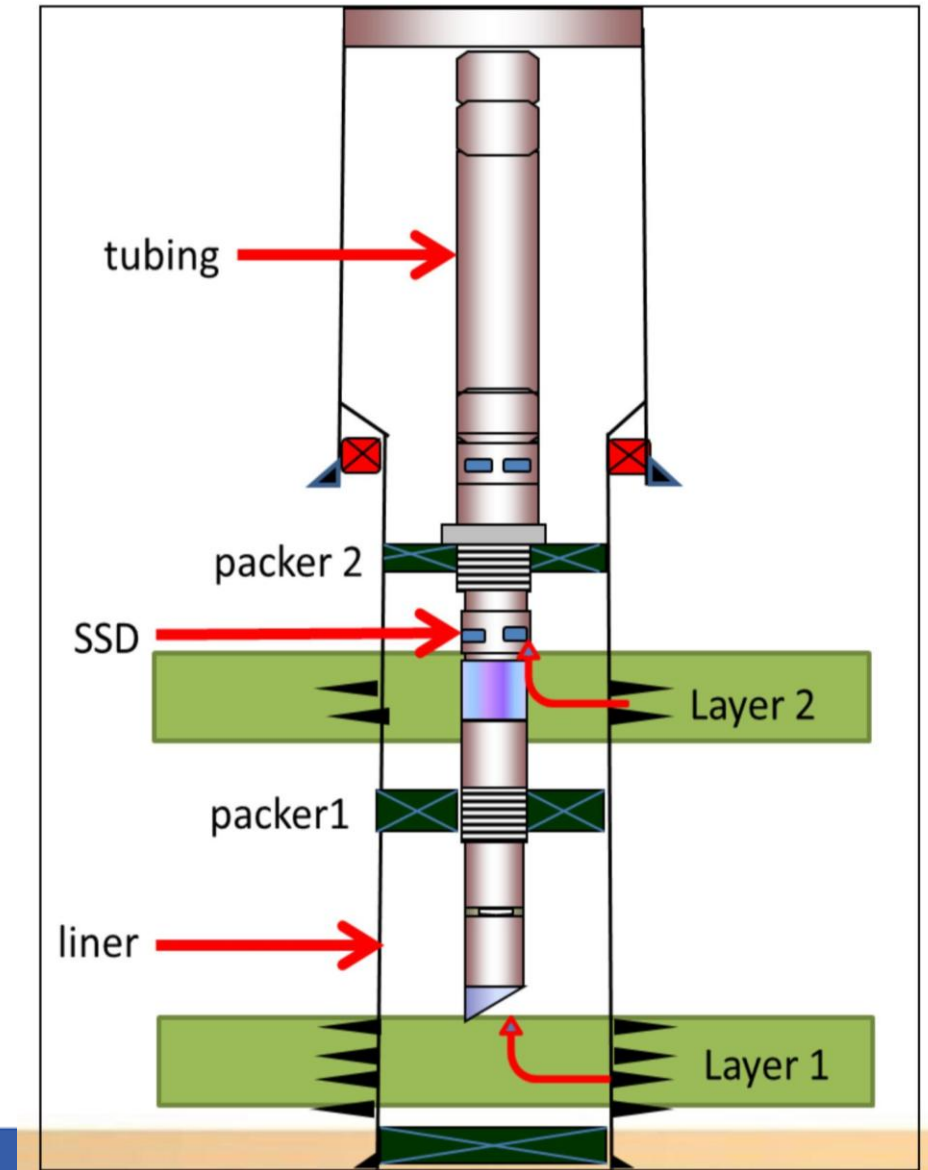
Single string completion with Liner

- It is the simplest way of completing the well.
- In this method well is completed for single zone with single tubing

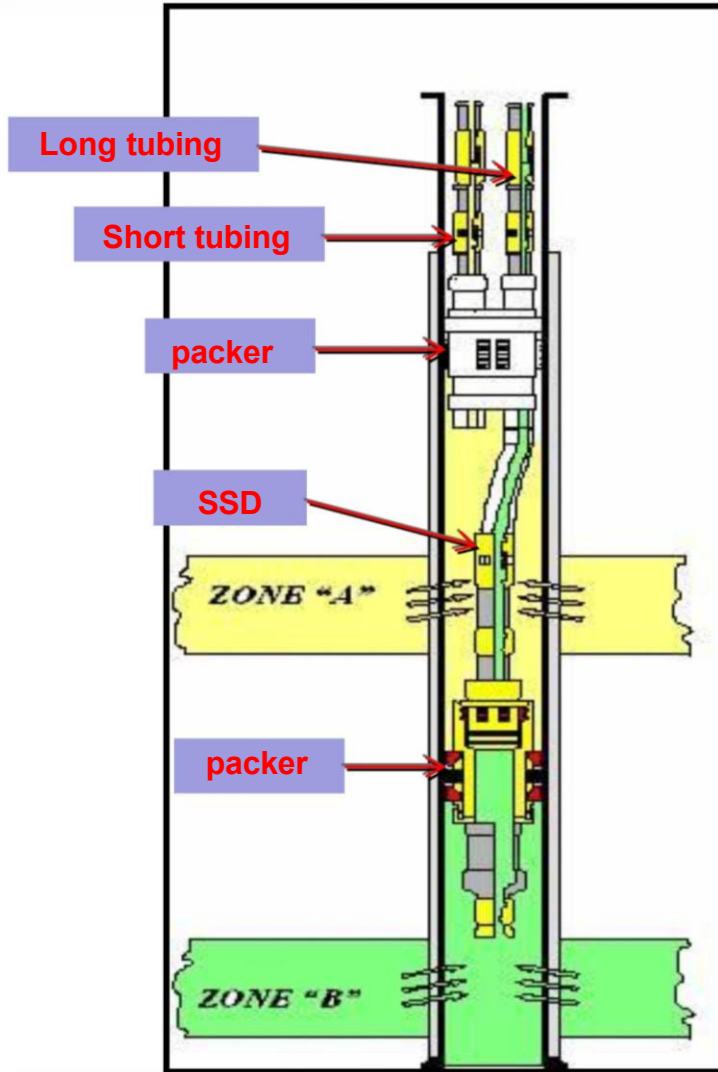


Single selective Completion

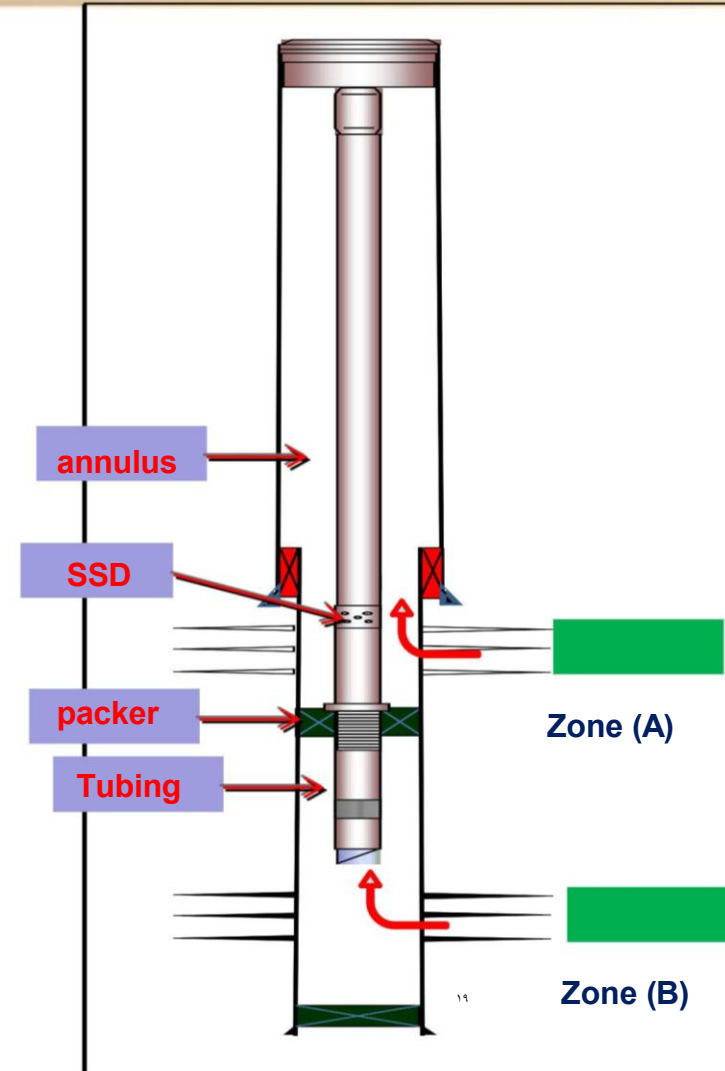
- Single string commingle completions.
 - All the reservoirs available in a well are produced simultaneously through single string.
 - Should be avoided if possible to eliminate cross-flow phenomena.
 - Monitoring of reservoir performance is extremely difficult



Single selective Completion

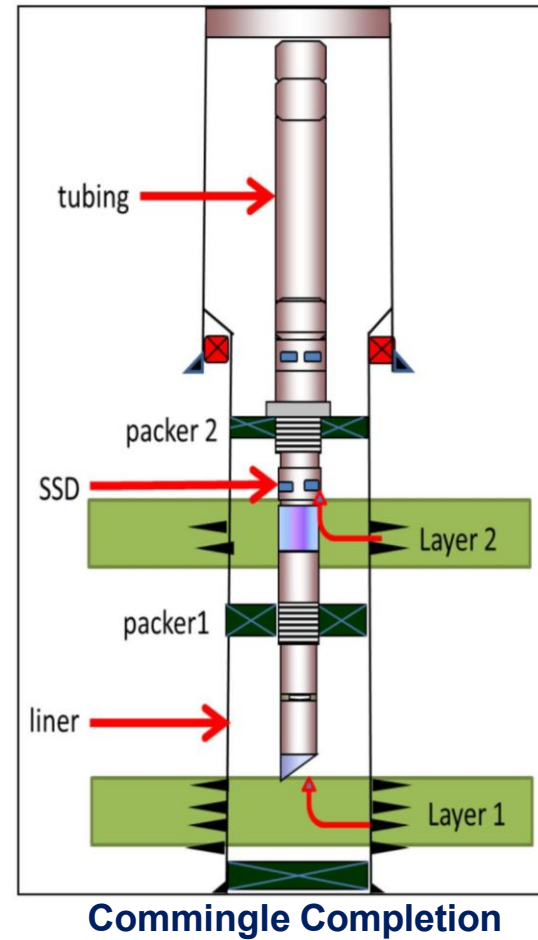
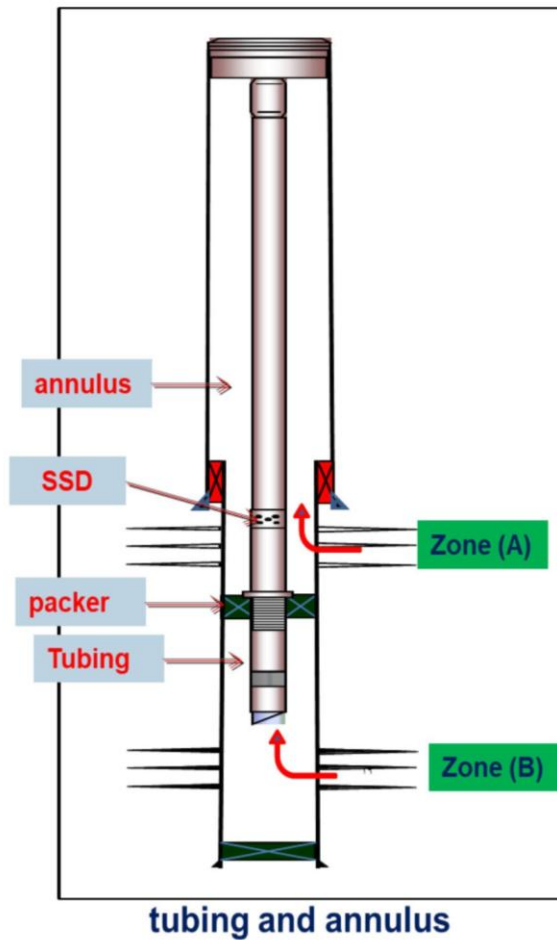
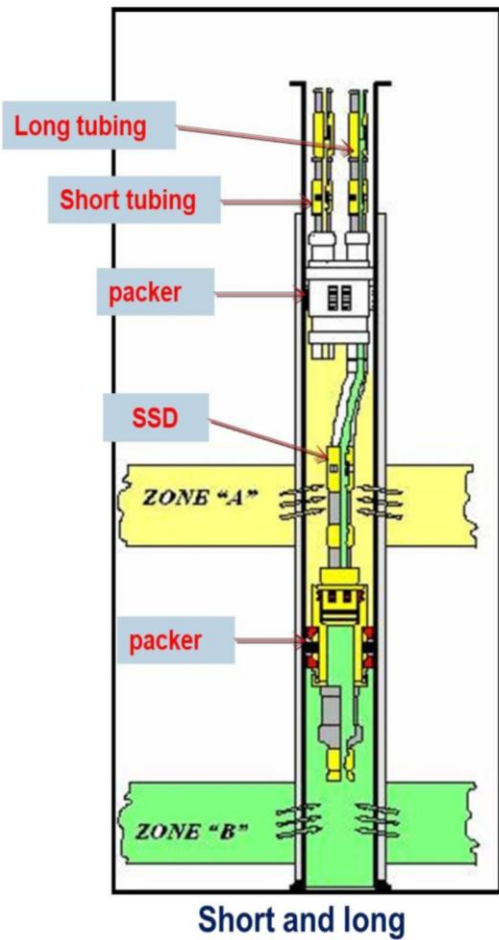


Short and long



tubing and annulus

Single selective Completion



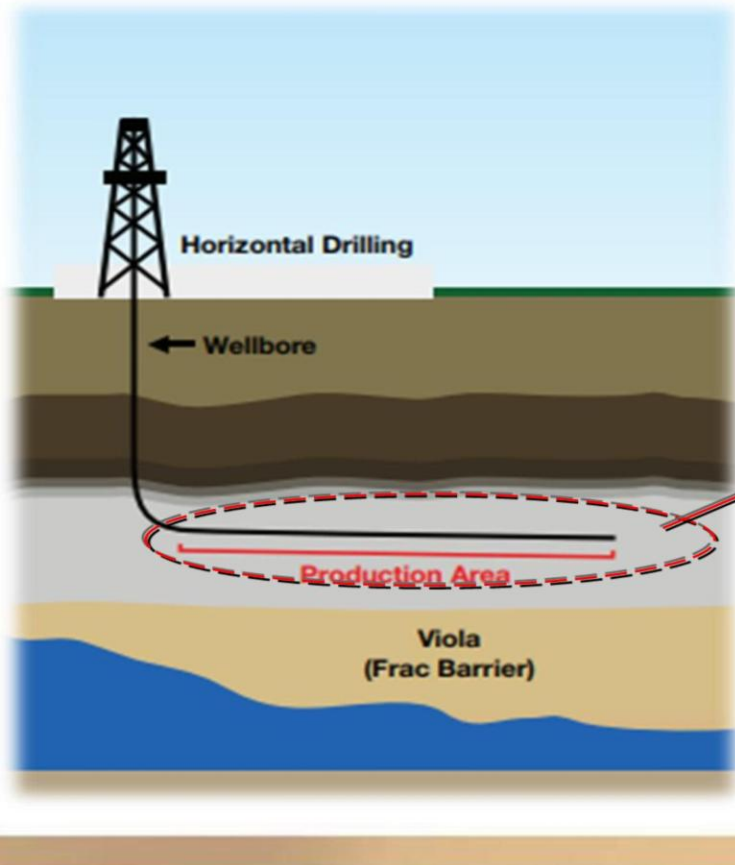
Key factors:

- Cost, N of wells, flow rate
- Flow assurance
- Well integrity
- Compatibility/interference between layers
- Selective control
- Complexity and installation risks; work-overs
- Well conversion
- Well and reservoir monitoring

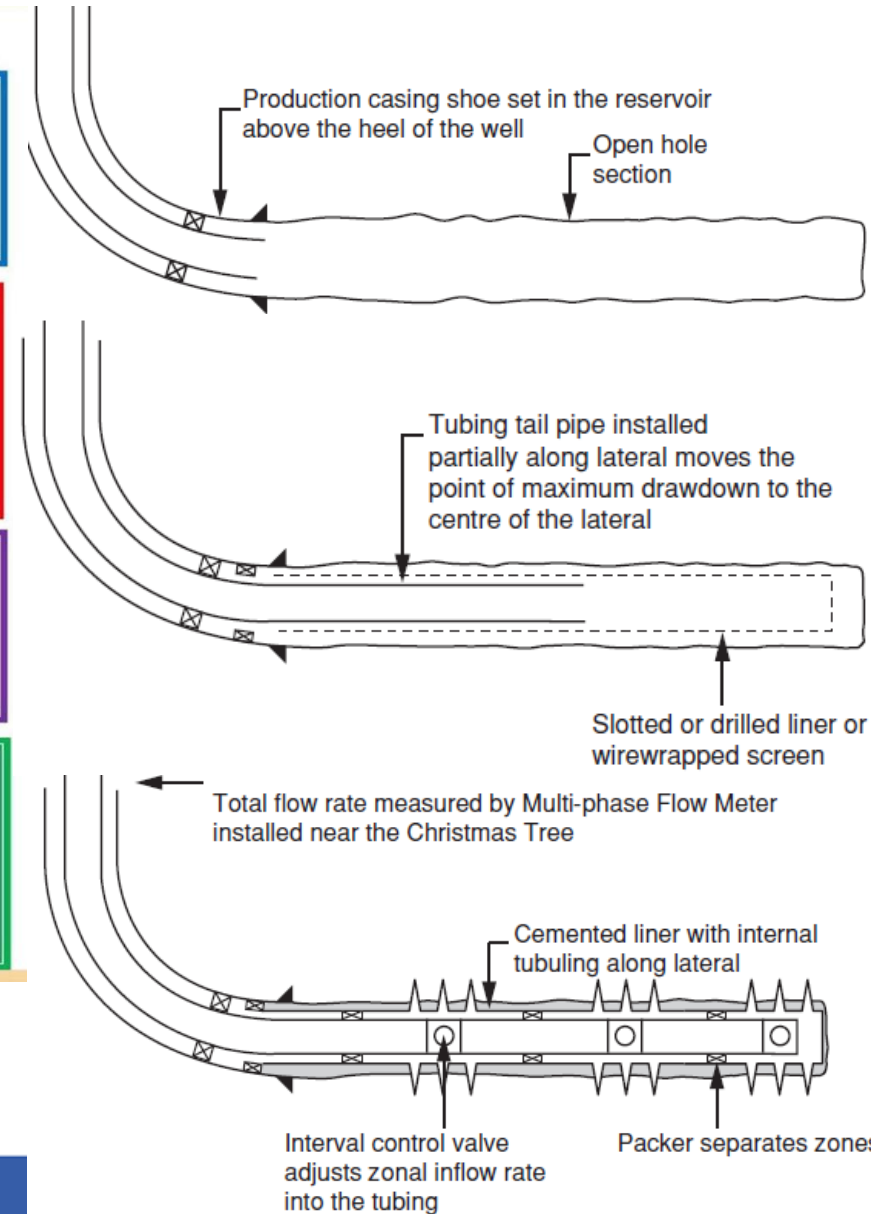
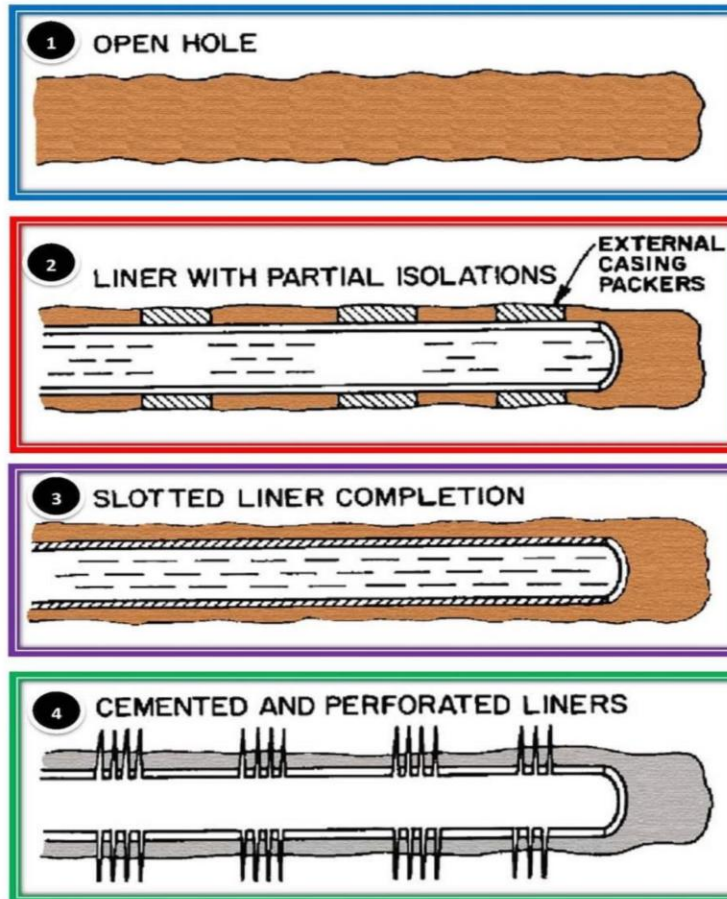
Choice also depends on:

- Offshore or Onshore Development
- Areal Size
- Fluids
- Well management

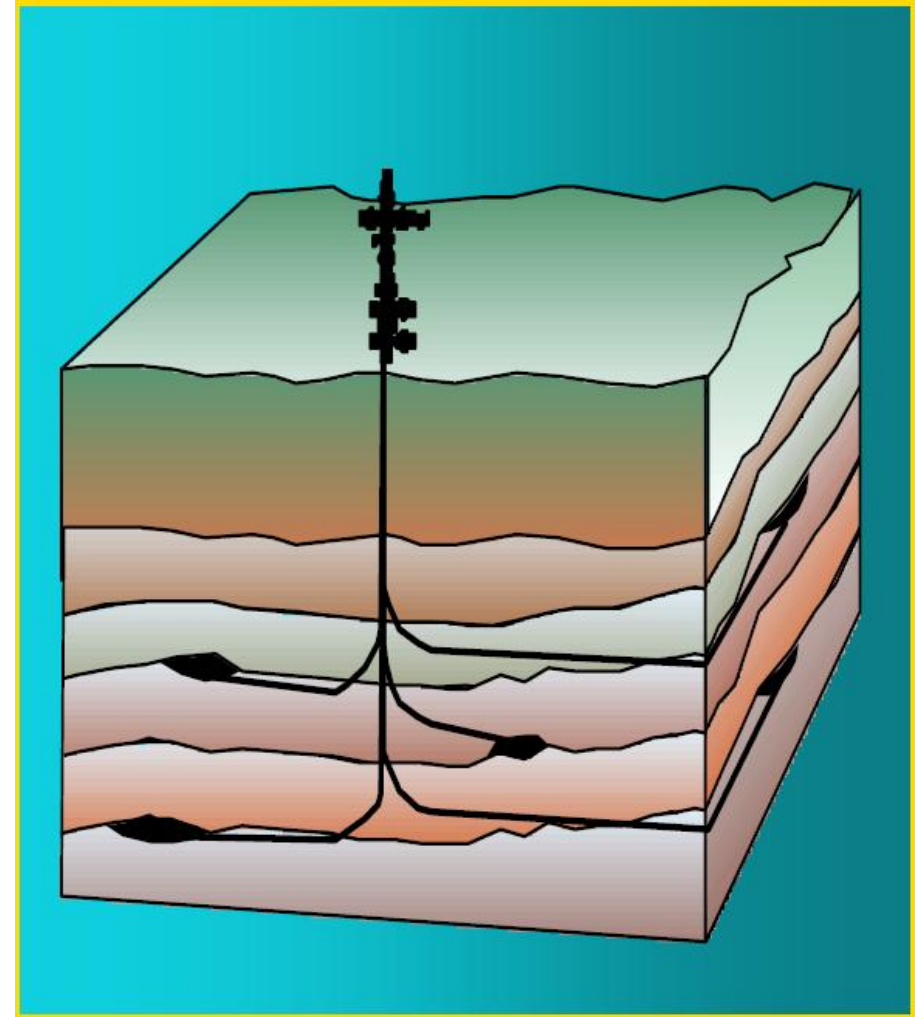
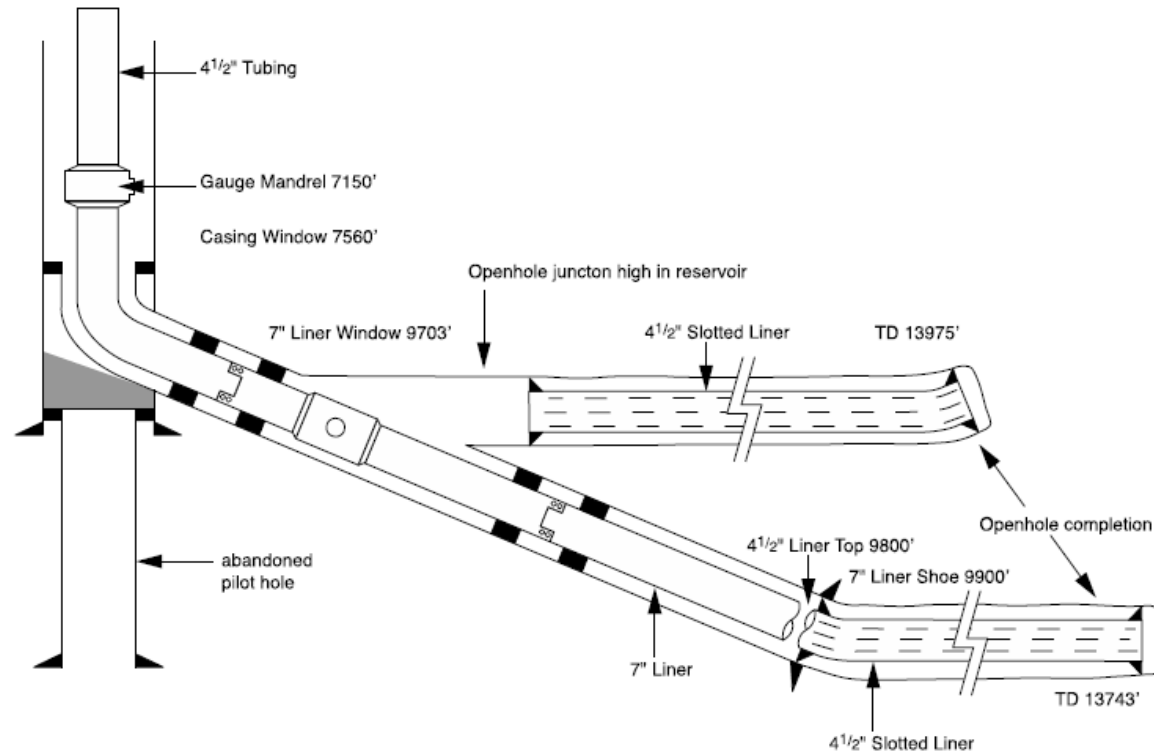
Horizontal Well Completion



Completion of Horizontal wells

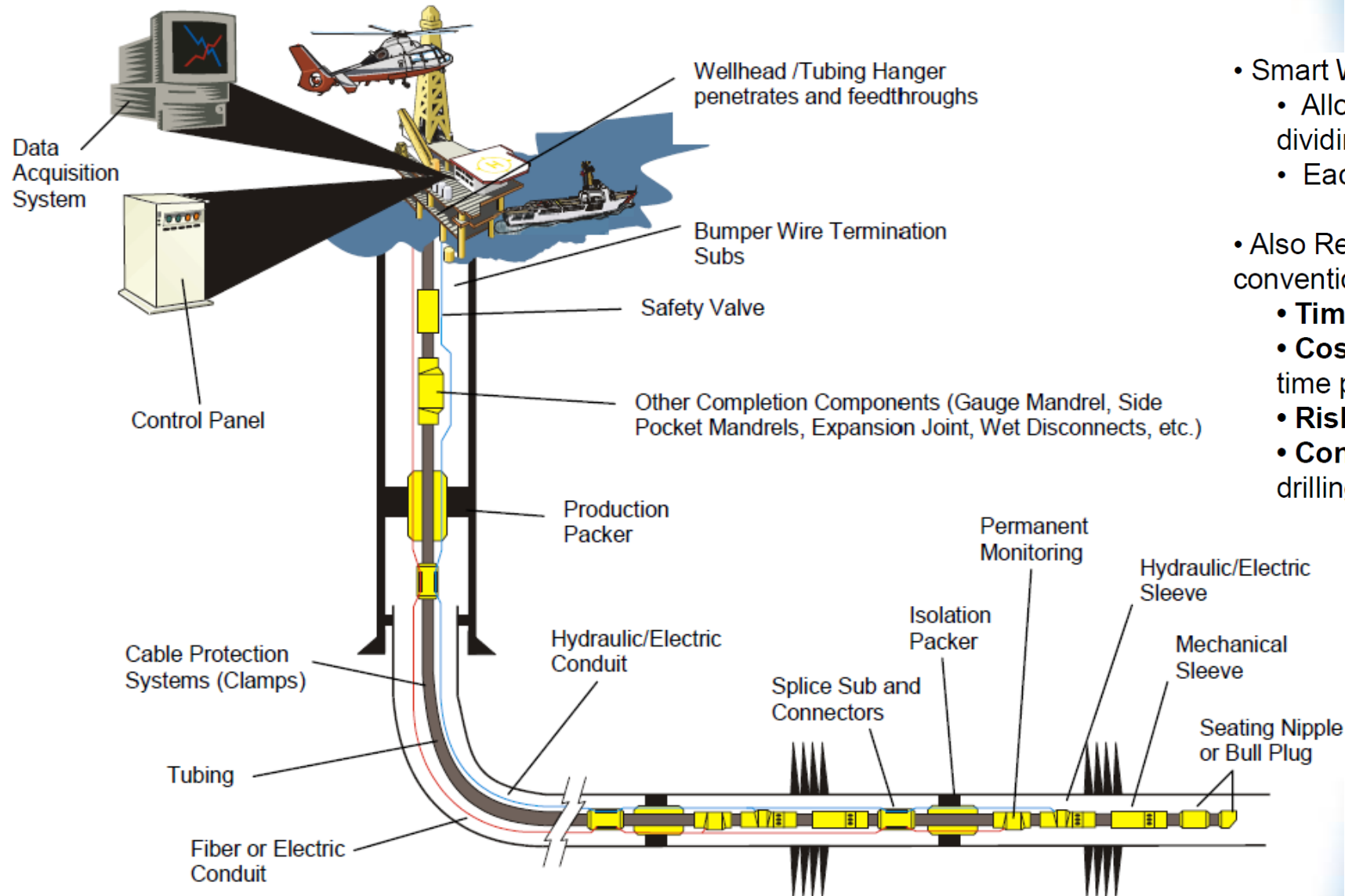


Multilateral Well Completion



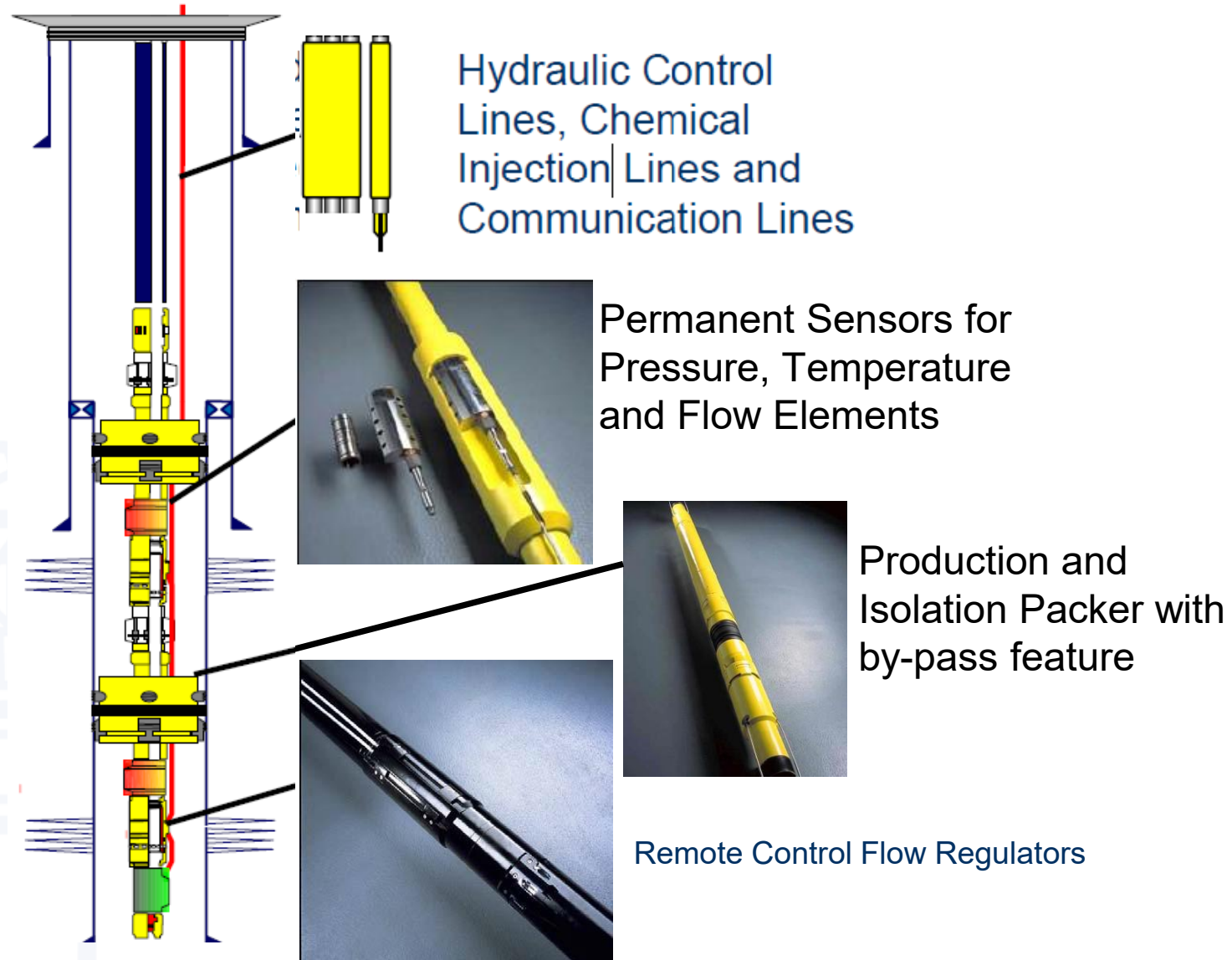
- Junction Complexity Levels (TAML)

The Intelligent (Smart) Well System

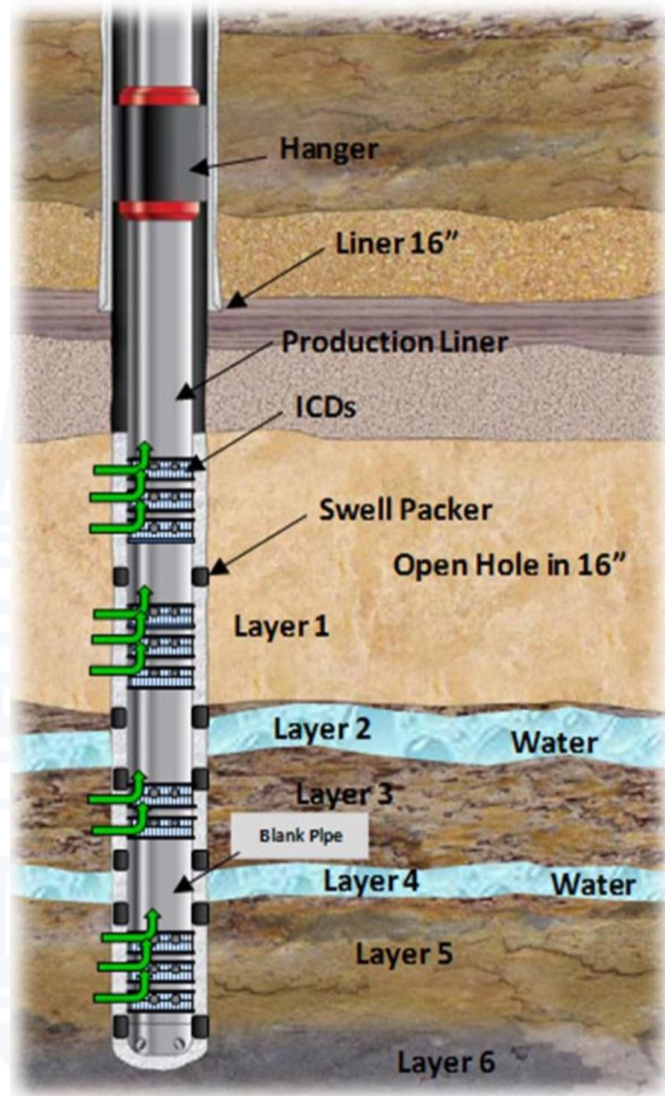


- Smart Wells are a completion technology that
 - Allows more efficient reservoir management by dividing the well into a number of segments
 - Each segment monitored & controlled separately.
- Also Reduced OPEX since the risks & costs of a conventional workover are avoided:
 - **Timing:** Waiting for (workover) rig availability
 - **Cost:** Completion zone change takes ~10 days rig time plus deferred oil. Subsea 10 X platform job cost.
 - **Risk:** Problems can occur during workover.
 - **Constraints:** Deepening the well completion requires drilling out plugs etc.

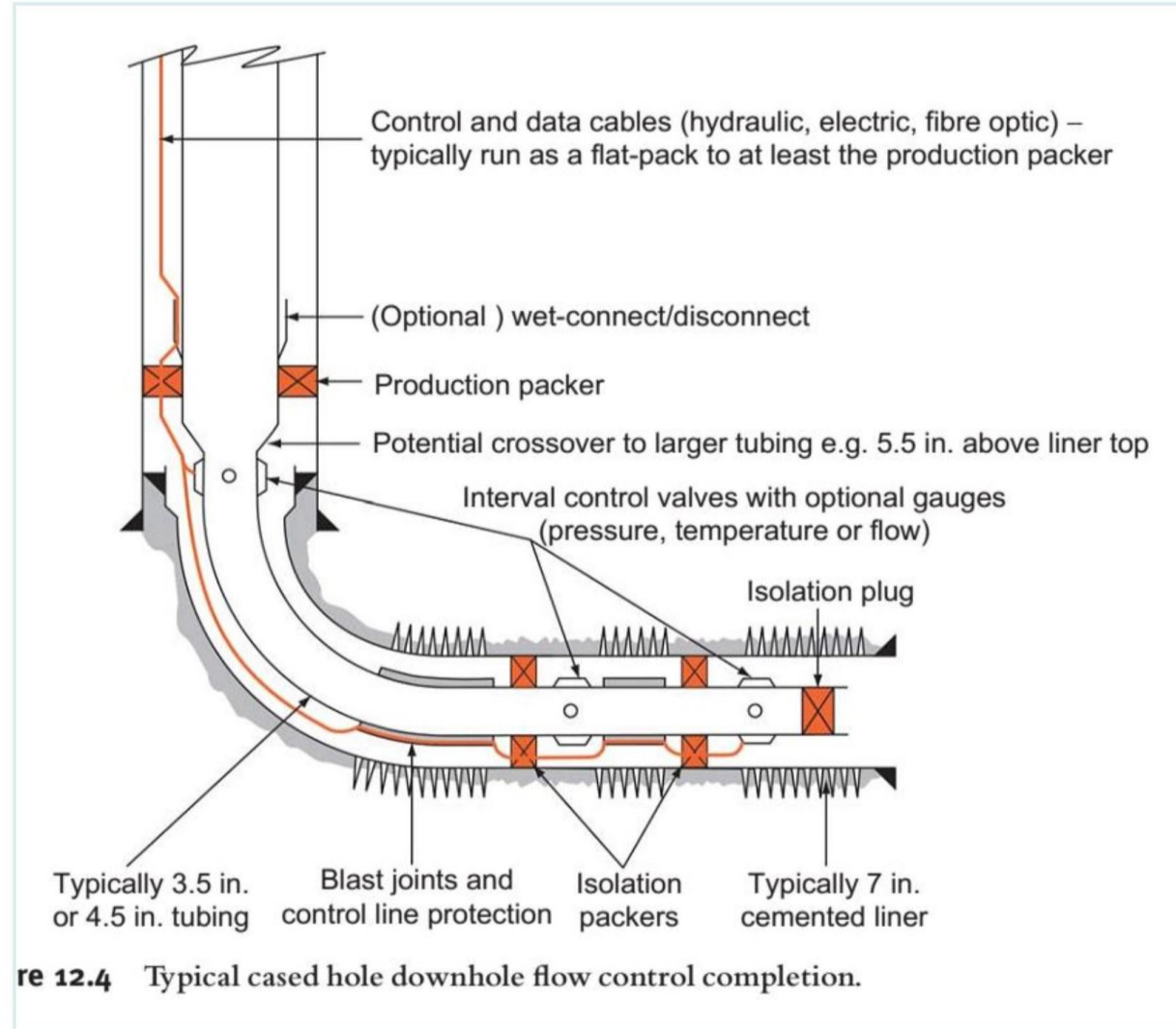
The Intelligent Well System – Basic Element



Intelligent Completion w/ ICD



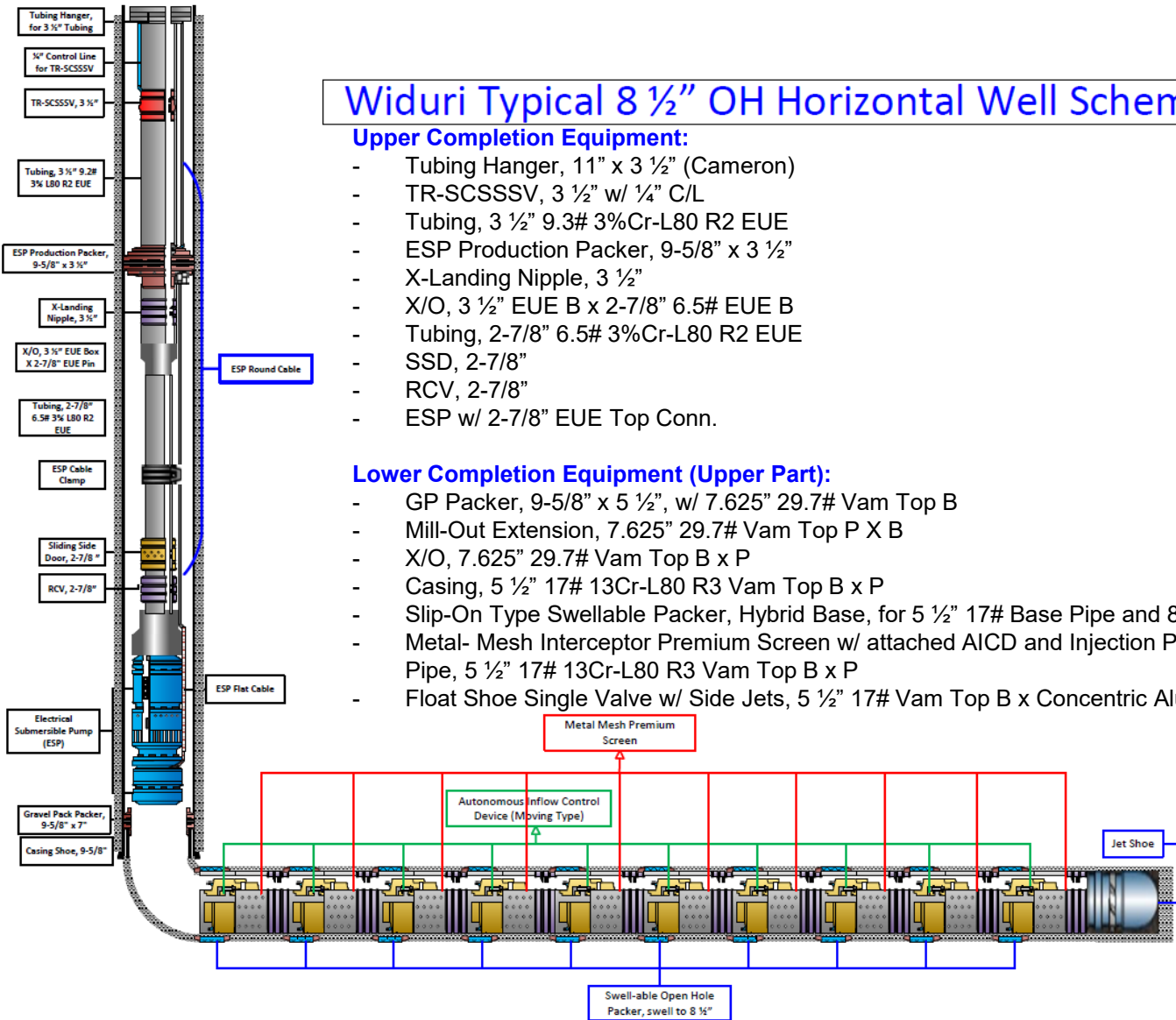
(ICD)-vertical well



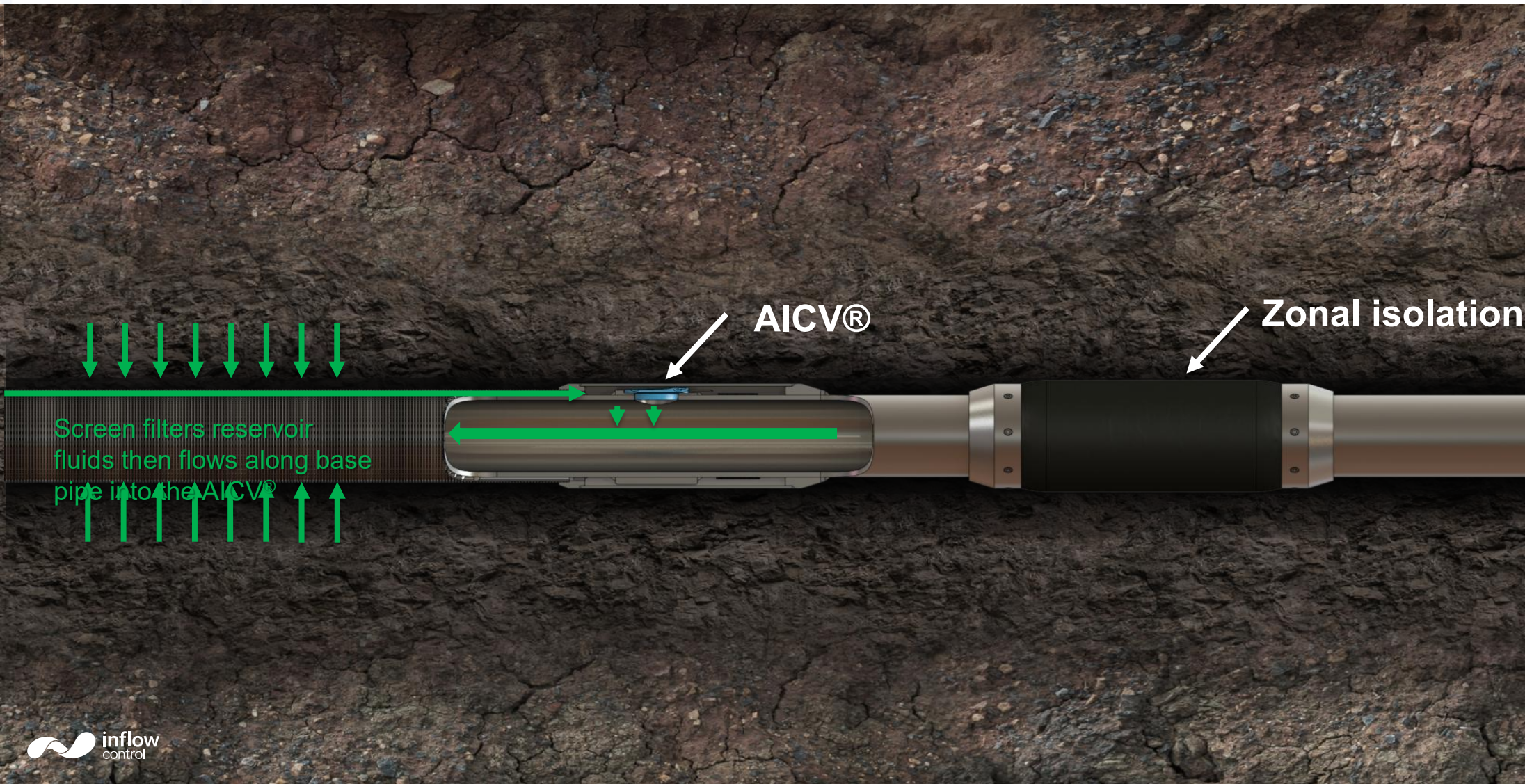
re 12.4 Typical cased hole downhole flow control completion.

(ICD)-Horizontal well

Intelligent Horizontal Well w/ AICD

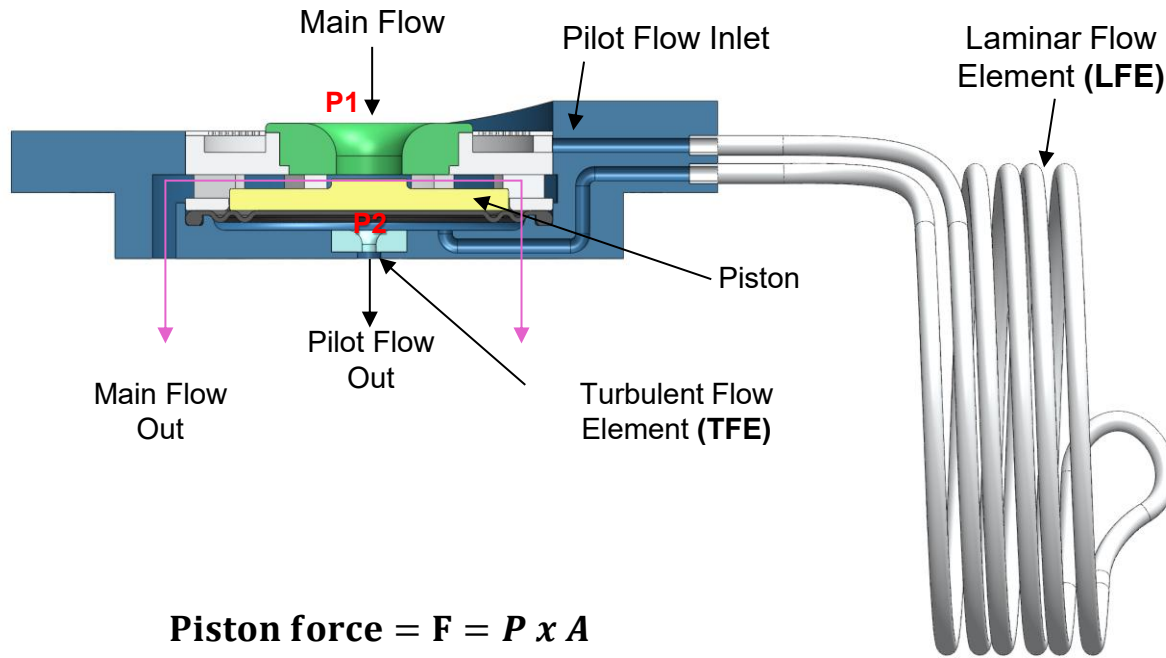
[illegible]

Fluid Flow Path: Reservoir → Screen → AICV®

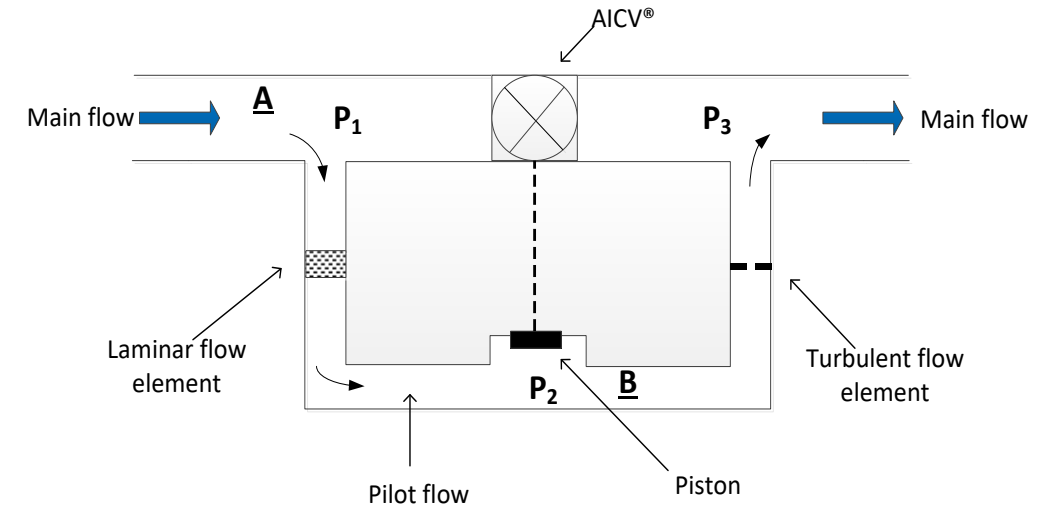


AICV® Principle (Moving Type – Active)

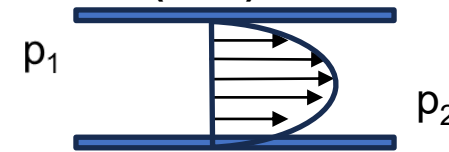
Viscosity & Density Sensitive



$$\text{Piston force} = F = P \times A$$



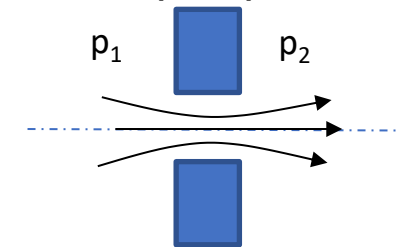
Laminar flow element (LFE):



$$\Delta p = \frac{32 \mu v L}{D^2}$$

$$\Delta p = f(\mu, v)$$

Turbulent flow element (TFE):



$$\Delta p = C \cdot \frac{1}{2} \rho v^2$$

$$\Delta p = f(\rho, v^2)$$

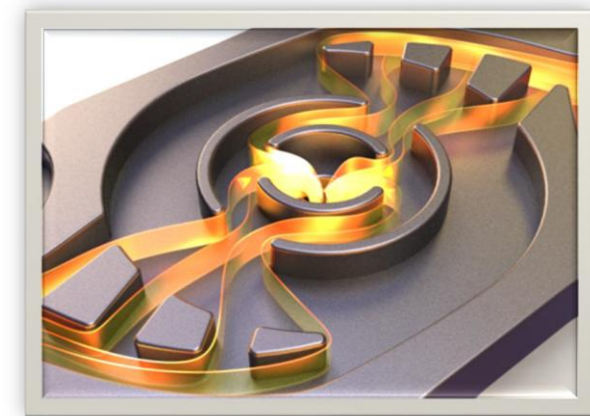
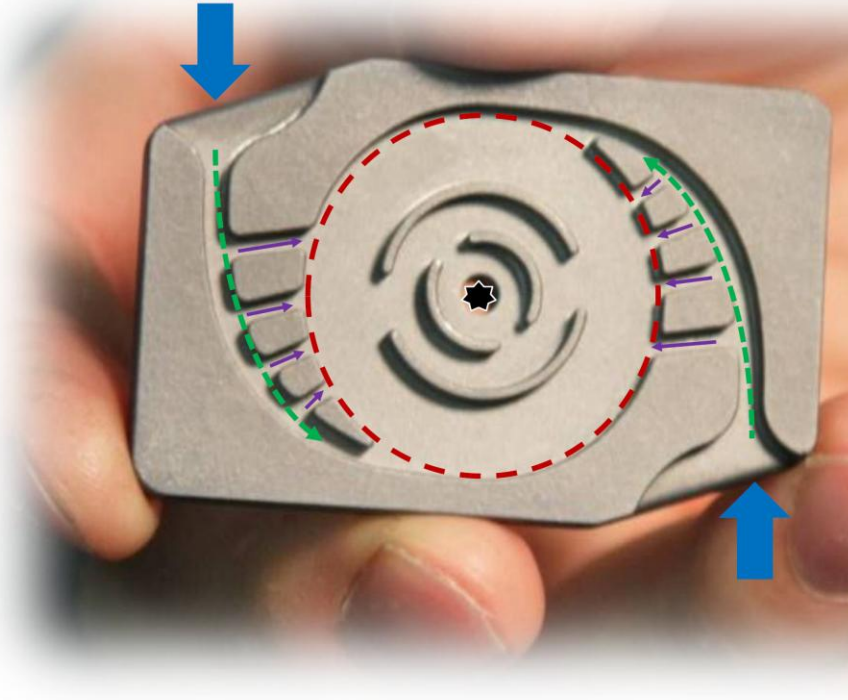
AICD Principle (Non-Moving Type – Passive)

Viscosity Sensitive

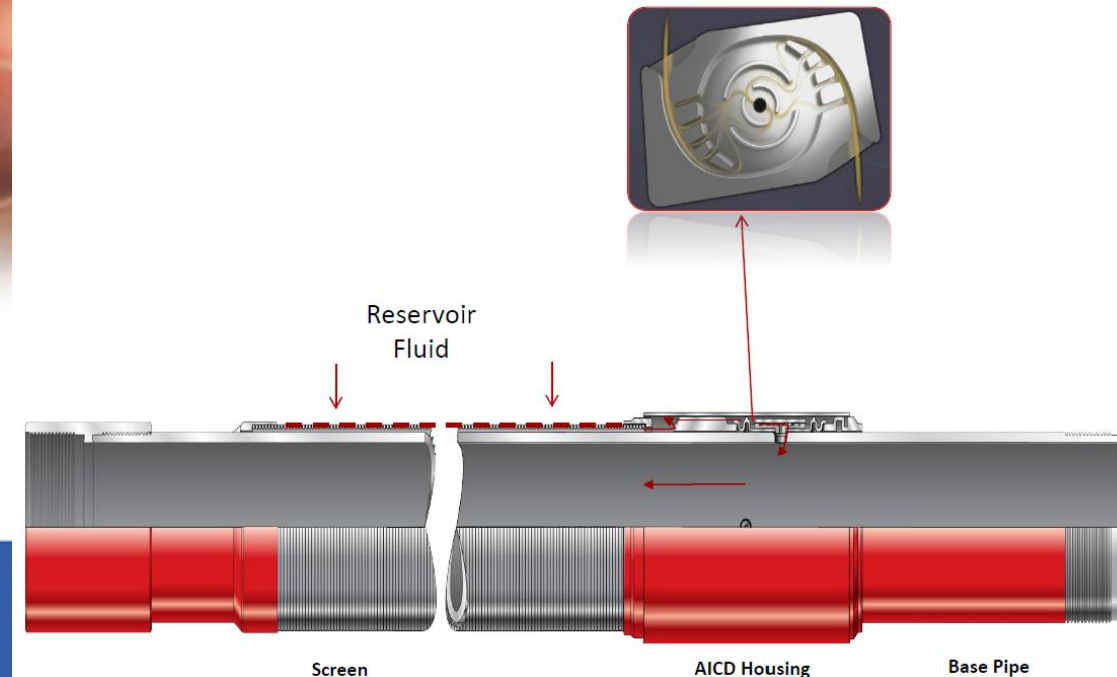
Functional Features

Design	Oil Viscosity	Fluid Restriction
Range 1	0.3 – 1.5 cP	Gas and Water
Range 2	1.5 – 10 cP	Gas and Water
Range 3	3 – 200 cP	Gas and Water
Range 4	150+ cP	Gas and Water

- A - Entrance
- B - Exit
- C – Tangential or Divergent Path
- D - Direct or Multiple Paths
- E Vortex Bowl



EquiFlow® Autonomous Inflow Control Device



ICD vs AICV®

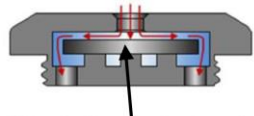
ICD



AICV®



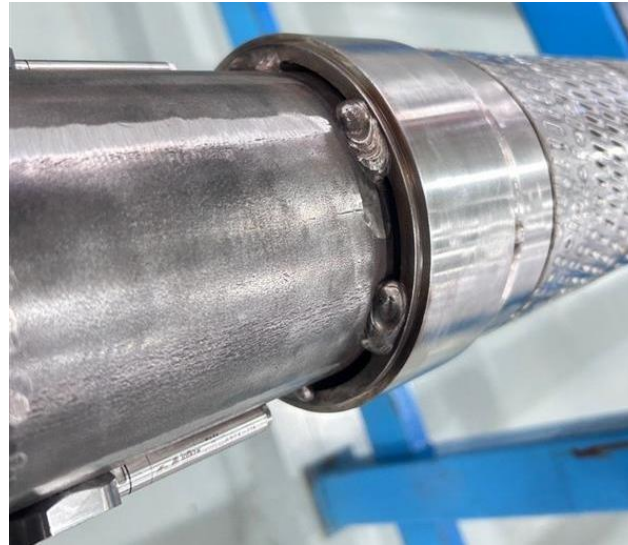
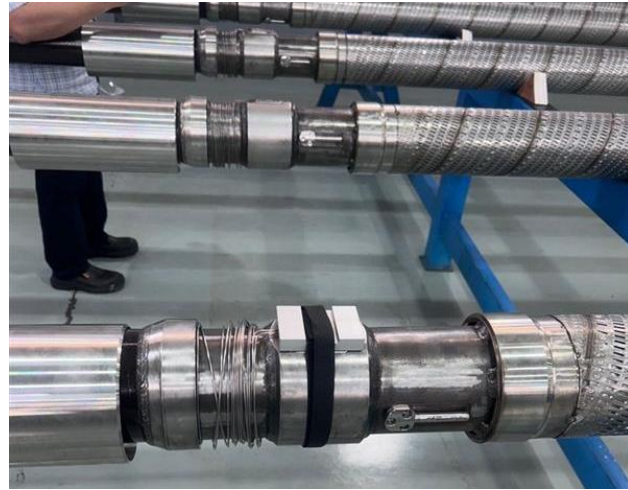
AICV Moving Type vs. AICD Non-Moving Type

No.	Description	AICV Moving Type	AICD Non Moving Type	Remarks
1	Basic Design	- Viscosity & Density Sensitivity - More complex engineering design - Higher risk of plugging within the fluid inflow media – moving parts no longer can be worked	- Viscosity Sensitivity - Simpler engineering design - No risk of plugging within the fluid inflow media - More resistance to erosion	- Both design equipped w/ outer screen jacket for sand control possibility and compartmentalize w/ Swellable Packer - When the well is stimulated with acid, the fine grains will be loose and produced to the surface. The AICD Non-Moving Parts utilizing the static nozzles will not cause plugging within the fluid inflow media when the fine grains are produced
2	Application	Suitable for Open Hole horizontal completion	- Suitable for Open Hole horizontal completion - Fit to highly depleted reservoir for very narrow pressure drawdown (ΔP up to only 150psi)	
3	Choking Capability	Moving part system enable to choke produce water up to 90%	Since there's no moving part (always open), it's enabled to choke produce water up to 50%	
4	Impact to Fluid Rate When More Water Produced	Fluids produced the more water, the more produced AICD Moving Parts X  Valve choking mechanism when more water produced that will directly reduce the production rate	Total fluids produced the same rate will remain at the same rate 	For AICD Non-Moving Type: - Vortex mechanism favors the flow of oil much easier quicker than the water flow through the nozzle - Total fluids through the nozzle remain at the

AICV Moving Type vs. AICD Non-Moving Type

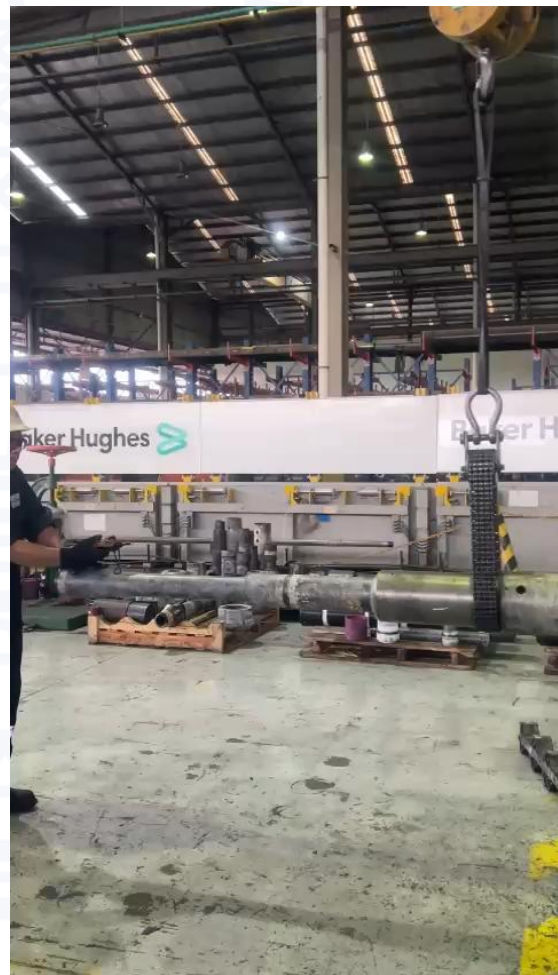
No.	Description	AICV Moving Type	AICD Non-Moving Type	Remarks
5	Installation	- Way simpler – no additional wash pipe (inner string) is required to establish circulation - Shorter installation time	- More complicated – need additional wash pipe to establish circulation - Longer time installation is needed to make up and run-in-hole wash pipe (inner string)	For non-moving type AICD, lower completion run c/w inner string
6	Additional job if Acid Stimulation is required	Need additional device to isolate end of lower completion string – post installation (additional cost if acid stimulation job is required – need full-cycle workover job, additional bridge plug (inflatable type) and coiled tubing unit)	No additional device is required to isolate lower completion string – acid stimulation job can be done in stages directly with wash pipe (inner string) still in the hole after set lower completion packer	Additional US\$600K – US\$1Mio is required for the well w/ Moving Type AICV for acid stimulation job
7	Cost	Cheaper for initial installation	More expensive for initial installation	As per current active contract

Manufacturing at Batam

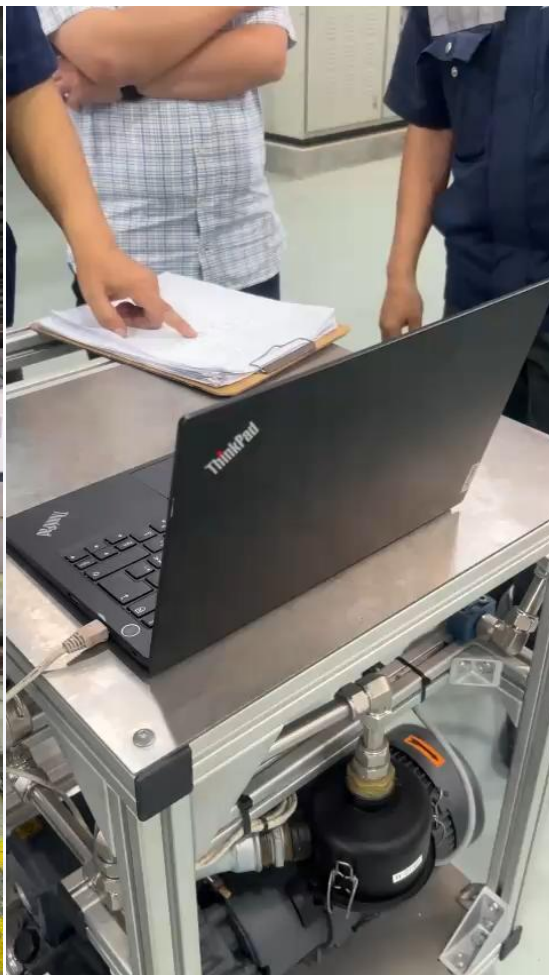


Workshop Preparation

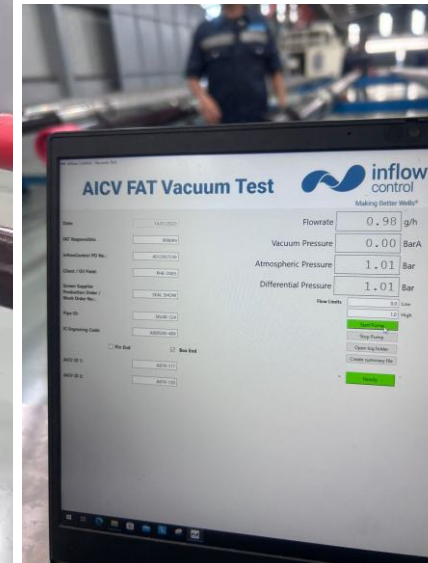
GP Packer



Flow Test



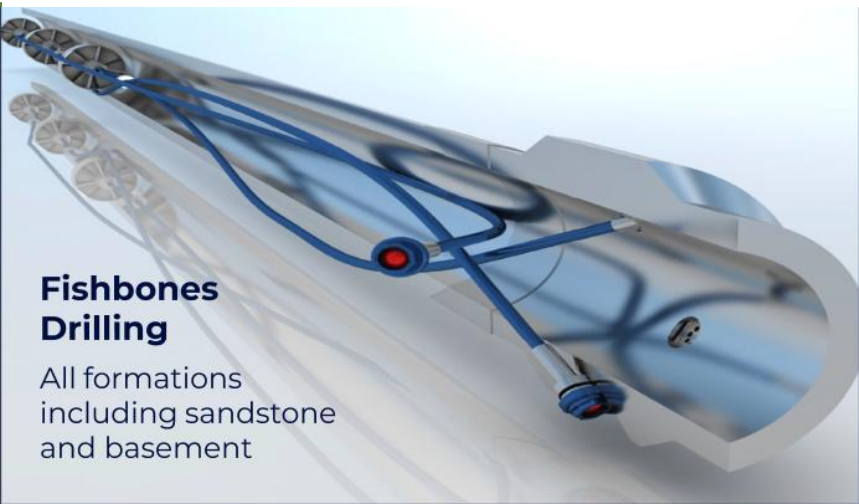
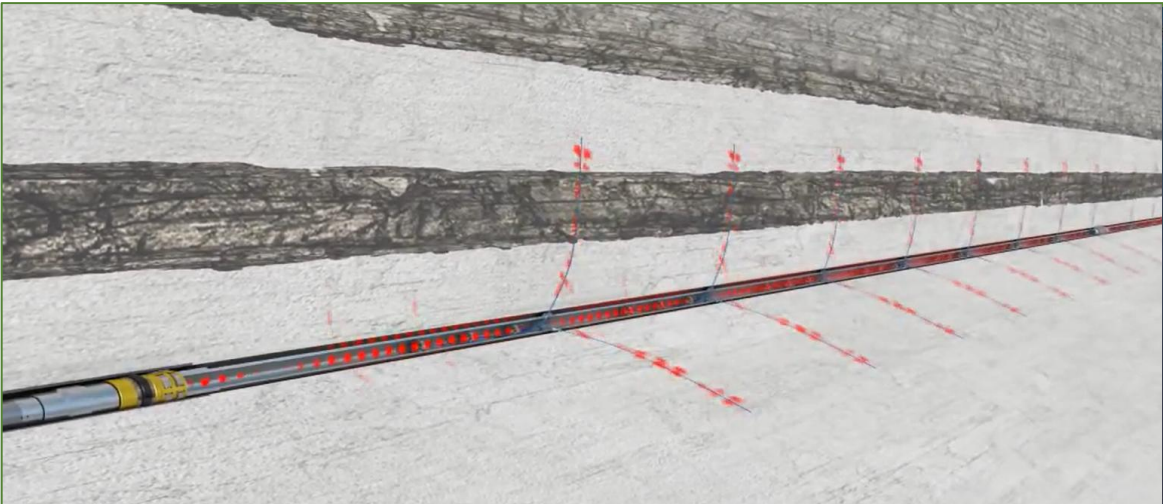
Vacuum/Seal Test



Field Execution



Upcoming Job: Fishbone® with AICD



Fishbones Drilling

All formations including sandstone and basement

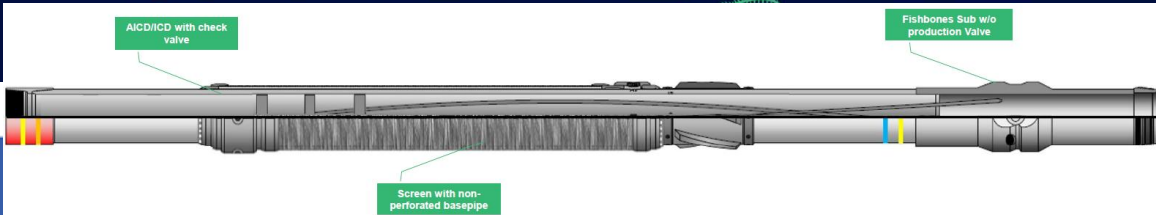
Technology Needle trajectory



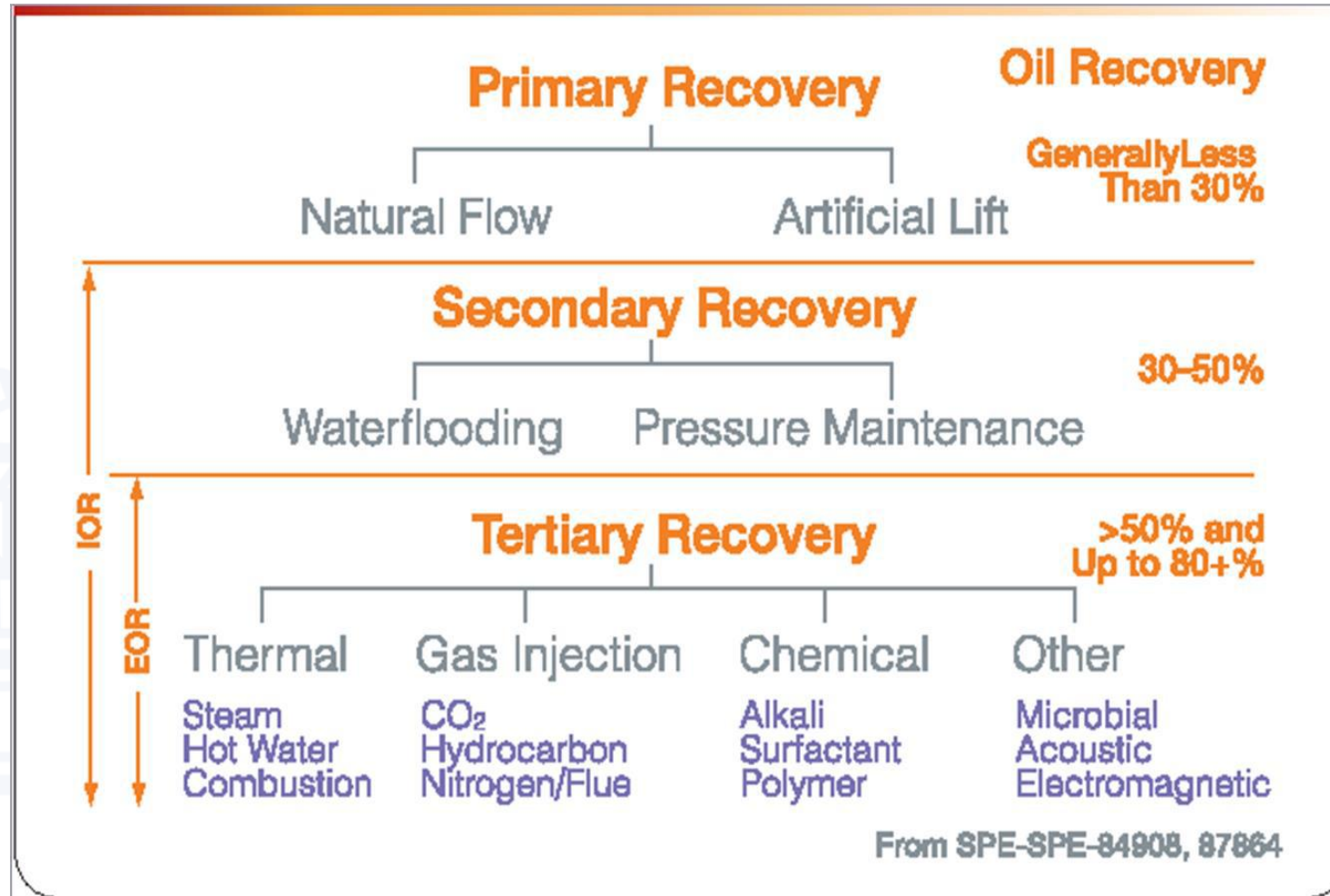
Technology Fishbones Drilling



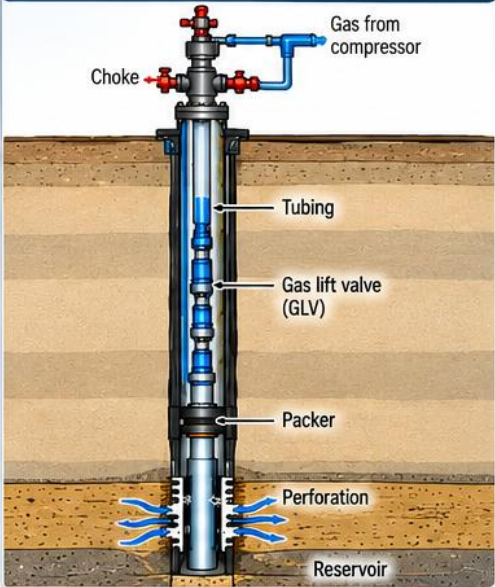
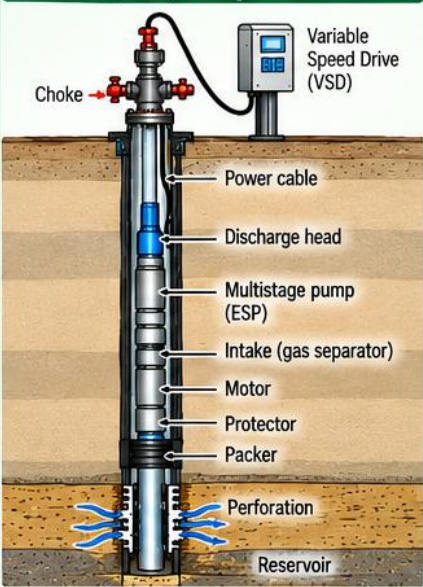
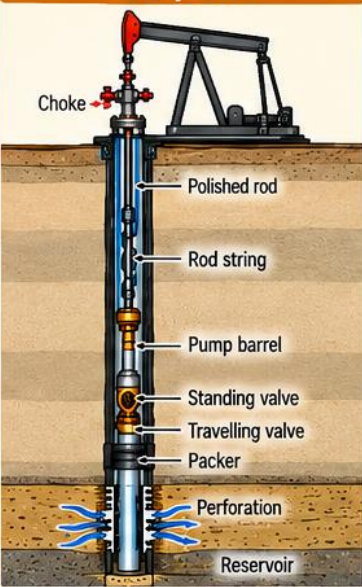
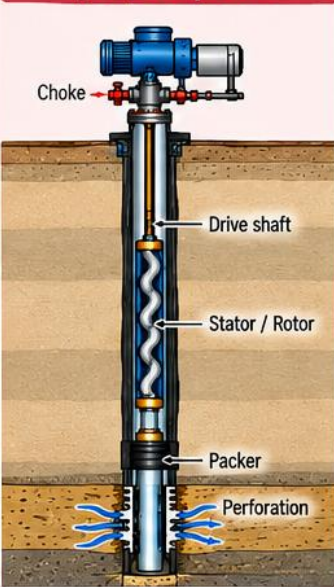
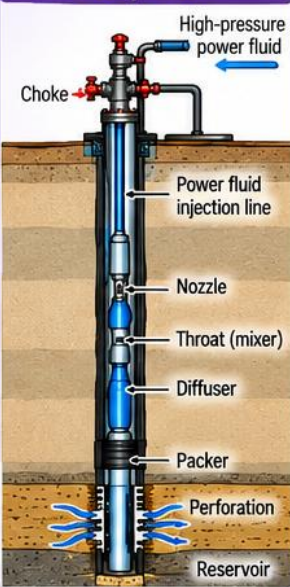
1. To repair (tight carbonates) reservoir permeability (stimulation) in combining w/ Autonomous Inflow Control Device (AICD) in one-run for oil producer well in horizontal section (eliminate rig time for separate run hydraulic fracturing job)
2. To maximize reservoir exposure, connect to naturally fractures (if any), to uniform distribution of acid (if required), to improve well productivity and at the end potentially increases ultimate recovery.
3. For more accurate stimulation zones
4. By combination w/ AICD application in horizontal well will balance inflow along horizontal section, mitigate early water breakthrough, reduce drawdown and water coning effect (by compartmentalize w/ swellable packer utilization)



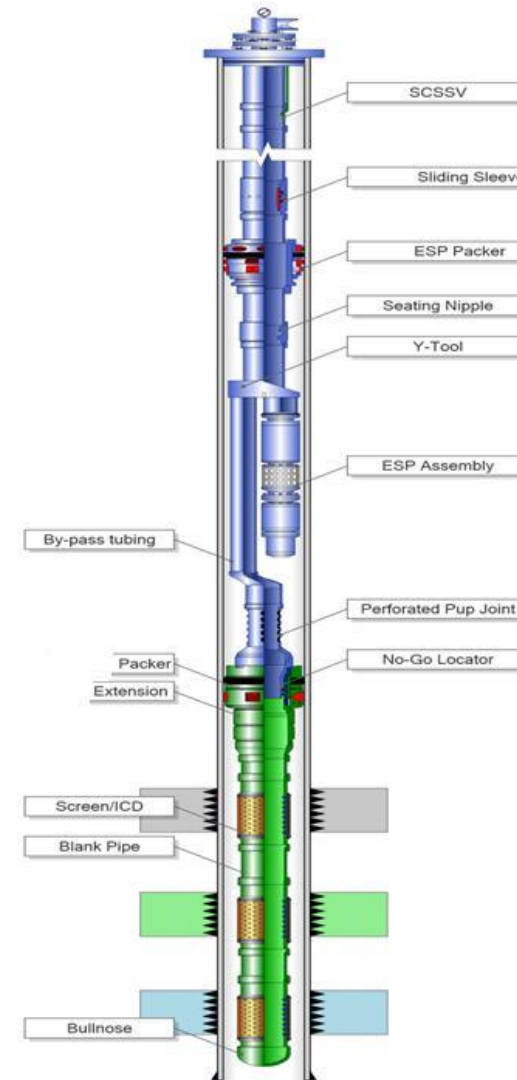
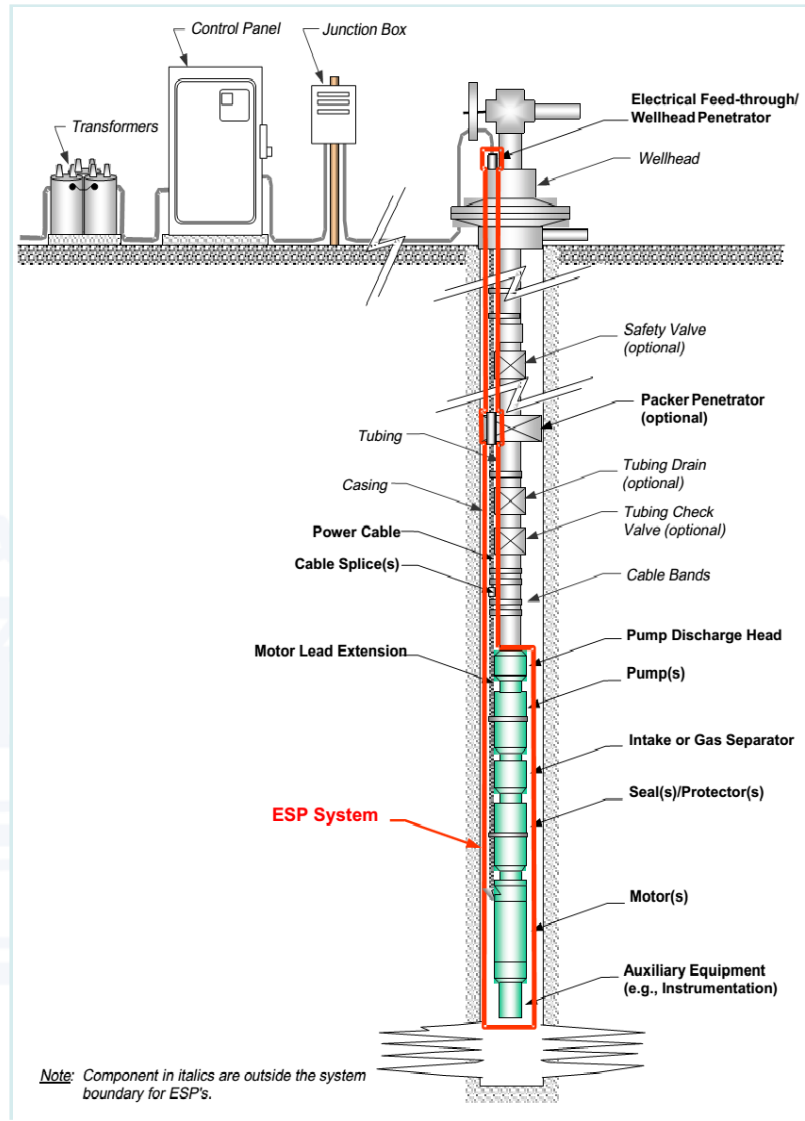
EOR- IOR



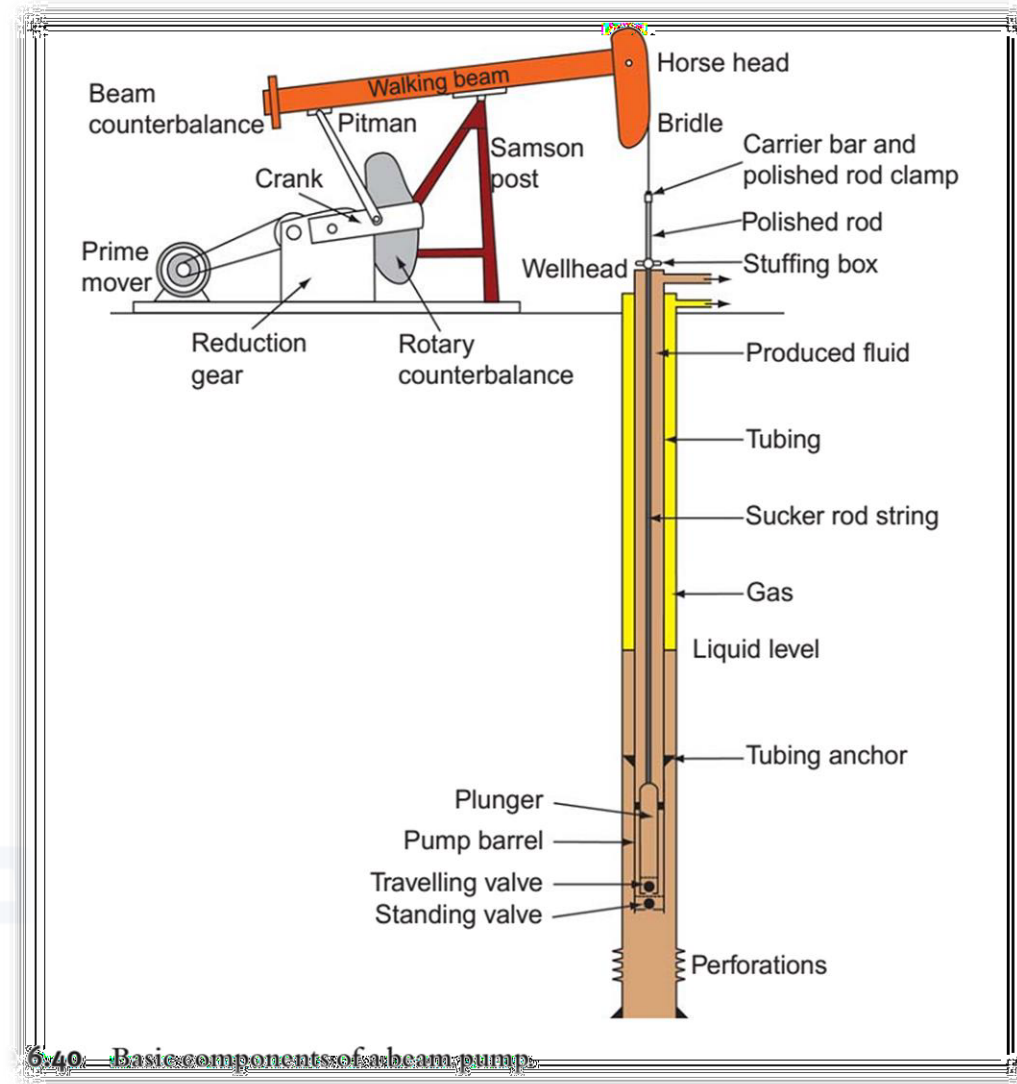
Type Of Well Completion according to Artificial lift

Gas Lift Completion	Electric Submersible Pump (ESP) Completion	Sucker Rod Pump (SRP) Completion	Progressing Cavity Pump (PCP) Completion	Hydraulic Pump (Jet Pump) Completion
				
How it works Injects high-pressure gas into tubing at various depths through gas lift valves to reduce fluid density and lift fluids to surface.	How it works Submersible multistage centrifugal pump installed downhole and driven by electric motor. Increases fluid pressure to lift fluids to surface.	How it works Reciprocating pump driven by surface pumpjack. Converts rotary motion to reciprocating motion to lift fluids by mechanical action.	How it works Helical rotor and elastomeric stator (positive displacement pump) rotate to lift fluids continuously and smoothly to surface.	How it works Injects high-pressure power fluid (usually treated water or produced fluid) through a nozzle to create suction and lift reservoir fluids (no moving parts downhole).
Typical application • Medium to high depth wells • High gas availability • Offshore/large fields • Variable rate or multiple zones	Typical application • Moderate to high liquid rate wells • Moderate depth (typically < 3,000 m) • Low gas content • Onshore and offshore	Typical application • Shallow to moderate depth wells • Low to moderate liquid rate • Low gas content • Onshore (widely used)	Typical application • High viscosity fluids • High water cut / sand content • Low to moderate gas content • Onshore and offshore (vertical or deviated wells)	Typical application • High gas/oil ratio wells (where ESP is not suitable) • High temperature or sand-prone wells • Remote/offshore wells • Deep wells
Pros • Simple and robust system • Flexible (can adjust injection rate and depth) • Suitable for high GOR wells • Lower CAPEX and OPEX vs ESP/PCP • Good for varying rates and multiple zones	Pros • High liquid handling capacity • Continuous and smooth production • Can handle moderate solids (with proper design) • Variable speed control to optimize rate • Suitable for stable and high-rate wells	Pros • Simple and proven technology • Lower CAPEX • Tolerant to some gas and solids • Easy to operate and maintain • Suitable for mature/declining wells	Pros • Handles high viscosity fluids • Good for high water cut and solids • Continuous and stable production • Lower vibration and noise • Can be run in deviated/horizontal wells (with design)	Pros • No moving parts downhole (simple, reliable) • Can handle high gas/oil ratio • Suitable for high temperature and harsh environment • Good for remote/offshore applications • Can be retrievable with wireline
Cons • Requires gas supply and compressor • Less efficient at very low rates • Lift gas reduces net production • Limited for high liquid rate or very deep/high-pressure wells	Cons • Higher CAPEX and OPEX • Sensitive to gas (gas lock) • Requires reliable power supply and electrical system • Limited for high temperature and very high H₂S/CO₂ environments • Workover required for retrieval/repair	Cons • Limited to shallow-moderate depth • Limited liquid rate • Requires surface space (pumpjack) • Less suitable for high gas, high viscosity, or deviated/horizontal wells • Rod/tubing wear and higher frequent maintenance	Cons • Sensitive to high gas content (gas lock) • Stator elastomer limitation (temperature and chemical compatibility) • Higher CAPEX than SRP • Stator wear requires replacement • Limited for very high rate or high temperature wells	Cons • Requires high-pressure power fluid and surface pumping system • Lower efficiency compared to ESP/PCP • Consumes power fluid (reduces net production). • Less suitable for low-rate wells • Needs appropriate surface facilities (water/gas supply and pumping system)

Completion with ESP

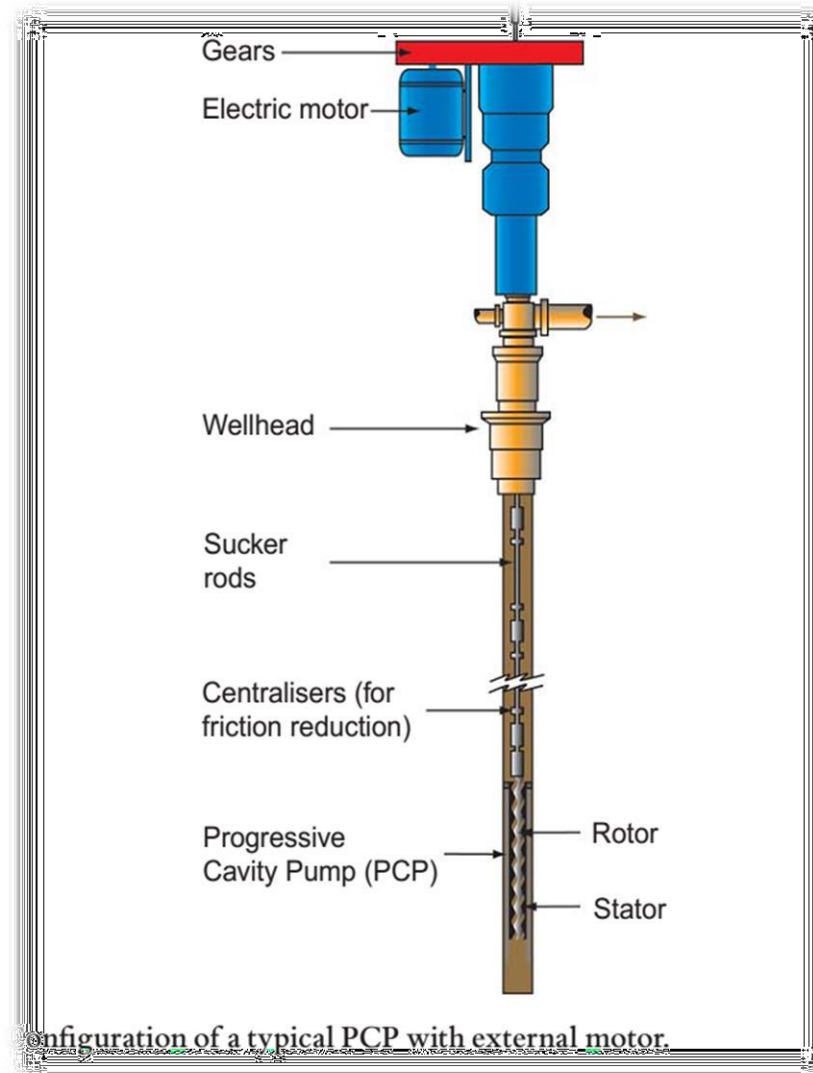


Completion with beam pump (Sucker Rods Pumps)



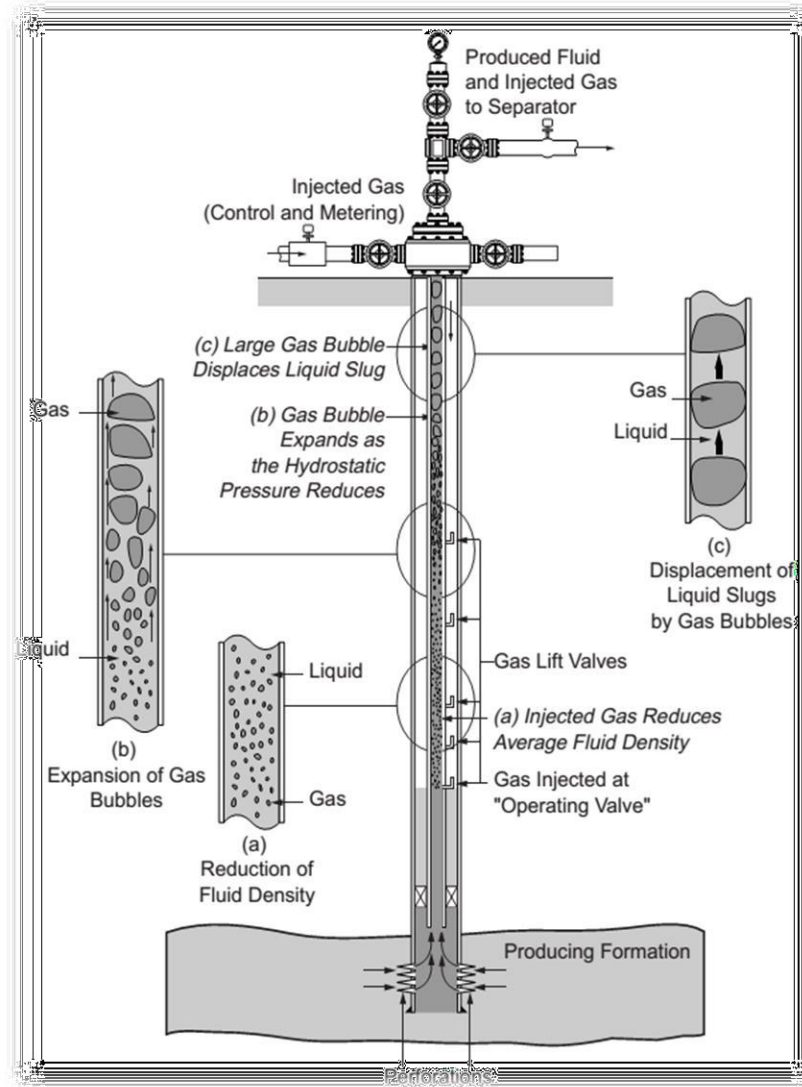
6.40 Basic components of beam pump

Completion with Progressing cavity Pump (PCP)

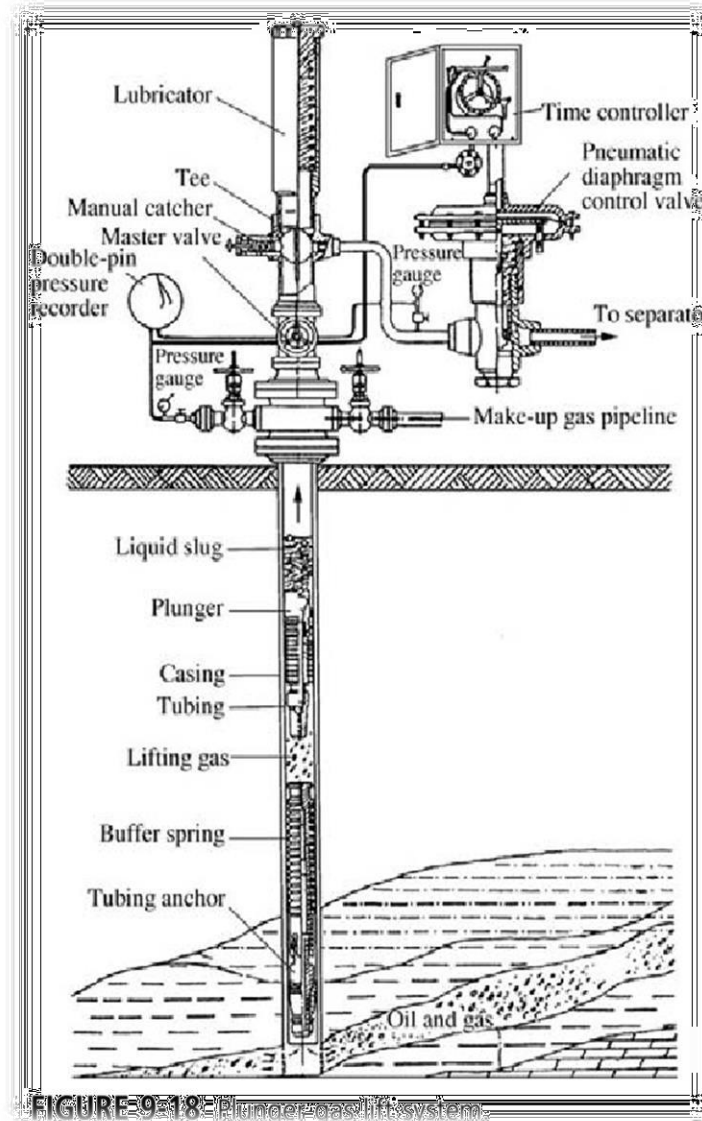


Configuration of a typical PCP with external motor.

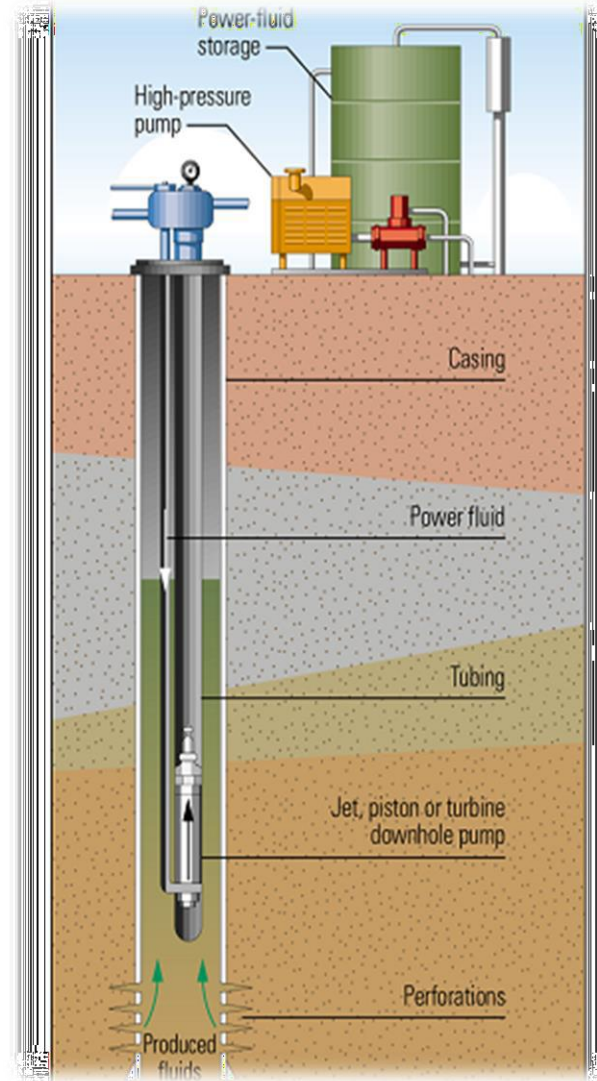
Completion with Gas-lift



Completion with (Plunger Gas-lift system)



Completion with Hydraulic-lift -Pump





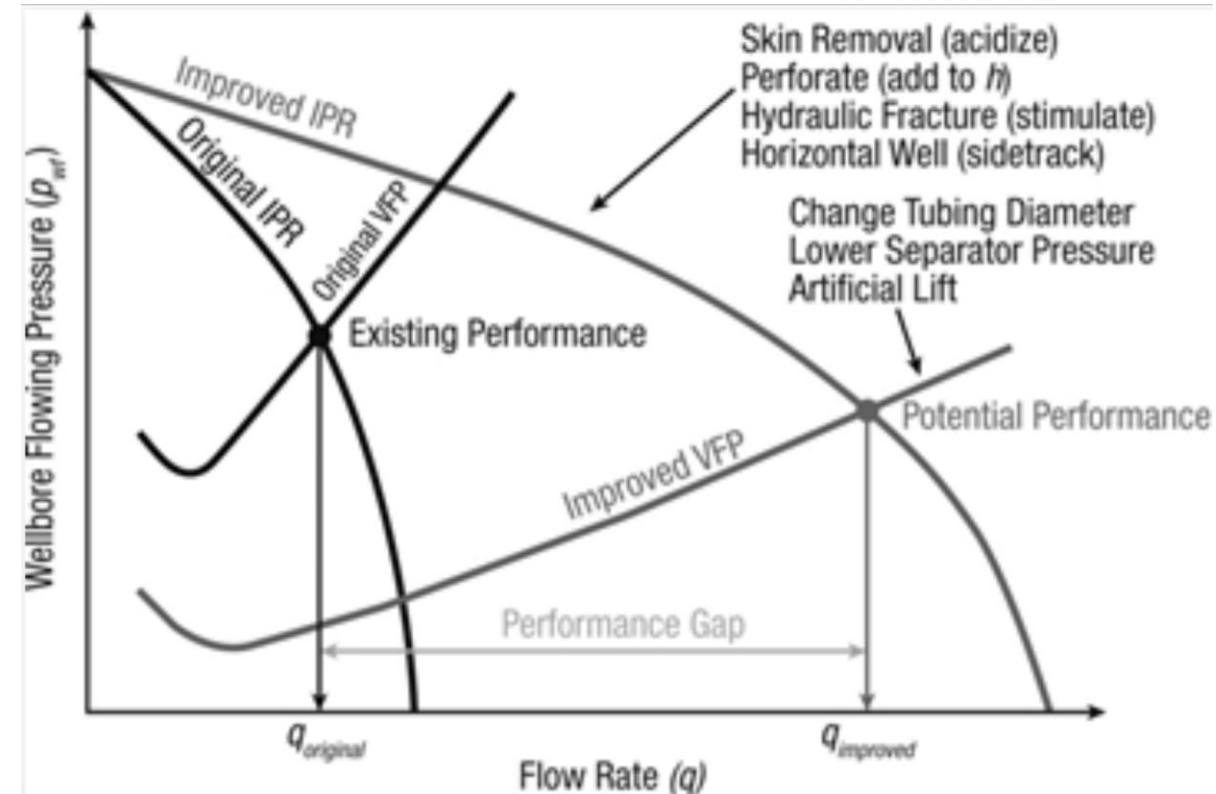
6. Well Completion Equipment and Accessories

Well Completion Equipment and Accessories

Production Technologist and Completions Engineer will determine the Tubing size sensitivity based on:

- IPR (Inflow Performance Relationship) with various Water Cut and GOR (Gas Oil Ratio) sensitivity.
- IPR with Tubing Head Pressure (Back Pressure) vs Reservoir Pressure sensitivity.

Use “NODAL Analysis” Software program



Completion Tubing Sizes

Tubing size ranges from 2 3/8"

2 7/8"

3 1/2"

4"

4 1/2"

5 1/2"

7"

To as large as 9 5/8"

Well completion equipment and Accessories

Completion accessories play a valuable role in customizing completions for well and reservoir specific requirements. Surveillance equipment design and reliability is dependent upon other peripheral equipment in the well. The accessories generally consist of tubing-mounted and flow-control equipment. Some of these are listed:

1-Tubing Mounted.

1. Sliding sleeves
2. Landing nipples
3. Expansion joints
4. Side-pocket mandrel
5. Pump out subs
6. Blast joints

2-Flow Control

This may be deployed inside tubing with slick-line.

1. Flow couplings
2. Blanking plugs
3. Equalizing standing valves
4. Circulating plugs

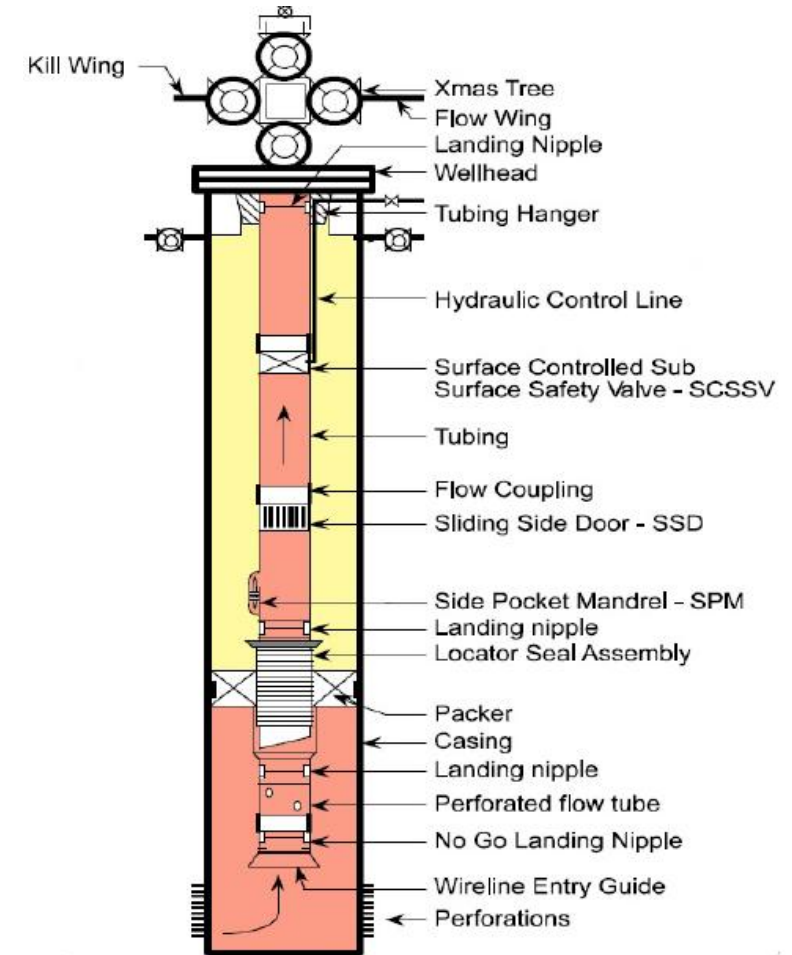
Well completion equipment and Accessories

- **A conventional completion typically contains components such as**

- Packers
- Subsurface Safety Valve
- Expansion and anchoring devices

- **Accessories includes:**

- Landing nipple
- Gauges
- Side Pocket Mandrel
- Sliding Side Doors
- Bottom hole chokes and regulators
- Flow coupling and Blast joints
- Perforated joint
- Wireline Re-Entry Guides
- Xmas tree & Tubing Hangers

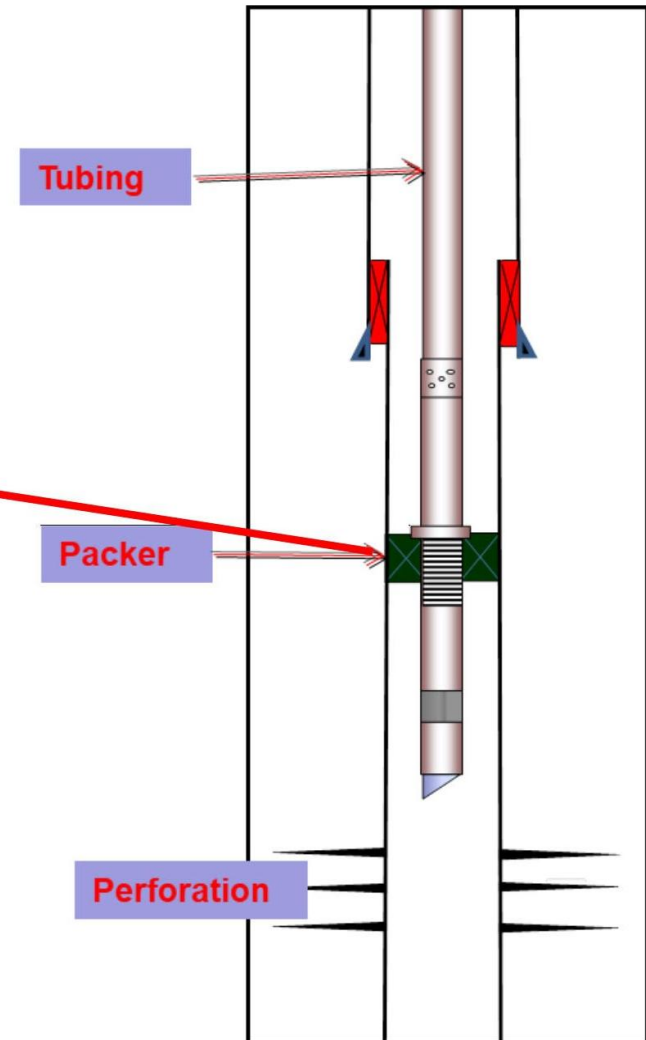
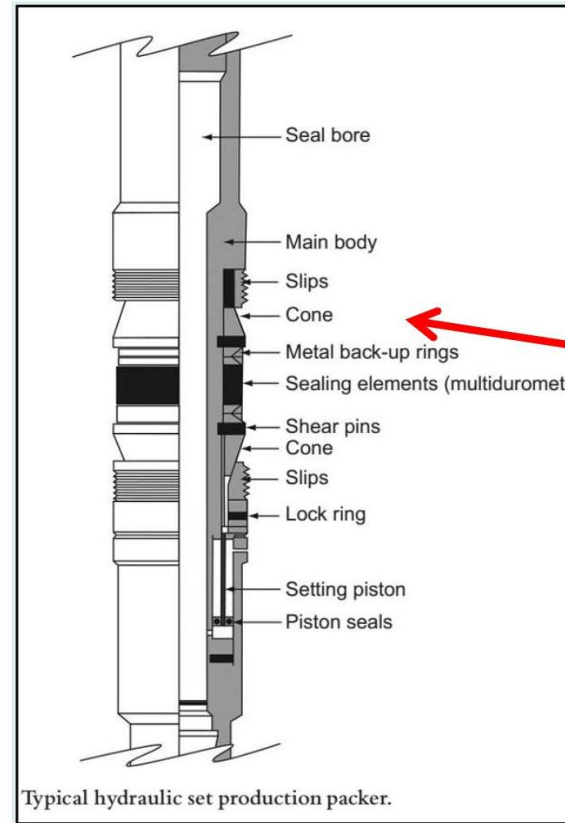


Typical Permanent Completion

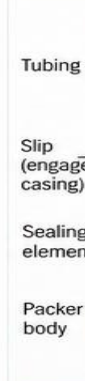
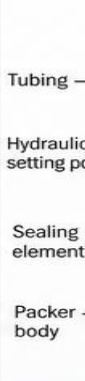
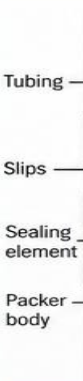
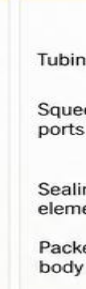
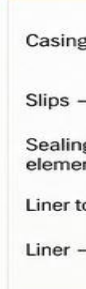
Packers

Packers. These are down hole devices used to provide a seal between the inside of the production casing and outside of the tubing string.

The packer must not restrict normal production flow. The tubing-mounted packers are either hydraulic or mechanical set and can be of the retrievable, isolation, or production type.



Packer Types Used in Oil & Gas Wells

1. Compression (Set-down) Packer  How it works Set by axial downward force (set-down weight). Compression of sealing element against casing. Typical use • Temporary zonal isolation • Testing • Workover jobs (e.g. squeeze) Pros • Simple design • Easy to set and retrieve • Widely used Cons • Lower pressure/temperature rating • Can move if tubing load changes	2. Tension (Set-up) Packer  How it works Set by upward tension. Slips engage casing and element seals by upward pull. Typical use • Zonal isolation • Production completion • Artificial lift (e.g. ESP, gas lift) Pros • Higher pressure rating • Good stability • Common for production and long-term Cons • Requires tubing pull to set • More complex than compression packer	3. Hydraulic Packer  How it works Set by applying hydraulic pressure from surface or downhole tool. Releases by hydraulic pressure (usually opposite direction). Typical use • Production completion • Smart well / ICV • ESP lower packer • High pressure applications Pros • High setting force • High pressure/temperature capability • Reliable for long-term Cons • Requires hydraulic control system • More expensive • Operationally more complex	4. Mechanical-Set Packer (Using Setting Tool)  How it works Set by mechanical tool (e.g. J-slot, bowl, or cam). Setting force compresses/slides element to seal against casing. Typical use • Workover and intervention • Zonal isolation • When controlled setting is required Pros • Controlled and secure setting • Good for higher loads • Can be retrieved with tool Cons • Requires setting tool and additional operations • Slightly higher cost	5. Permanent (Production) Packer  How it works Set mechanically (often hydraulic or mechanical) and designed for long-term, permanent zonal isolation. Typical use • Production completion • Artificial lift (ESP, gas lift) • Multi-zone completion Pros • High pressure/temperature • Long-term reliability • Suitable for production environment Cons • Cannot be easily retrieved • Higher cost • Requires careful design and setting
6. Retrievable Packer  How it works Set mechanically, hydraulically or by set-down/up. Can be retrieved later using a specific retrieval tool (e.g. pulling, jar, or hydraulic release). Typical use • Workover / testing • Temporary zonal isolation • Intervention operations Pros • Can be retrieved • Flexible for multiple operations • Available in various ratings Cons • More complex than permanent packer • Requires retrieval tool • Higher cost	7. Cement Packer (Squeeze Packer)  How it works Set mechanically or hydraulically to isolate a zone. Has ports to pump cement through after packer is set. Typical use • Cement squeeze operations • Zonal isolation for remedial cementing (e.g. behind casing or across perforations) Pros • Allows cement placement through packer • Good isolation for squeeze jobs • Can be retrievable or permanent Cons • Limited to squeeze applications • Requires pumping and pressure control • May have lower pressure rating than production packers	8. Bridge Plug (Mechanical Packer)  How it works Set by mechanical tool. Expands element to seal against casing. Usually does not have throughbore (solid plug). Typical use • Permanent zonal isolation • Abandonment (P&A) • Blocking off zones • Often combined with cement on top Pros • High integrity seal • Simple and reliable • Common for P&A Cons • Not retrievable (most types) • No throughbore for production • Requires milling if removal needed	9. Liner Packer  How it works Set to seal liner to casing. Can be mechanical or hydraulic set. Typical use • Liner hanger system • Creating seal between liner and casing • Often used in deep or complex wells Pros • Provides secure liner-casing seal • High pressure/temperature rating • Used in many completion designs Cons • Requires running liner string • Higher cost • Not typically used for temporary isolation	

Sliding Sleeves

- **Side Sliding Door (SSD)**

- The capability of establish tubing / annulus communication in order to equalize pressures and for circulation is a desirable feature in many completion designs.
- SSD or sliding sleeves provide communication between the well production conduit and various annulus.

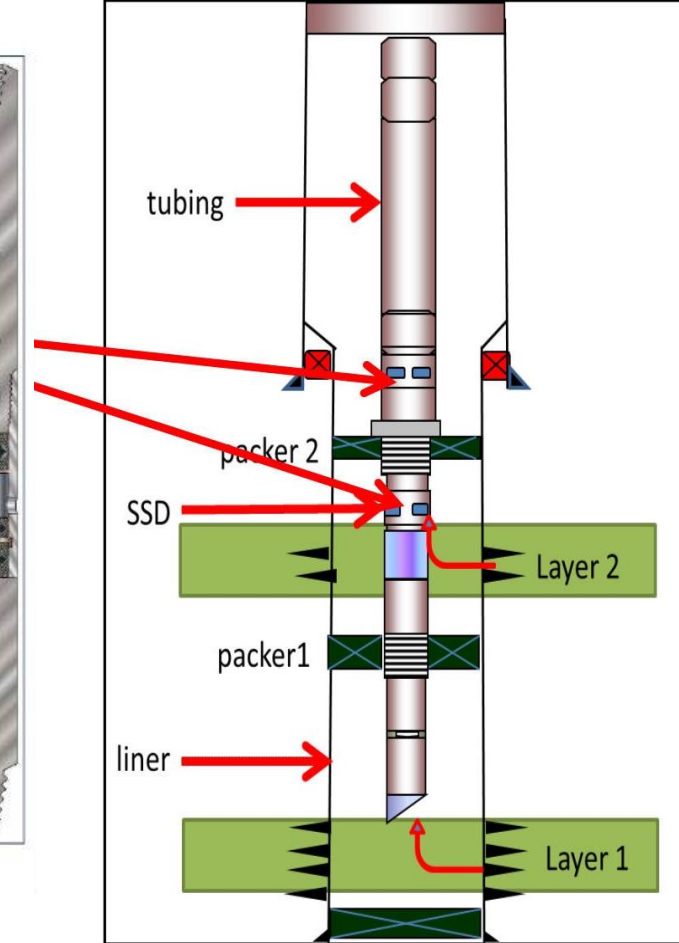
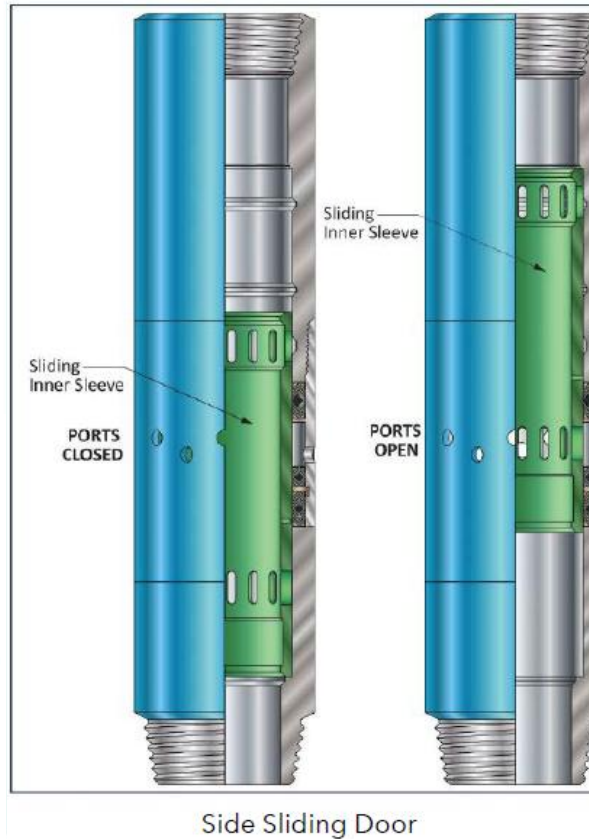
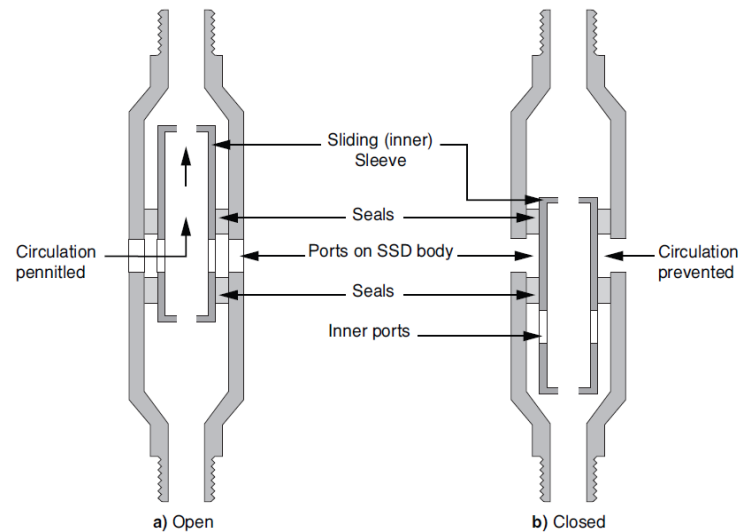
- **Various applications include:**

- Fluid displacement
- Well killing (by fluid circulation)
- Commingled production
- Kicking off wells (gas lift)
- Pressure equalization
- Selective testing or producing multiple zones
- Chemical injection

- **Various applications include:**

- Wireline / coiled tubing methods
- Pressure applied to the tubing after dropping or running a shifting dart
- Pressure applied to the annulus acting on the differential piston area

- **They may be selected in the either the shift down to open or shift up to open versions.**

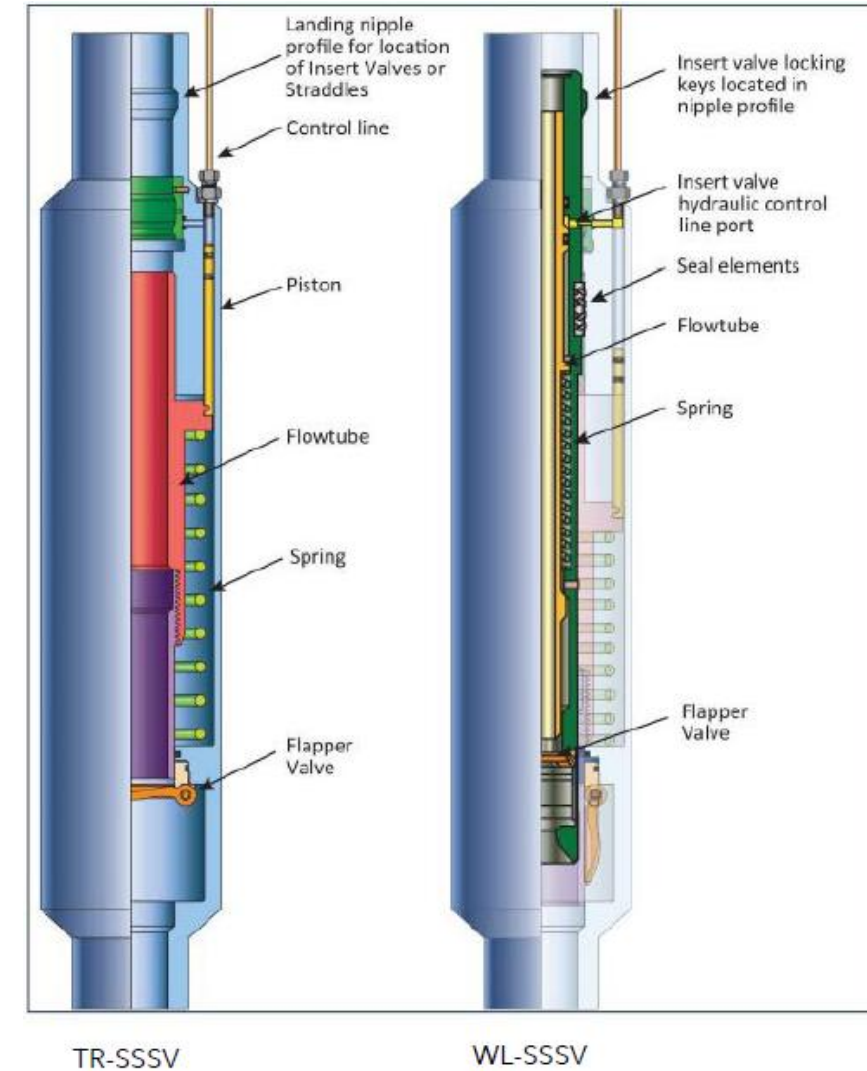


Surface-controlled subsurface safety valves (SCSSVs)

The SCSSVs are also installed in the tubing string below the surface tubing hanger. They are controlled by hydraulic pressure through a capillary (control) line that connects to a surface control panel. Most SCSSV designs today use a flapper to form a seal. Both elastomeric and metal-to-metal seal designs are available.

The SCSSV is a normally closed (failsafe) valve and requires continuous hydraulic pressure on the control line to keep it open.

- The prime purpose of the subsurface safety valve (SSSV) is to protect people, property and the environment against uncontrolled production while safeguarding the hydrocarbon resource asset.
- The uncontrolled situations may arise as a result of sabotage, collision, equipment failure, human errors, fire or leaks.
- The incorporating of a SSSV in the well completion depends on a number of factors:
 - Capacity of the hydrocarbon source
 - Characteristics of the hydrocarbon reservoir
 - Government regulations
 - Operator requirements
- **The general recommendation in the light of the potential effect of oil and gas spills on the environment, is that all wells capable of SUSTAINED NATURAL FLOW be equipped with subsurface safety valve.**



Impact of failure SCSSSV or not available of SCSSSV in offshore



Wireline Entry Guide (WEG)

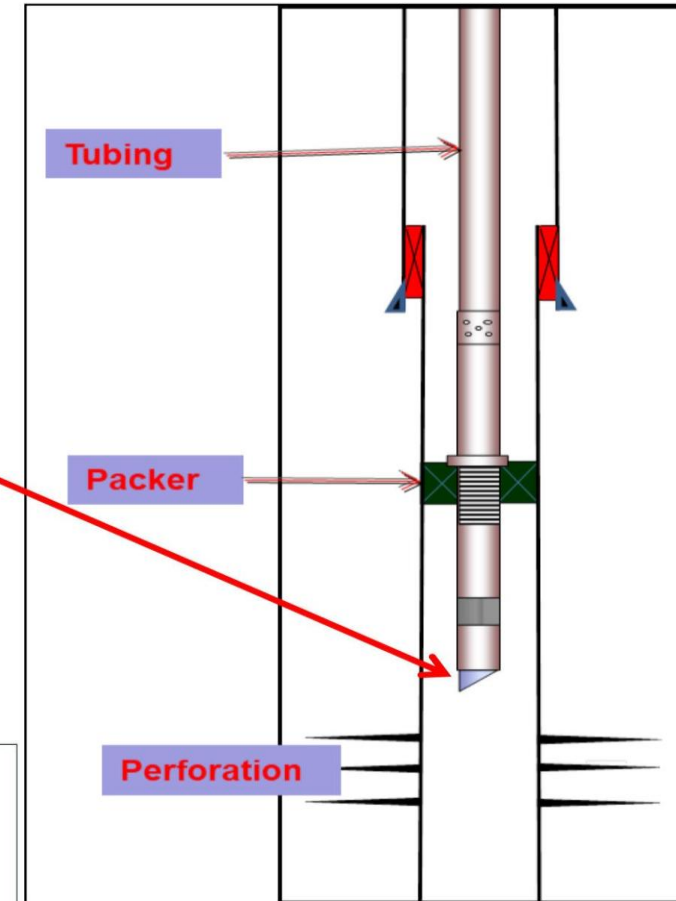
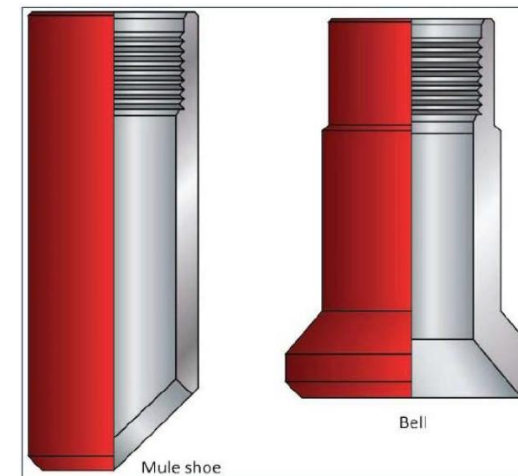
- There are two types: (The mule shoe guide - The bell guide)

➤ **The mule shoe:**

- Most commonly used guide in completion installations.
- Generally it is 1 ft and consists of a female thread at its upper end and a flat bottom with 45 degrees cut in it at the lower end.
- The 45 degrees taper eases the running of the tubing completion through restrictions, such as liner hangers.
- Also while pulling a wireline tool string back into the tubing, a 45 degrees chamfer on the taper will guide the rope socket and tool string back into the tail pipe assembly.

➤ **The bell guide:**

- It's shaped like a bell with 45 degrees taper cut in its lower end.
- The upper end is like the mule shoe type, has a female thread cut in it.
- The 45 degrees taper allows easy re-entry of the wireline tool string when it is pulled back into the tail pipe assembly.
- This guide generally used in completion where the end of the tubing does not need to bypass the top of a liner hanger.



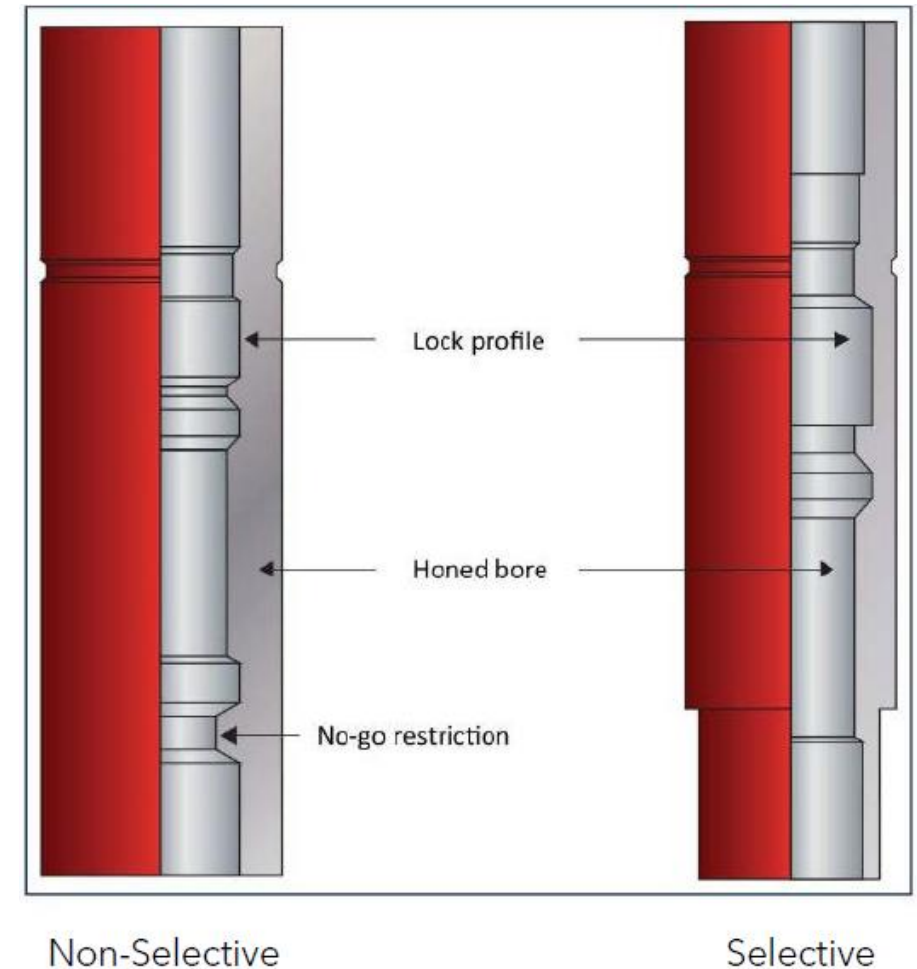
Landing Nipple

- **Landing Nipples:**

- A landing nipple is a short tubular component machined with tubing thread top & bottom, compatible with completion tubing.
- It is internally machined to provide a locking recess and sealing bore into which the locking device is installed
- **Types:** Non-Selective and Selective

- **Advantages of using landing nipple in a single string completion:**

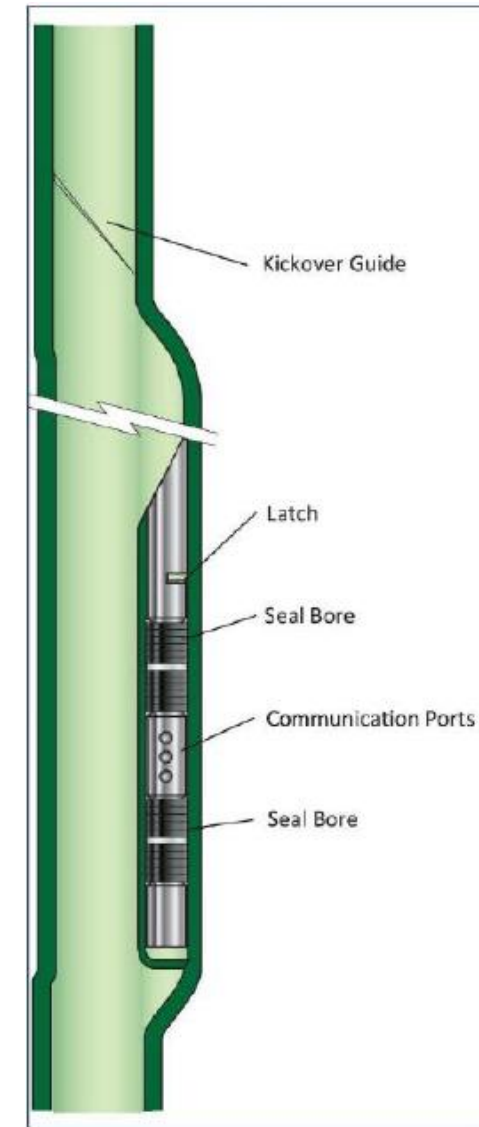
- The well may be plugged from above or below or from both directions
- The tubing string may be pressure tested
- SSSV, Check valves, ... may be safely set in the tubing string
- Bottom hole pressure and temperature gauges may be installed
- The nipples may be used as a reference point for checking depth



Side Pocket Mandrel

- **Side Pocket Mandrel (SPM):**

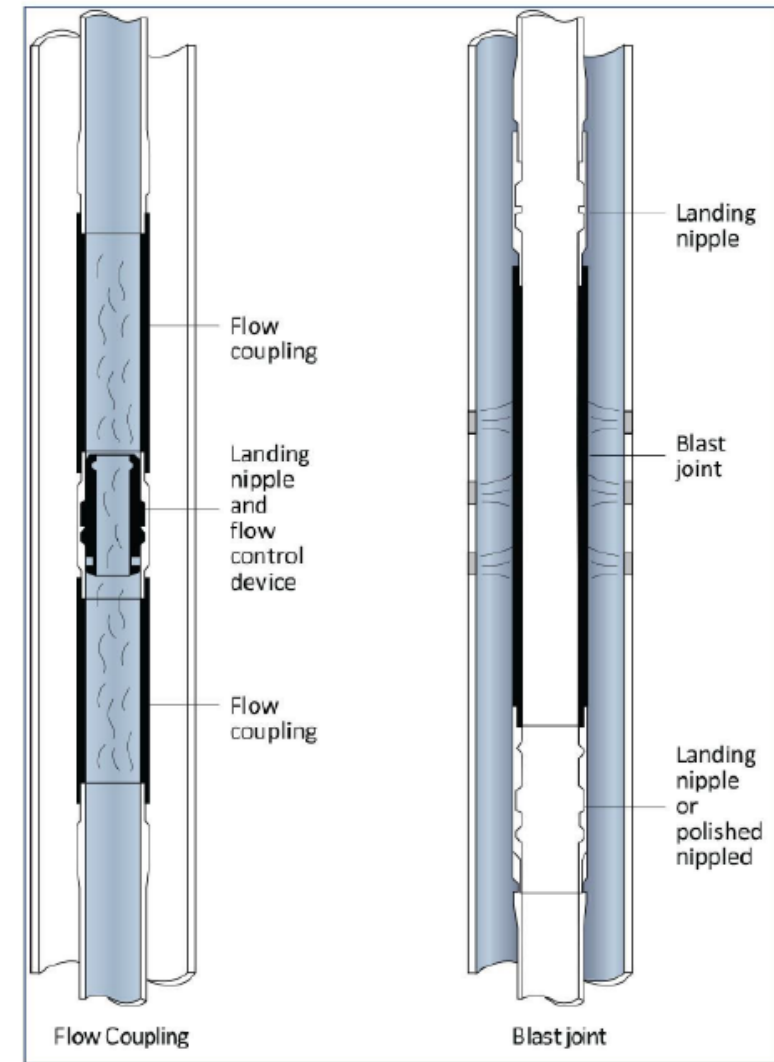
- Originally used for gas lift, but can be used as an alternative circulation device.
- The SPM, uses valves that can be set or retrieved on wireline using a kickover tool, which positions the device in the side pocket mandrel
- When no communication is required between tubing and annulus, a dummy valve is located in the SPM.



Side Pocket Mandrel

Blast Joint and Flow Coupling

- These are important aspects of life-of-the-well completion planning.
- They are designed to inhibit the effects of erosion / corrosion caused by flow turbulence and jetting actions.
- **Flow Coupling:**
 - Flow couplings are short sections of thick walled pipe.
 - Flow couplings should be considered above and below any major ID change (i.e., Crossovers, TR-SSSV, Landing nipples,...)
 - The purpose of this is to withstand any internal erosion caused by turbulent flow through differing tubing IDs and therefor promotes enhanced completion reliability of the production life of the well.
- **Blast Joints:**
 - Blast joint withstand erosion externally, and are normally positioned at perforated production zones where the jetting action of the fluid can erode the outside of the tubing.
 - A blast joint is a joint of tubing with enhanced wall thickness, and usually manufactured from a heat treated alloy, also can be coated by a tungsten carbide.



Blanking Plug and Circulating Plug

Blanking Plug. . These are designed to seal tubing pressure from either above or below the plug . Used for applications that require either bleeding off pressure in the tubing or circulating above the plugs.



Circulating Plugs

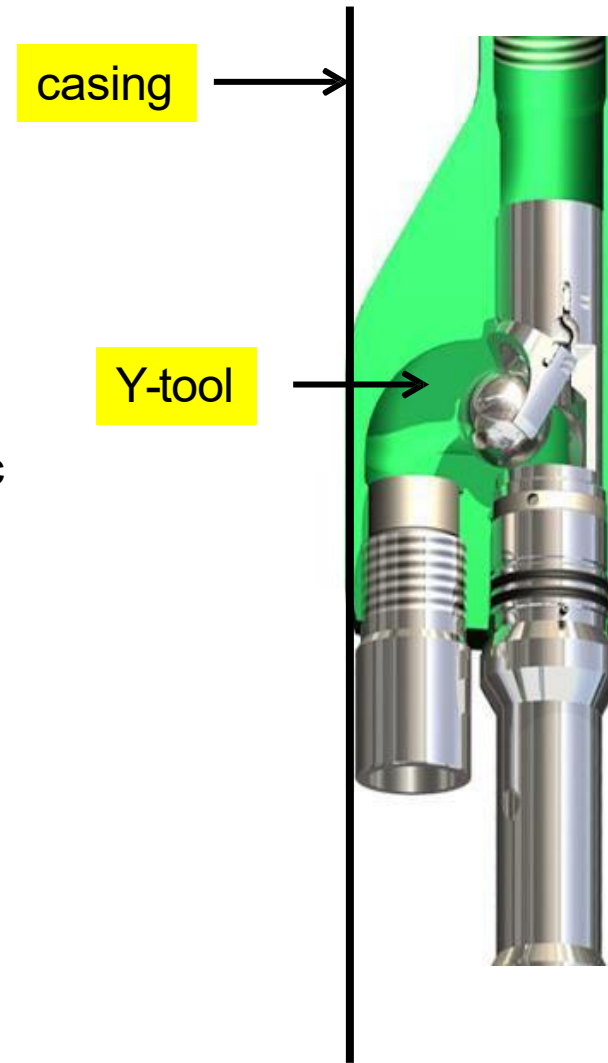
Hold pressure from below and allow flow from above. Used to plug formation pressure in the tubing string while providing adequate flow area for pumping into the formation.

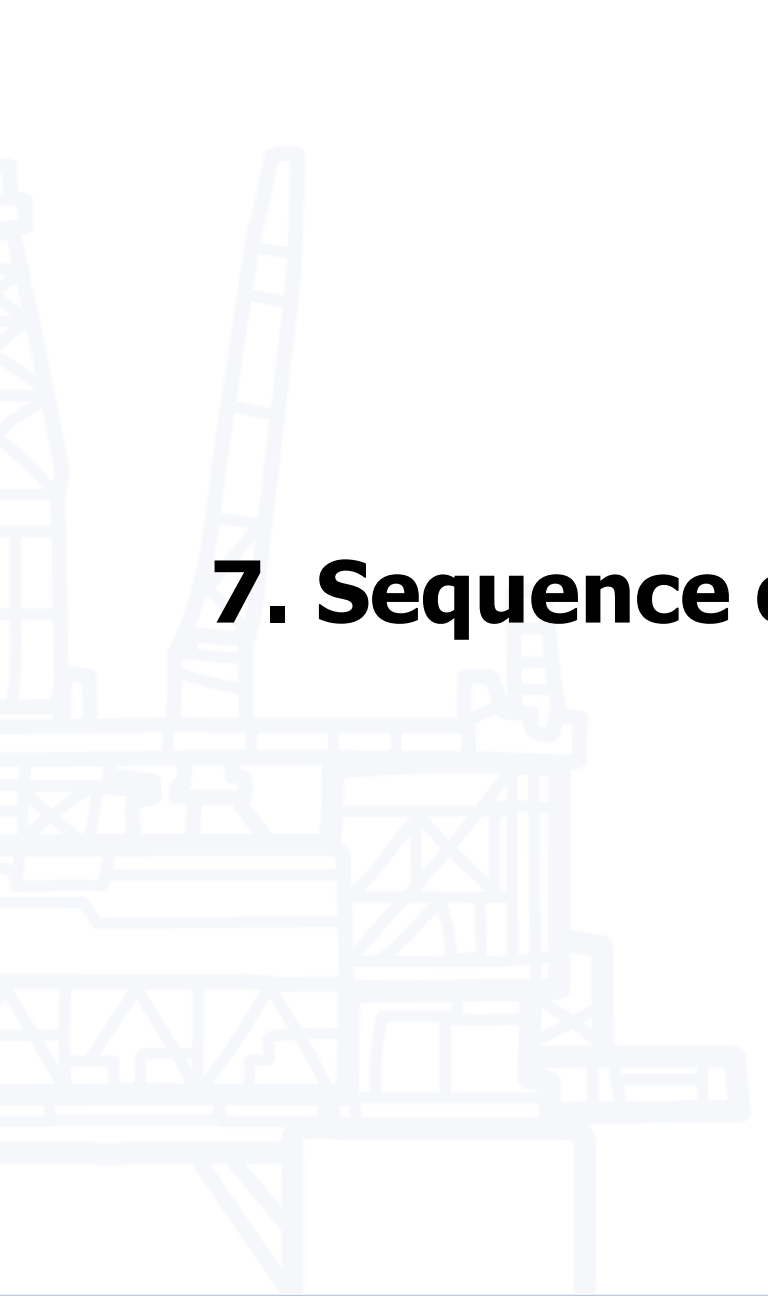


Well completion equipment and Accessories

Y-tool

The Y-tool is a solution to enable production-logging and well intervention below a working ESP at any point in time during production without pulling the completion string. The Y-tool is installed on the production tubing, providing two separate conduits. One conduit concentric with the production tubing and enables access to the reservoir below the ESP. The second conduit is offset and used to support the ESP system. Flow rates in different perforation intervals and other valuable geophysical information could be collected for production optimization and enhanced recovery plans.



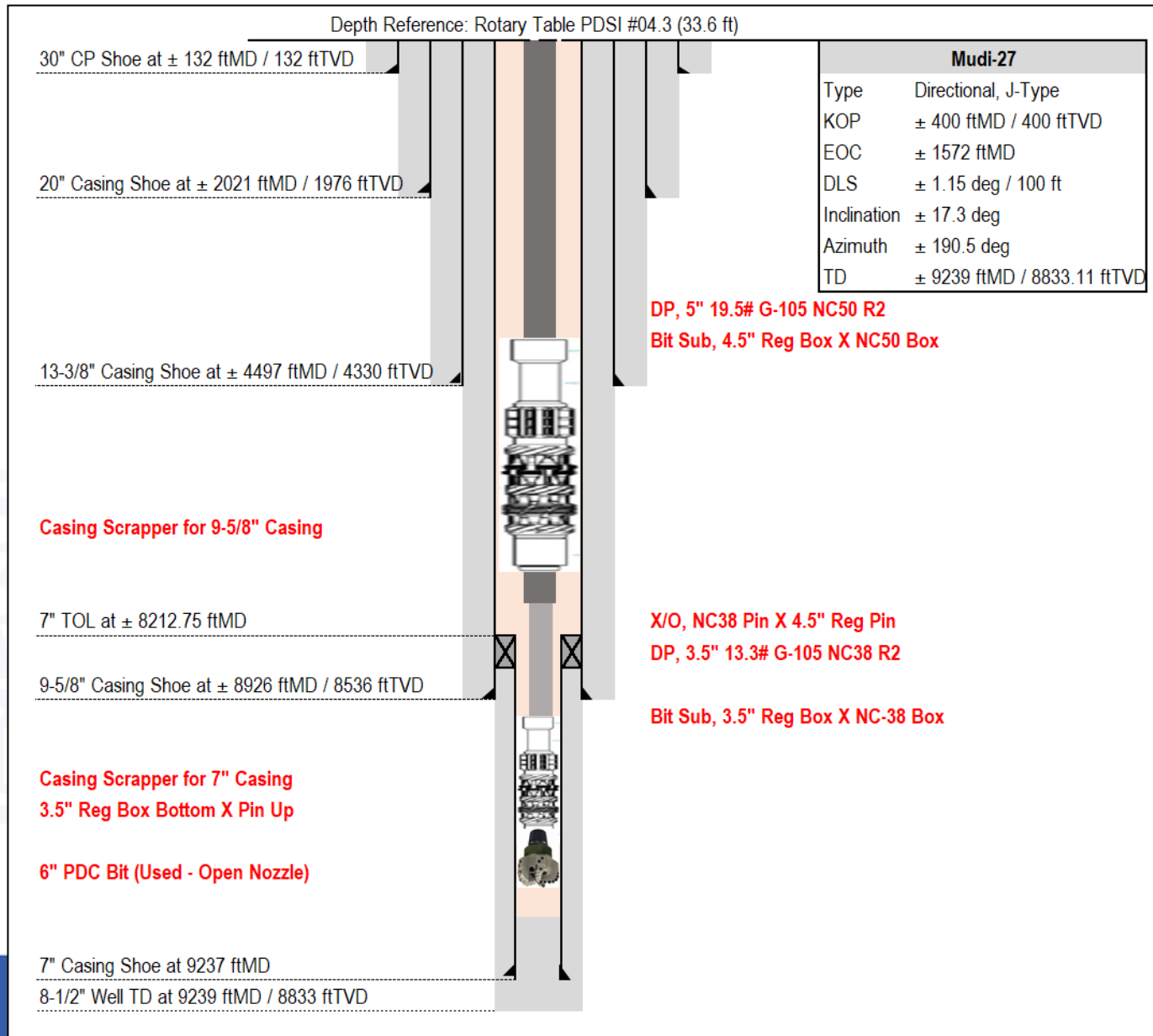


7. Sequence of Completion Operations

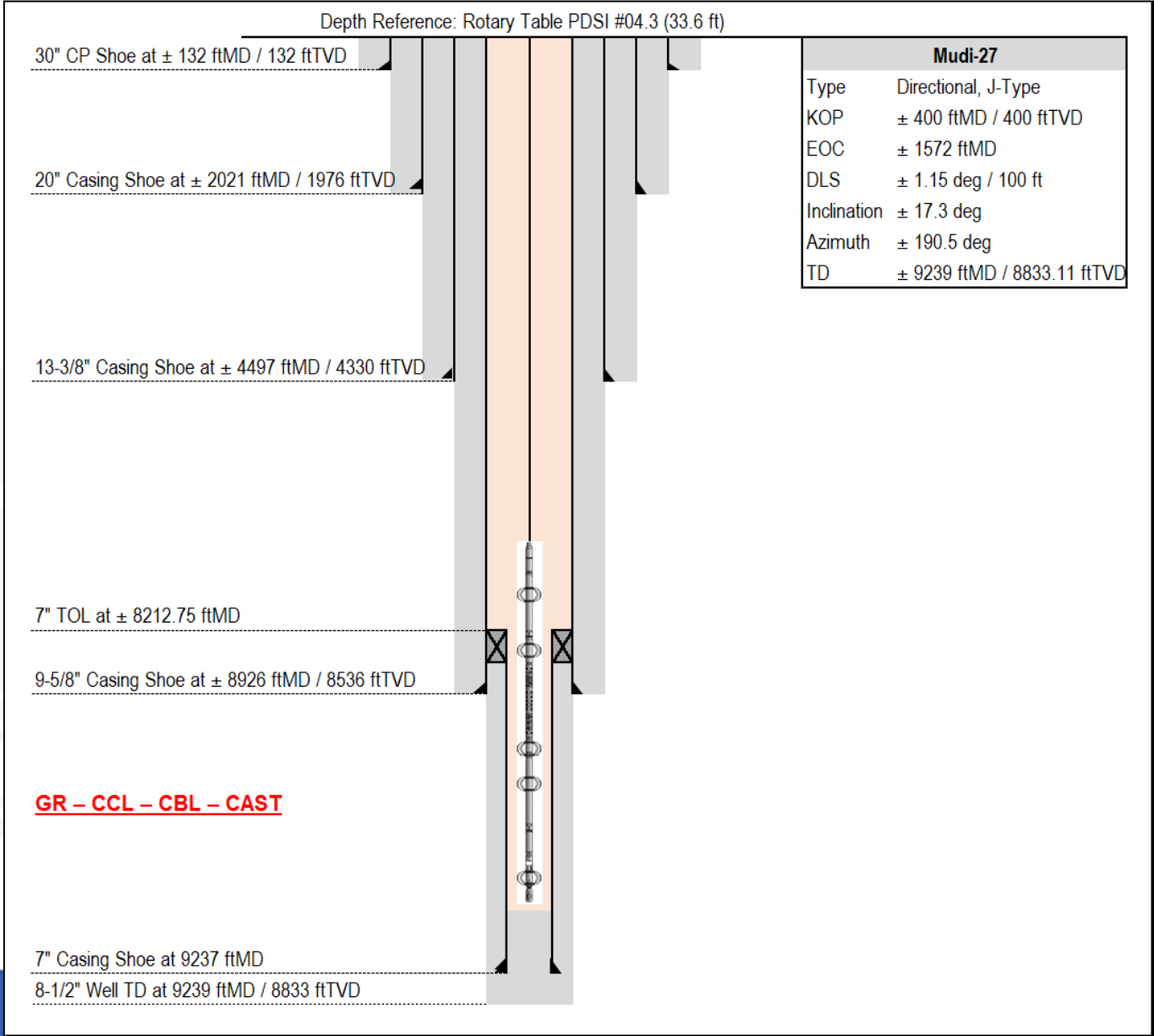
Sequence of Completion Operations:

- **Well cleanout**
- **Perforate (overbalance)**
- **(Sand exclusion - if required)**
- **Run and set completion string**
- **Install X-mas tree**
- **Flowline Tie-in**
- **Unloading well**
- **Handover to production**

Well Preparation And Cleaning (Well Bore Clean Out)



CBL – CCL – VDL / Sonic Run



Well Perforating

MUDI-27 WELLBORE DIAGRAM

Depth Reference: Rotary Table PDSI #04.3 (33.6 ft)

30" CP Shoe at ± 132 ftMD / 132 ftTVD

20" Casing Shoe at ± 2021 ftMD / 1976 ftTVD

13-3/8" Casing Shoe at ± 4497 ftMD / 4330 ftTVD

7" TOL at ± 8212.75 ftMD

9-5/8" Casing Shoe at ± 8926 ftMD / 8536 ftTVD

4.5" 5SPF HSD Gun

7" Casing Shoe at 9237 ftMD

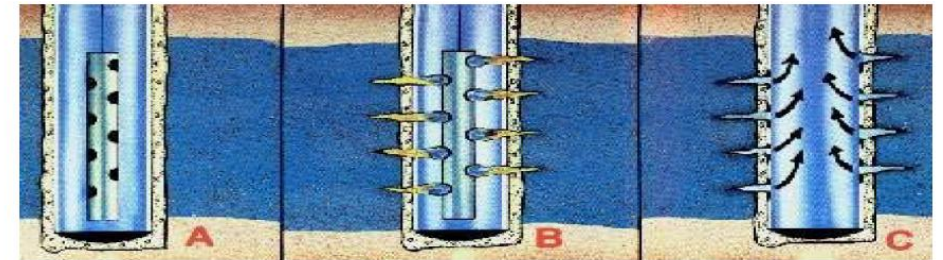
8-1/2" Well TD at 9239 ftMD / 8833 ftTVD

Mudi-27

Type	Directional, J-Type
KOP	± 400 ftMD / 400 ftTVD
EOC	± 1572 ftMD
DLS	± 1.15 deg / 100 ft
Inclination	± 17.3 deg
Azimuth	± 190.5 deg
TD	± 9239 ftMD / 8833.11 ftTVD

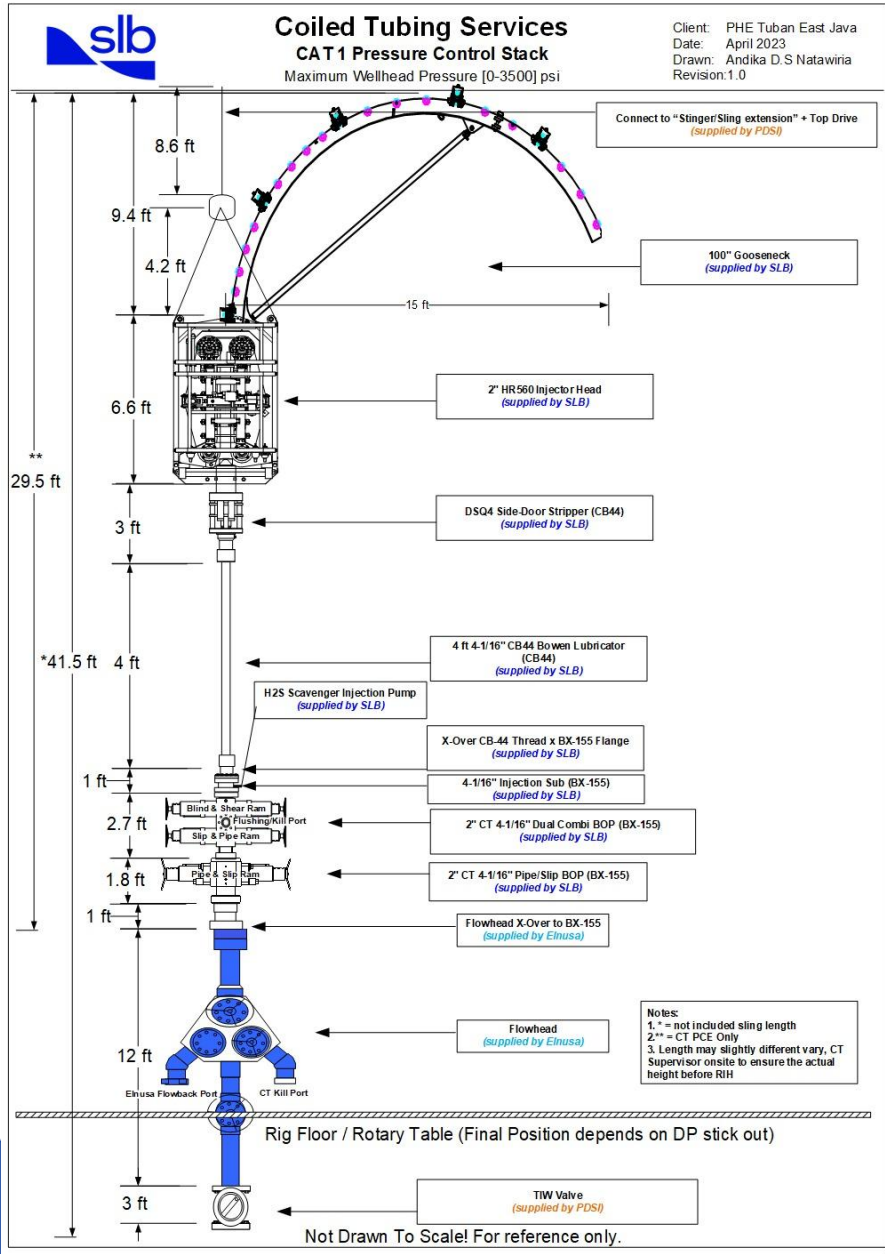
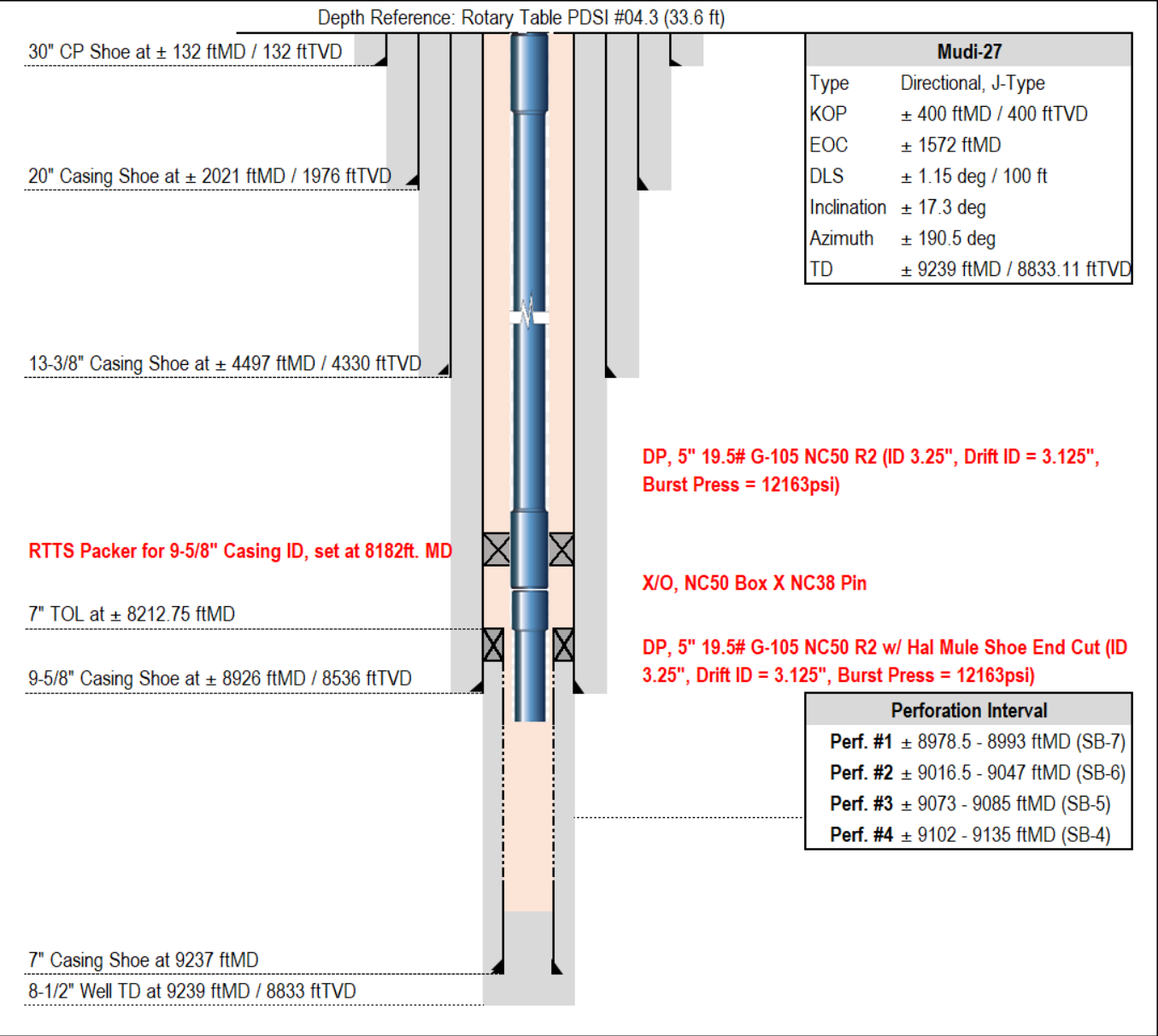
Perforation Interval

Perf. #1	$\pm 8978.5 - 8993$ ftMD (SB-7)
Perf. #2	$\pm 9016.5 - 9047$ ftMD (SB-6)
Perf. #3	$\pm 9073 - 9085$ ftMD (SB-5)
Perf. #4	$\pm 9102 - 9135$ ftMD (SB-4)



- Perforating gun (WCP type) is lowered into the hole at the depth where the oil or gas formation is found (A).
- After the gun is lined up properly, powerful explosive charges are fired (B) from the control panel at the surface. These explosives blast a hole in the steel casing and cement, up to several feet out into the rock.
- Finally, the oil and gas fluids flow into the holes and up the well to the surface (C).

Stimulation Job w/ dedicated stimulation string



Running Completion w/ ESP

Depth Reference: Rotary Table PDSI #04.3 (33.6 ft)

30" CP Shoe at ± 132 ftMD / 132 ftTVD

20" Casing Shoe at ± 2021 ftMD / 1976 ftTVD

13-3/8" Casing Shoe at ± 4497 ftMD / 4330 ftTVD

3-1/2" Tubing at ± 5090 ftMD

ESP (Bottom) at ± 6050 ftMD

7" TOL at ± 8212.75 ftMD

9-5/8" Casing Shoe at ± 8926 ftMD / 8536 ftTVD

7" Casing Shoe at 9237 ftMD

8-1/2" Well TD at 9239 ftMD / 8833 ftTVD

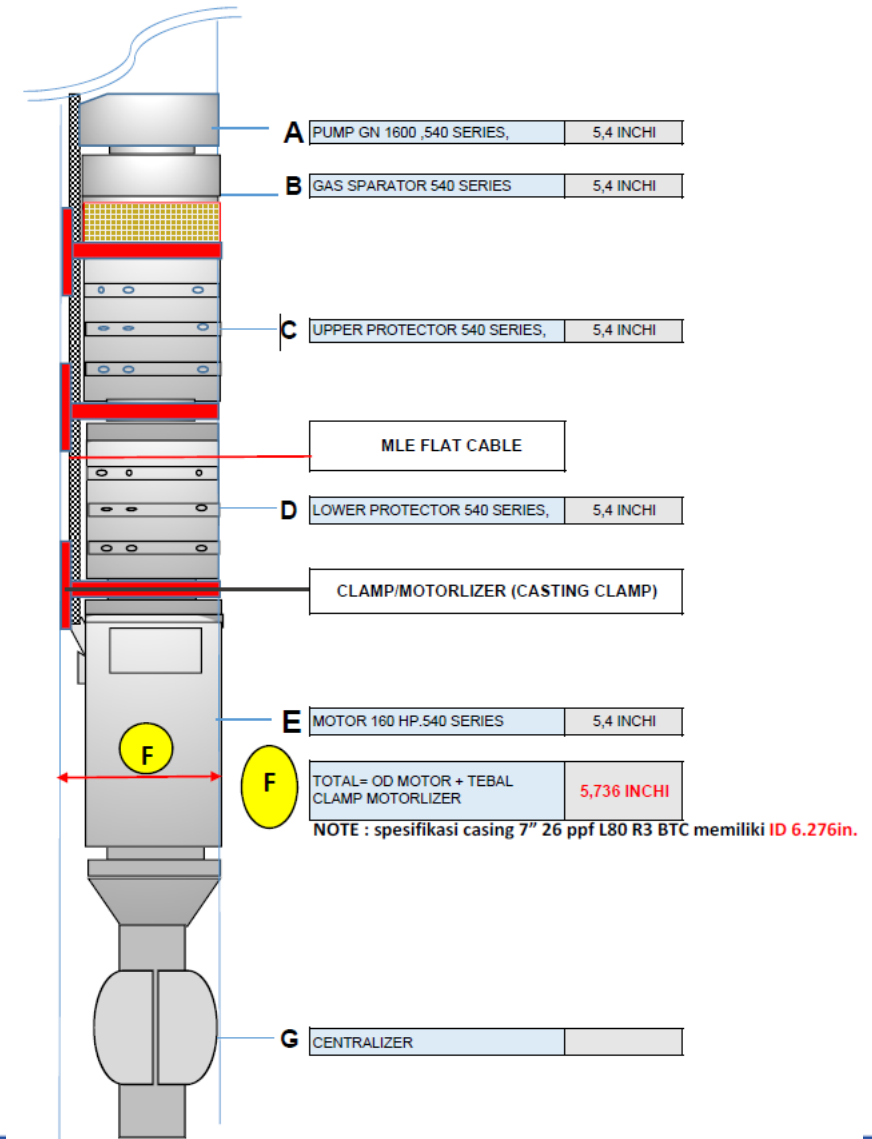
Mudi-27

Type	Directional, J-Type
KOP	± 400 ftMD / 400 ftTVD
EOC	± 1572 ftMD
DLS	± 1.15 deg / 100 ft
Inclination	± 17.3 deg
Azimuth	± 190.5 deg
TD	± 9239 ftMD / 8833.11 ftTVD

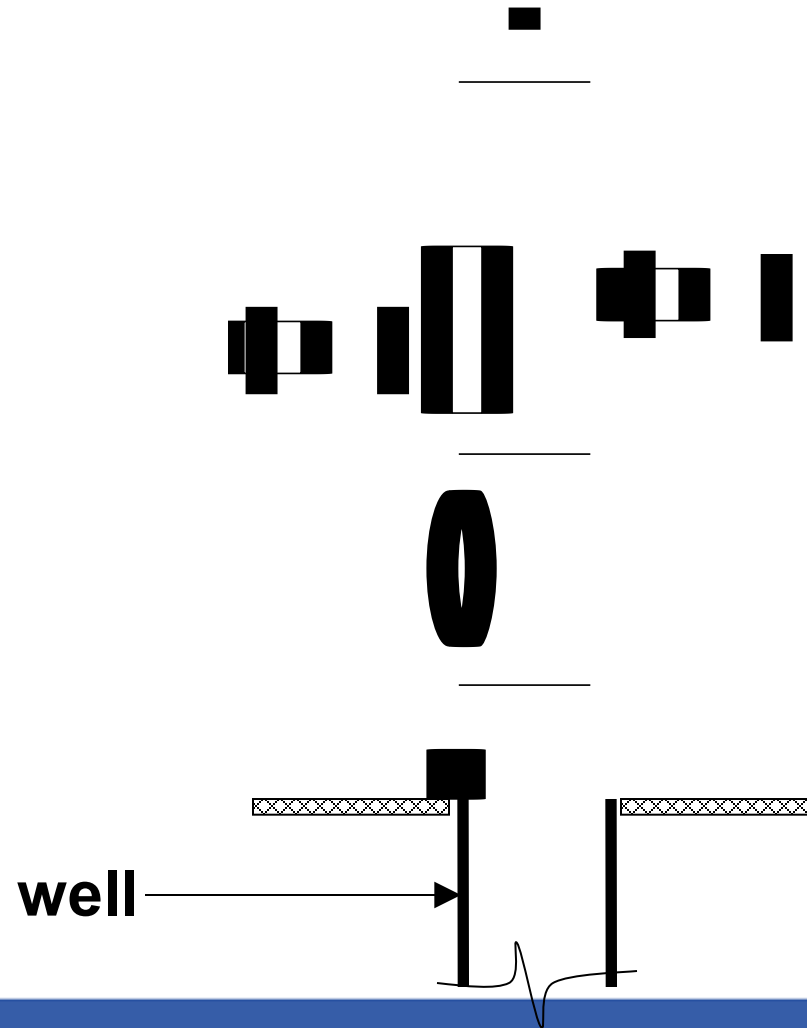
Bottom 3-1/2" Tubing, L-80, 9.2 ppf, Vam Top, R2
ESP 1600 Pump Intake Depth at ± 6000 ftMD

Perforation Interval

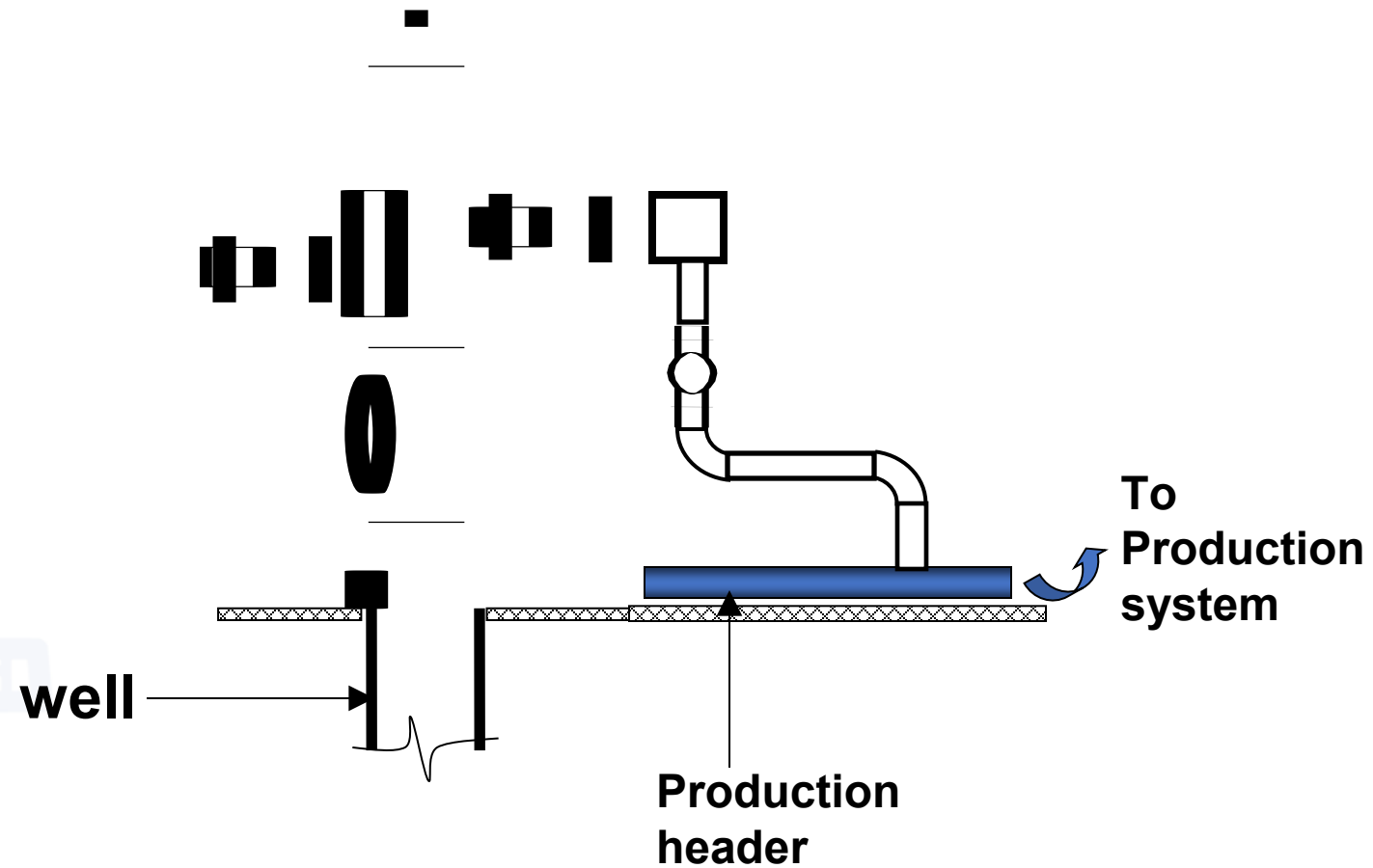
Perf. #1	± 8978.5 - 8993 ftMD (SB-7)
Perf. #2	± 9016.5 - 9047 ftMD (SB-6)
Perf. #3	± 9073 - 9085 ftMD (SB-5)
Perf. #4	± 9102 - 9135 ftMD (SB-4)



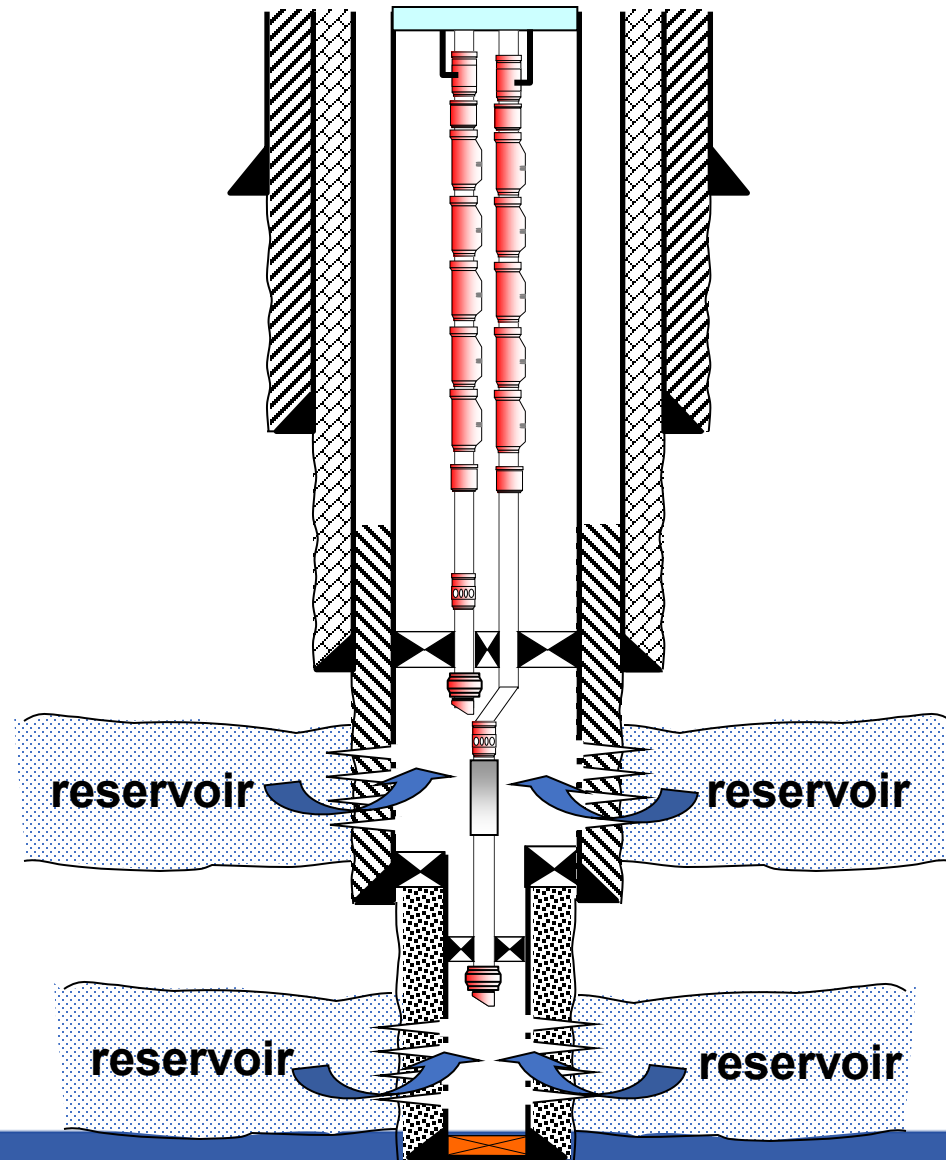
Install Christmas Tree



Production Flow Line Tie-In

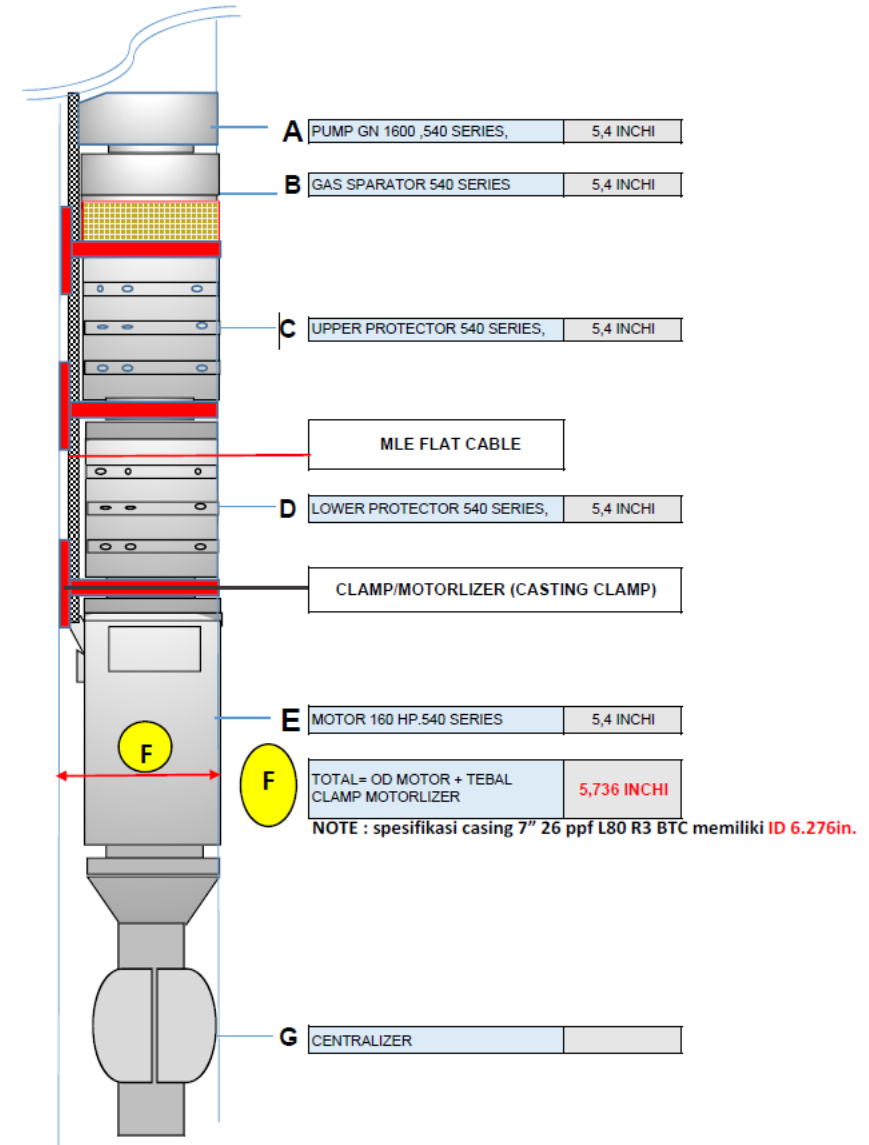
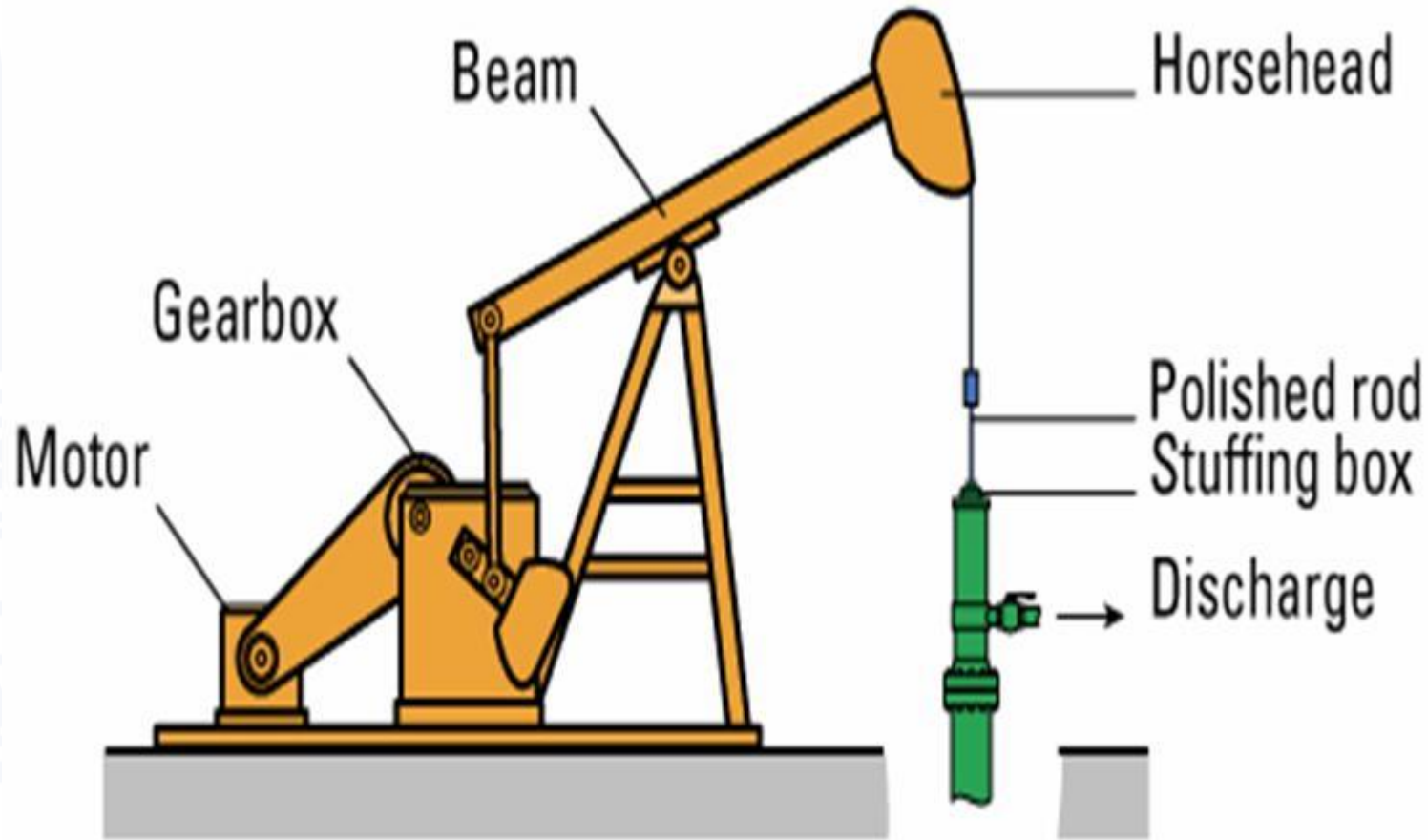


Unloading Well and Handover To Production Operations

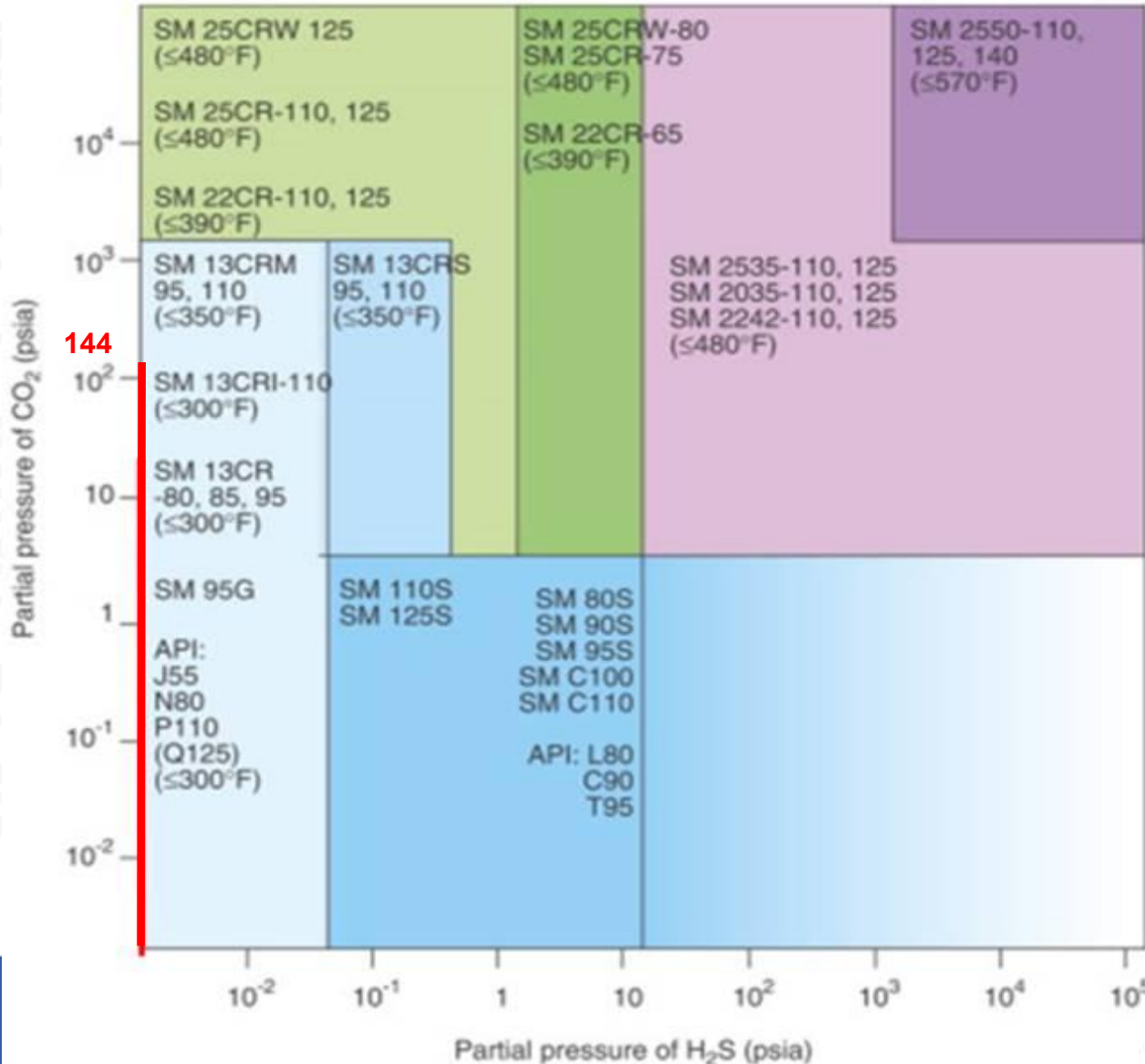


Artificial Lift

Sucker rod (Bean Pump)



Material Selection for Tubular (Casing & Tubing) (based on Sumitomo Chart)



- **Pres** = 1,800psi
- **BHT** = 250F
- **CO₂** = 8%
- **H₂S** = 0%
- **CO₂ Partial Pressure** = Est. Res Pressure X CO₂ Concentration (ppm)
= 1,800psi X 8% X 10,000 X 10⁻⁶
= **144psia**
- **H₂S Partial Pressure** = 0psia

Recommendation:

Based on CO₂ and H₂S Partial Pressure Calculation, YVOA has CO₂ Partial Pressure 144psia and no H₂S, so min. **SM13Cr-80**, will be highly recommended to apply for all tubulars (Casing and Tubing) which directly interface to CO₂ gas productions

Well Head & Tree Design Criteria (API 6A 21st Ed.)

Based on API 6A, 21st Edition, there are 4 (four) criteria of Wellhead and Tree Design:

- Temperature Class
- Material Class
- Product Specification Level (PSL)
- Performance Requirement (PR)

Note: When ordering, please specify at least the first two of above four criteria

• Temperature Class

Based on Temperature Class, the criteria are shown as follows:

- K = temperature range between -75 to 180 deg F (-60 to 82 deg C)
- L = temperature range between -50 to 180 deg F (-46 to 82 deg C)
- N = temperature range between -50 to 140 deg F (-46 to 60 deg C)
- P = temperature range between -20 to 180 deg F (-29 to 82 deg C)
- R = Room Temperature
- S = temperature range between 0 to 140 deg F (-18 to 60 deg C)
- T = temperature range between 0 to 180 deg F (-18 to 82 deg C)
 - S or T apply for most standard duty slip casing hangers
- U = temperature range between 0 to 250 deg F (-18 to 121 deg C)
 - U is common for generally available wellheads and valves
- V = temperature range between 35 to 250 deg F (2 to 121 deg C)
- X = temperature range between 0 to 350 deg F (-18 to 180 deg C)
- Y = temperature range between 0 to 650 deg F (-18 to 345 deg C)

Alternatives also apply if several wells need for combination temperature class.

Example:

- PU = temperature range between -20 to 250 deg F (-29 to 121 deg C)

Recommendation:

- YVOA has temperature range between 0 – 250 deg F, so **Temperature Class U** will be highly recommended to apply for Wellhead and Tree Design.

• Material Class

Material Class tabulation criteria can be shown as follows

	H ₂ S Partial Pressure			
	0 to < 0.05 psia	0.05 to 0.5 psia	> 0.5 to 1.5 psia ¹	> 1.5 psia ¹
CO ₂ Partial Pressure	0 to < 7 psia	AA Non-Sour Non-Corrosive	DD-NL ⁴ or EE-0,5 Sour Non-Corrosive	DD-NL ⁴ or EE-NL ³ Sour Non-Corrosive
	7 psia to 30 psia	BB Non-Sour Slightly Corrosive	EE-0,5 Sour Slightly Corrosive	EE-1,5 ² Sour Slightly Corrosive
	> 30 psia to < 200 psia	CC Non-Sour Moderately to Highly Corrosive	FF-0,5 Sour Moderately to Highly Corrosive	FF-1,5 ² Sour Moderately to Highly Corrosive
	200 psia and up	CC or HH Non-Sour Highly Corrosive	FF-0,5 or HH-0,5 Sour Highly Corrosive	FF-1,5 or HH-1,5 Sour Highly Corrosive
Corrosion resistant alloys may be needed.		GE Oil & Gas Pressure Control recommends that Corrosion Resistant Alloys be considered beginning at CO ₂ partial pressures of 200 psia. Consult with Engineering for material selection. Additional factors should be considered.		

¹ Material Class ZZ may be used when materials and trims previously used for sour service will not comply with current NACE MR0175/ISO 15156 standards.

² 17-4 stainless steel is not allowed for stems or tubing hangers above 0.5 psia H₂S.

³ 410 stainless steel is not allowed for stems or tubing hangers above 1.5 psia H₂S.

⁴ Components described and marked DD-NL may not have SS internal parts.

Recommendation:

- Based on CO₂ and H₂S Partial Pressure Calculation, YVOA has CO₂ Partial Pressure range >30psia to <200psi and no H₂S, so **Material Class CC** will be recommended to apply for Wellhead and Tree Design

Note: Material Classes for some components (Tubing Heads, Valves or Mandrel Hangers) may be the same or higher than the minimum required, due to specific product design requirements and/or NACE MR0175/ISO 15156 materials requirements.

Well Head & Tree Design Criteria (API 6A 21st Ed.)

Product Specification Level (PSL)

PSL flow chart below showed how to decide the PSL level for Wellhead and Tree Design

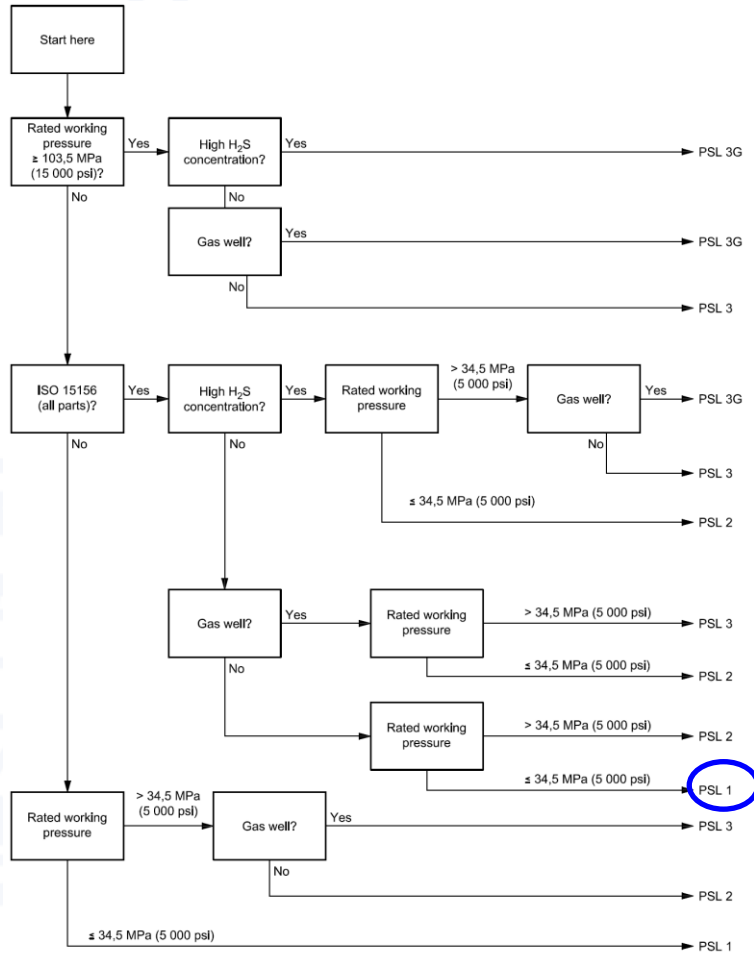


Figure A.14 — Recommended minimum PSL for primary parts of wellhead and christmas tree equipment

Performance Requirements (PR)

Cycle testing of designs:

- Different cycle tests for valves, casing hangers or other types of products
- PR1 API minimum required cycle testing
 - Minimum testing to qualify any design to be API Monogrammed
- PR2 additional test cycles
 - Manufactures option or costumer requirement
- Annex F Severe standardized testing includes mechanical, pressure and temperature cycling
 - Manufactures option or costumer requirement
 - Annex F has PR1 or PR2 (additional cycles) option

Recommendation:

PR1 API minimum required cycle testing will be recommended for Wellhead and Tree Design

Xmas Tree Design recommendation for YVOA:

Based on above four criteria, Xmas Tree design recommendation for YVOA is **U – CC – PSL1 – PR1**

YVOA is “assume as” Non-Gas Well with Reservoir Pressure 1,800psi and no H₂S content

Recommendation:

Based on the reservoir pressure, fluid type and H₂S concentration of YVOA, **PSL1** will be recommended for Wellhead and Tree Design



terima kasih